

stat_block_orbit_conformal_certificate

Specialization: v1

TL;DR. For iid replicated matched-block experiments with a known invariant assignment law, equality of calibration and target mass on every member orbit is necessary and sufficient for universal armwise score-law equality. Observable selectors are normalized and supported on the requested realized arm for assignments with positive design mass, with auxiliary draws conditional on assignment; assignments outside that support receive zero integrated weight. A finite stacked linear program finds a common two-arm selector under exact balance, quotient likelihood-ratio weighting gives armwise conformal validity under orbit overlap, and ordinary ranks admit exact-balance, total-variation, disjoint-support, and matched-pair certificates. Separately, a preorder and memoized four-cell algorithm exactly evaluates $\Gamma_{M,\alpha,\epsilon}(w)$ only for the deterministic bounded orbit-constant-score audit submodel. It retains one coherent target and within-replicate cross-arm dependence but is not a full-model joint-inclusion or causal effect-set coverage theorem.

1 Project justification

Gap. Finite-group testing and group-symmetry prediction establish validity from invariance, while causal and cluster-randomized conformal work studies counterfactual or treatment-effect targets under its own assignment regimes. Those scopes do not themselves decide whether an assignment-based calibration selector and a deployment target induce the same score law on a declared finite orbit quotient. The computational literature treats general discrete rectangular probabilities, simultaneous conformal error control, multi-target error allocation, or structured-set volume, all of which differ from this paper’s restricted deterministic orbit-score audit functional.

Niche. The note separates four layers for iid replicated matched-block experiments: universal calibration/deployment score-law equality on a finite orbit quotient; common-selector feasibility; quotient-weighted and ordinary-rank causal guarantees; and exact computation of a deliberately restricted deterministic joint rank-deficit audit. The selector conditions are normalization, realized-arm support on the positive-design support, and conditional auxiliary draws; assignments outside that support receive zero integrated weight.

Fill. The orbit-mass equivalence supplies a necessity witness as well as sufficiency, the stacked linear program supplies a computable common selector, quotient likelihood ratios supply weighted armwise validity despite member-level singularity, and clipping plus Bonferroni produces one common-member effect guarantee. The asymmetric preorder evaluator preserves the coherent common target and within-replicate cross-arm dependence and reports exact time and memory charges for the paper-defined deterministic audit functional, while explicitly excluding stochastic conditional residual scores from that frontier.

2 Related work

Hemerik and Goeman and Ramdas, Barber, Candès, and Tibshirani study finite permutation or group-testing reference mechanisms [2, 3]; Dobriban and Mengxin Yu study predictive inference under group symmetry [7]. The orbit-balance theorem instead concerns iid replicated invariant latent arrays, assignment-based calibration selection, equality of calibration and deployment score laws, and a universal finite-orbit mass equivalence with an orbit-separating necessity witness. Lei and Candès and Wang, Li, and Yu are causal and cluster-randomized conformal comparators for target and design scope [1, 5]. Tibshirani, Barber, Candès, and Ramdas supply the inherited likelihood-ratio weighted-exchangeability theorem with the test-point mass at $+\infty$ [4]; this manuscript’s weighted design step is the finite orbit quotient, quotient overlap despite member-level singularity, and two armwise guarantees combined for one common causal target by clipping and Bonferroni. For computation, Lebrun treats arbitrary rectangles for several discrete count laws [11]; Timans et al. study dependence-aware simultaneous conformal family-wise error control [12]; Schlembach, Smirnov, and Winands optimize multi-target hyperrectangle volume under joint coverage [13]; and Gao et al. optimize structured prediction-set volume [10]. This manuscript instead exactly evaluates the explicitly restricted deterministic bounded orbit-constant joint rank-deficit functional with asymmetric ranks, one coherent target, and within-replicate cross-arm dependence. The comparison asserts neither priority nor a general coverage theorem.

3 Setup

Target (estimand / structural parameter): $\tau_T = Y_{0T}(1) - Y_{0T}(0)$. Primary functional / estimator / diagnostic:

$$\mathcal{C}_w^W = I_{1,w}^W - I_{0,w}^W, \quad \Pr_P(\tau_T \in \mathcal{C}_w^W \mid \mathcal{T}) \geq 1 - \alpha$$

- B — integer; $\{1, 2, \dots\}$; block count

Definition. B is the number of matched blocks in a replicate.

- b — block index; $\{1, \dots, B\}$; block index

Definition. b indexes a matched block.

- $\mathcal{V}$ — finite set; base members; member set

Definition. $\mathcal{V} = \bigsqcup_{b=1}^B \mathcal{V}_b$ is the base-member set.

- $\mathcal{V}_b$ — finite set; $\mathcal{V}$; block members

Definition. $\mathcal{V}_b = \{v \in \mathcal{V} : \beta(v) = b\}$ is block b .

- β — map; $\beta : \mathcal{V} \rightarrow \{1, \dots, B\}$; block membership

Definition. $\beta(v)$ is the block containing v .

- $v_{b,r}$ — base member; $\mathcal{V}_b$; pair member

Definition. For a pair block, $v_{b,0}$ and $v_{b,1}$ are its members, with $r \in \{0, 1\}$.

- $\mathcal{U}$ — finite set; assignment-labelled member states; state set

Definition. $\mathcal{U} = \mathcal{V} \times \{0, 1\}$, where (v, a) records member v 's potential-outcome arm.

- $\mathcal{U}_a$ — finite subset; $\mathcal{U}$; arm state set

Definition. $\mathcal{U}_a = \mathcal{V} \times \{a\}$.

- ι_a — map; $\iota_a : \mathcal{V} \rightarrow \mathcal{U}_a$; arm lift

Definition. $\iota_a(v) = (v, a)$.

- $\mathcal{G}$ — finite group; within-block member permutations; declared symmetry group

Definition. Each $g \in \mathcal{G}$ induces a permutation π_g satisfying $\beta(\pi_g(v)) = \beta(v)$.

- π_g — permutation; $\pi_g : \mathcal{V} \rightarrow \mathcal{V}$; member action

Definition. π_g is the within-block member permutation induced by g .

- $g \cdot u$ — group action; $\mathcal{U}$; arm-preserving state action

Definition. $g \cdot (v, a) = (\pi_g(v), a)$.

- S_2 — finite group; two elements; conditional local pair-swap group

Definition. S_2 is the abstract identity-and-swap group. For a pair block b , the notation s_b denotes its nonidentity swap and $\langle s_b \rangle = \{e, s_b\} \cong S_2$ is invoked as a subgroup of $\mathcal{G}$ only when $\mathcal{V}_b = \{v_{b,0}, v_{b,1}\}$ and $\langle s_b \rangle \leq \mathcal{G}$.

- $O_{b,a}$ — pair cell; $2^{\mathcal{U}}$; arm-specific two-state pair cell

Definition. $O_{b,a} = \{(v_{b,0}, a), (v_{b,1}, a)\}$ denotes an arm-specific two-state pair cell unconditionally. If $\mathcal{V}_b = \{v_{b,0}, v_{b,1}\}$ and the pure local swap subgroup $\langle s_b \rangle$ is contained in $\mathcal{G}$, then $O_{b,a}$ is a global $\mathcal{G}$ -orbit. Absent that sufficient audit, the notation alone makes no global-orbit claim; another subgroup, including a coupled-swap subgroup, may nevertheless generate the same orbit.

- $\mathcal{O}$ — partition; $2^{\mathcal{U}}$; orbit quotient

Definition. $\mathcal{O} = \mathcal{U}/\mathcal{G}$ is the partition into arm-preserving member orbits.

- K — constant; $(0, \infty)$; bounded witness scale

Definition. K is a known outcome envelope.

- X_{iv} — covariate; $\mathbf{X}$; prediction input

Definition. X_{iv} is member v 's baseline covariate in replicate i .

- $Y_{iv}(a)$ — potential outcome; $[-K, K]$; latent response

Definition. $Y_{iv}(a)$ is member v 's potential outcome in arm a in replicate i .

- $Y_v(a)$ — generic potential outcome; $[-K, K]$; law-class coordinate

Definition. $Y_v(a)$ denotes a potential-outcome coordinate in a generic latent array.

- L_i — latent array; complete potential-outcome arrays; proof-only array

Definition. $L_i = \{(X_{iv}, Y_{iv}(0), Y_{iv}(1)) : v \in \mathcal{V}\}$.

- Z_i — assignment vector; $\{0, 1\}^{\mathcal{V}}$; randomization output

Definition. $Z_i = (Z_{iv} : v \in \mathcal{V})$ is the realized matched-block assignment.

- Y_{iv}^{obs} — observed outcome; $[-K, K]$; observed response

Definition. $Y_{iv}^{\text{obs}} = Y_{iv}(Z_{iv})$.

- O_i — observed array; observed matched-block records; algorithmic record

Definition. $O_i = \{(X_{iv}, Z_{iv}, Y_{iv}^{\text{obs}}) : v \in \mathcal{V}\}$.

- D_i — complete record; latent array with assignment; complete record

Definition. $D_i = (L_i, Z_i)$ is used only in probability statements and proofs.

- gL_i — latent-array action; latent arrays; latent relabelling

Definition. $(gL_i)_{\pi_g(v)} = (X_{iv}, Y_{iv}(0), Y_{iv}(1))$.

- gD_i — complete-record action; complete records; complete-record relabelling

Definition. $(gD_i)_{\pi_g(v)} = (X_{iv}, Y_{iv}(0), Y_{iv}(1), Z_{iv})$.

- gO_i — observed-array action; observed arrays; observed relabelling

Definition. $(gO_i)_{\pi_g(v)} = (X_{iv}, Z_{iv}, Y_{iv}^{\text{obs}})$.

- gz — assignment action; $\{0, 1\}^{\mathcal{V}}$; assignment relabelling

Definition. $(gz)_{\pi_g(v)} = z_v$.

- P — probability law; laws of latent arrays; latent law

Definition. P is the superpopulation law of a generic latent matched-block array.

- ν — probability mass function; matched-block assignments; randomization law

Definition. ν is the known assignment law.

- $\mathcal{T}$ — training sample; observed auxiliary data; fixed conditioning information

Definition. $\mathcal{T}$ is external observed training data.

- $\hat{m}_a$ — predictor; $\hat{m}_a = \hat{m}_a(\mathcal{T}) : \mathbf{X} \rightarrow \mathbb{R}$; working predictor

Definition. $\hat{m}_a$ is an arbitrary measurable map trained only from $\mathcal{T}$.

- q_{tar} — probability mass function; $\Delta(\mathcal{V})$; target selector law

Definition. q_{tar} is fixed before scored data are observed.

- T — random base member; $\mathcal{V}$; common target

Definition. $T \sim q_{\text{tar}}$ is the one common target member.

- T^a — state; $\mathcal{U}_a$; target arm lift

Definition. $T^a = \iota_a(T)$.

- X_{0T} — selected covariate; $\mathcal{X}$; target predictor input

Definition. X_{0T} is observed after target selection.

- $Y_{0T}(a)$ — selected potential outcome; $[-K, K]$; target potential outcome

Definition. $Y_{0T}(a)$ is proof-only except at the realized arm.

- $q_{\text{tar},a}$ — probability mass function; $\Delta(\mathcal{U})$; target lifted law

Definition. $q_{\text{tar},a}(\iota_a(v)) = q_{\text{tar}}(v)$, with zero mass outside $\mathcal{U}_a$.

- $\mathcal{H}$ — finite set; selector kernels; selector library

Definition. $\mathcal{H}$ indexes allowed observable selector kernels.

- $\kappa_{h,a}$ — Markov kernel; $\kappa_{h,a}(v \mid z)$; observable selector kernel

Definition. $\kappa_{h,a}$ selects an observed arm- a member under assignment z .

- $q_{h,a}^{\text{obs}}$ — probability mass function; $\Delta(\mathcal{U})$; integrated selector law

Definition. $q_{h,a}^{\text{obs}}(v, a) = \sum_z \nu(z) \kappa_{h,a}(v \mid z)$, with zero mass outside $\mathcal{U}_a$.

- w — weight vector; $\Delta(\mathcal{H})$; calibration design

Definition. When $\mathcal{H}$ is finite and nonempty, $w = (w_h : h \in \mathcal{H})$ is one fixed or prespecified mixture shared by both arms. No such w exists when $\mathcal{H} = \emptyset$.

- $\kappa_{w,a}$ — Markov kernel; $\kappa_{w,a} = \sum_{h \in \mathcal{H}} w_h \kappa_{h,a}$; implemented selector

Definition. $\kappa_{w,a}$ is the implemented mixture kernel.

- $q_{w,a}^{\text{obs}}$ — probability mass function; $\Delta(\mathcal{U})$; integrated calibration law

Definition. $q_{w,a}^{\text{obs}} = \sum_{h \in \mathcal{H}} w_h q_{h,a}^{\text{obs}}$.

- $J_{i,a}$ — random base member; $\mathcal{V}$; selected observed member

Definition. $J_{i,a} \sim \kappa_{w,a}(\cdot \mid Z_i)$.

- $U_{i,a}$ — random state; $\mathcal{U}_a$; selected score state

Definition. $U_{i,a} = (J_{i,a}, a)$.

- $S_{i,a}$ — score; $[0, \infty)$; observable calibration score

Definition. $S_{i,a} = |Y_{iJ_{i,a}}^{\text{obs}} - \hat{m}_a(X_{iJ_{i,a}})|$.

- $S_{0,a}$ — score; $[0, \infty)$; proof-only target residual

Definition. $S_{0,a} = |Y_{0T}(a) - \hat{m}_a(X_{0T})|$.

- A — matrix; $\mathbb{R}^{(2|\mathcal{O}|) \times |\mathcal{H}|}$; stacked orbit matrix

Definition. $A_{(a,O),h} = \sum_{u \in O} q_{h,a}^{\text{obs}}(u)$.

- b_{LP} — vector; $\mathbb{R}^{2|\mathcal{O}|}$; target orbit vector

Definition. $(b_{\text{LP}})_{a,O} = \sum_{u \in \mathcal{O}} q_{\text{tar},a}(u)$.

- $\delta_{a,O}(w)$ — scalar; $[-1, 1]$; signed orbit gap

Definition. $\delta_{a,O}(w) = q_{w,a}^{\text{obs}}(O) - q_{\text{tar},a}(O)$.

- $\Delta_a(w)$ — functional; $[0, 1]$; total orbit imbalance

Definition. $\Delta_a(w) = \frac{1}{2} \sum_{O \in \mathcal{O}} |\delta_{a,O}(w)|$.

- M — integer; $\{1, 2, \dots\}$; calibration size

Definition. M is the number of calibration replicates.

- α — constant; $(0, 1)$; coverage level

Definition. α is the effect miscoverage level.

- k — integer; $\{1, \dots, M+1\}$; rank index

Definition. $k = \lceil (M+1)(1 - \alpha/2) \rceil$.

- $r_{a,(k)}$ — extended scalar; $[0, \infty]$; unweighted threshold

Definition. $r_{a,(k)}$ is the k -th order statistic of the M calibration scores when $k \leq M$, and $+\infty$ when $k = M+1$.

- $I_{a,w}$ — interval; subsets of $\mathbb{R}$; clipped unweighted arm interval

Definition. $I_{a,w} = [\hat{m}_a(X_{0T}) - r_{a,(k)}, \hat{m}_a(X_{0T}) + r_{a,(k)}] \cap [-K, K]$.

- $\mathcal{C}_w$ — set; subsets of $\mathbb{R}$; clipped unweighted effect set

Definition. $\mathcal{C}_w = I_{1,w} - I_{0,w}$, where both arm intervals are clipped to $[-K, K]$.

- $p_{a,O}$ — probability; $[0, 1]$; target orbit mass

Definition. $p_{a,O} = q_{\text{tar},a}(O)$.

- $q_{a,O}$ — probability; $[0, 1]$; calibration orbit mass

Definition. $q_{a,O} = q_{w,a}^{\text{obs}}(O)$.

- $\lambda_{a,O}$ — weight; $[0, \infty)$; orbit likelihood ratio

Definition. Under orbit overlap, $\lambda_{a,O} = p_{a,O}/q_{a,O}$ when $q_{a,O} > 0$, and $\lambda_{a,O} = 0$ when $p_{a,O} = q_{a,O} = 0$.

- $O_{i,a}$ — orbit; $\mathcal{O}$; calibration orbit

Definition. $O_{i,a}$ is the orbit containing $U_{i,a}$.

- $O_{T,a}$ — orbit; $\mathcal{O}$; target orbit

Definition. $O_{T,a}$ is the orbit containing T^a .

- $r_{a,w}^{\text{W}}$ — extended scalar; $[0, \infty]$; asymmetric weighted threshold

Definition. $r_{a,w}^{\text{W}}$ is the arm- a orbit-weighted left-continuous $(1 - \varepsilon_a)$ -quantile in `def:weighted-rank-inversion`.

- $I_{a,w}^{\text{W}}$ — interval; subsets of $\mathbb{R}$; asymmetric clipped weighted arm interval

Definition.

$$I_{a,w}^{\text{W}} = [\hat{m}_a(X_{0T}) - r_{a,w}^{\text{W}}, \hat{m}_a(X_{0T}) + r_{a,w}^{\text{W}}] \cap [-K, K]$$

, where $r_{a,w}^{\text{W}}$ uses level ε_a .

- $\mathcal{C}_w^W$ — **set**; subsets of $\mathbb{R}$; **asymmetric clipped weighted effect set**

Definition. $\mathcal{C}_w^W = I_{1,w}^W - I_{0,w}^W$, with armwise levels ε_1 and ε_0 , respectively.

- τ_T — **causal estimand**; $\mathbb{R}$; **common-member effect**

Definition. $\tau_T = Y_{0T}(1) - Y_{0T}(0)$.

- $\mathcal{P}_{\mathcal{G},K}$ — **class**; latent-array laws; **invariant law class**

Definition. $\mathcal{P}_{\mathcal{G},K}$ is defined by **def:invariant-laws**.

- Q_k — **order-statistic operator**; $Q_k : \mathbb{R}^M \rightarrow \mathbb{R} \cup \{+\infty\}$; **frontier rank operator**

Definition. $Q_k(x_1, \dots, x_M)$ is the k -th order statistic if $k \leq M$, and $+\infty$ if $k = M + 1$.

- s_a — **score map**; $s_a : \mathcal{O} \rightarrow [0, K]$; **frontier decision variable**

Definition. s_a assigns an orbit-constant score in arm a .

- $\Gamma_{M,\alpha}(w)$ — **functional**; $[0, 1]$; **symmetric joint-inclusion certificate frontier**

Definition. $\Gamma_{M,\alpha}(w)$ is the equal-split exact finite worst joint-armwise-inclusion certificate deficit defined by **def:coverage-frontier**; it is not the actual effect-set coverage deficit.

- $\rho_{M,\alpha}$ — **probability**; $[0, 1]$; **finite-rank staircase error**

Definition. $\rho_{M,\alpha} = (M + 1 - k)/(M + 1) = \lfloor (M + 1)\alpha/2 \rfloor / (M + 1)$ is the exact one-arm finite-rank error.

- ε — **prespecified arm-level vector**; $\{(e_0, e_1) \in (0, 1)^2 : e_0 + e_1 \leq \alpha\}$; **asymmetric error budget**

Definition. $\varepsilon = (\varepsilon_0, \varepsilon_1)$ allocates the effect miscoverage budget across the two armwise procedures.

- k_a — **integer**; $\{1, \dots, M + 1\}$; **arm-specific rank index**

Definition. $k_a = \lceil (M + 1)(1 - \varepsilon_a) \rceil$.

- ρ_{M,ε_a} — **probability**; $[0, 1]$; **arm-specific finite-rank error**

Definition. $\rho_{M,\varepsilon_a} = (M + 1 - k_a)/(M + 1) = \lfloor (M + 1)\varepsilon_a \rfloor / (M + 1)$.

- Q_{k_a} — **order-statistic operator**; $Q_{k_a} : \mathbb{R}^M \rightarrow \mathbb{R} \cup \{+\infty\}$; **arm-specific frontier rank operator**

Definition. Q_{k_a} is the k_a -th order statistic when $k_a \leq M$, and $+\infty$ when $k_a = M + 1$.

- $r_{a,(k_a)}$ — **extended scalar**; $[0, \infty]$; **asymmetric unweighted threshold**

Definition. $r_{a,(k_a)}$ is the k_a -th arm- a calibration-score order statistic when $k_a \leq M$, and $+\infty$ otherwise.

- $I_{a,w}^\varepsilon$ — **interval**; subsets of $\mathbb{R}$; **asymmetric clipped arm interval**

Definition.

$$I_{a,w}^\varepsilon = [\hat{m}_a(X_{0T}) - r_{a,(k_a)}, \hat{m}_a(X_{0T}) + r_{a,(k_a)}] \cap [-K, K]$$

- $\mathcal{C}_w^\varepsilon$ — **set**; subsets of $\mathbb{R}$; **asymmetric clipped effect set**

Definition.

$$\mathcal{C}_w^\varepsilon = I_{1,w}^\varepsilon - I_{0,w}^\varepsilon$$

- $\Gamma_{M,\alpha,\epsilon}(w)$ — **functional**; $[0, 1]$; **asymmetric joint-inclusion certificate frontier**

Definition. $\Gamma_{M,\alpha,\epsilon}(w)$ is the exact finite worst joint-armwise-inclusion certificate deficit over bounded orbit-constant scores defined by `def:coverage-frontier`; it is not the actual effect-set coverage deficit.

- D_* — **scalar**; $[0, 2]$; **optimal imbalance value**

Definition. When $\mathcal{H}$ is finite and nonempty, D_* is the minimum total two-arm orbit imbalance in `def:minimum-imbalance-common-selector`; it is undefined for the empty library.

- $\mathcal{W}_*$ — **optimizer set**; $2^{\Delta(\mathcal{H})}$; **minimum-imbalance selectors**

Definition. When $\mathcal{H}$ is finite and nonempty, $\mathcal{W}_*$ is the nonempty set of mixtures attaining D_* ; it is undefined for the empty library.

- w_* — **weight vector**; $\mathcal{W}_*$; **optimal calibration design**

Definition. When $\mathcal{H}$ is finite and nonempty, w_* is any prespecified minimum-imbalance common selector in $\mathcal{W}_*$; it is undefined for the empty library.

- $\mathcal{A}_\epsilon$ — **active-arm set**; $2^{\{0,1\}}$; **finite-rank active arms**

Definition. $\mathcal{A}_\epsilon = \{a \in \{0, 1\} : k_a \leq M\}$ is the set of arms whose rank threshold is finite under the prespecified error allocation.

- $D_{*,\mathcal{A}}$ — **scalar**; $[0, 2]$; **optimal active-arm imbalance**

Definition. When $\mathcal{H}$ is finite and nonempty, $D_{*,\mathcal{A}}$ is the minimum orbit imbalance over rows belonging to $\mathcal{A}_\epsilon$ in `def:minimum-imbalance-common-selector`; it is undefined when $\mathcal{H} = \emptyset$.

- $\mathcal{W}_{*,\mathcal{A}}$ — **optimizer set**; $2^{\Delta(\mathcal{H})}$; **minimum active-arm imbalance selectors**

Definition. When $\mathcal{H}$ is finite and nonempty, $\mathcal{W}_{*,\mathcal{A}}$ is the nonempty set of mixtures attaining $D_{*,\mathcal{A}}$; if $\mathcal{A}_\epsilon = \emptyset$, it equals $\Delta(\mathcal{H})$. It is undefined when $\mathcal{H} = \emptyset$.

- $w_{*,\mathcal{A}}$ — **weight vector**; $\mathcal{W}_{*,\mathcal{A}}$; **optimal active-arm calibration design**

Definition. When $\mathcal{H}$ is finite and nonempty, $w_{*,\mathcal{A}}$ is any deterministically tie-broken, prespecified member of $\mathcal{W}_{*,\mathcal{A}}$; it is undefined when $\mathcal{H} = \emptyset$.

4 Assumptions

Assumption 1 (`ass:finite-group-latent-invariance`). $P \circ (gL_i)^{-1} = P \quad \forall g \in \mathcal{G}$

Standard (*finite-group array-law invariance*, [2]).

Assumption 2 (`ass:bounded-potential-outcomes`). $\max_{v \in \mathcal{V}, a \in \{0,1\}} |Y_{iv}(a)| \leq K \quad \forall i \in \{0, \dots, M\}$

Novel. *A prespecified bounded clinical, educational, or proportion scale is realistic and permits auditable finite orbit-separation and zero-coverage constructions.*

Assumption 3 (`ass:known-matched-randomization`). $\mathcal{L}(Z_0, \dots, Z_M \mid L_0, \dots, L_M) = \prod_{i=0}^M \nu, \quad \nu(gz) = \nu(z)$

Standard (*known invariant matched-block randomization*, [8]).

Assumption 4 (`ass:independent-replicated-arrays`). $L_0, \dots, L_M \stackrel{\text{iid}}{\sim} P$

Novel. *This proposal-specific technical superpopulation condition imposes iid matched-block experiments to obtain split-conformal exchangeability; it is not an assumption or theorem of Wang, Li, and Yu's Statistica Sinica preprint.*

Assumption 5 (`ass:external-observed-training`). $\mathcal{T} \perp (L_0, \dots, L_M, Z_0, \dots, Z_M, T, (J_{i,a})_{1 \leq i \leq M, a \in \{0,1\}})$

Novel. *External observed training is an implementable split construction and permits exact statements conditional on the trained predictor maps.*

Assumption 6 (ass:prespecified-target). $T \sim q_{\text{tar}}, \quad T \perp (\mathcal{T}, L_0, \dots, L_M, Z_0, \dots, Z_M, (J_{i,a}))$

Novel. A target member fixed independently of scored data is a transparent deployment rule for an externally selected facility and preserves one coherent target across both arms.

Assumption 7 (ass:observable-selector-kernel). $\kappa_{h,a}(v \mid z) \geq 0, \quad \sum_{v \in \mathcal{V}} \kappa_{h,a}(v \mid z) = 1, \quad z_v \neq a \Rightarrow \kappa_{h,a}(v \mid z) = 0 \quad \forall h \in \mathcal{H}, a$

Novel. These kernels select only a member whose realized arm is observed on every assignment that can occur under the known randomization law.

Assumption 8 (ass:selector-auxiliary-randomization). $\mathcal{L}((J_{i,a})_{1 \leq i \leq M}, a \in \{0,1\} \mid (L_i)_{0 \leq i \leq M}, (Z_i)_{0 \leq i \leq M},$

Novel. Independent auxiliary draws from observable kernels are implementable and prevent calibration selection from using latent arrays or the target.

Assumption 9 (ass:orbit-overlap). $p_{a,O} > 0 \Rightarrow q_{a,O} > 0 \quad \forall a \in \{0,1\}, O \in \mathcal{O}$

Standard (finite-support overlap, [4]).

5 Definitions

Definition 1 (def:group-action). Defined object. *arm-preserving complete-record action*

Construction. For every $g \in \mathcal{G}$, $g \cdot (v, a) = (\pi_g(v), a)$, $\beta(\pi_g(v)) = \beta(v)$, $(gL_i)_{\pi_g(v)} = (X_{iv}, Y_{iv}(0), Y_{iv}(1))$, $(gD_i)_{\pi_g(v)} = (X_{iv}, Y_{iv}(0), Y_{iv}(1), Z_{iv})$, $(gO_i)_{\pi_g(v)} = (X_{iv}, Z_{iv}, Y_{iv}^{\text{obs}})$, and $(gz)_{\pi_g(v)} = z_v$.

Inputs. $\mathcal{G}; \pi_g; \beta; L_i; D_i; O_i; gz$.

Definition 2 (def:pair-action). Defined object. *local pair-swap specialization*

Construction. For named pair members define $O_{b,a} = \{(v_{b,0}, a), (v_{b,1}, a)\}$ as a two-state pair cell, without thereby asserting that it is a global orbit. If $\mathcal{V}_b = \{v_{b,0}, v_{b,1}\}$ and the pure local swap subgroup $\langle s_b \rangle = \{e, s_b\}$ is contained in $\mathcal{G}$, then $s_b(v_{b,r}, a) = (v_{b,1-r}, a)$ for $r, a \in \{0,1\}$, s_b fixes members outside block b , and it relabels complete records by *def:group-action*. Because every group element preserves blocks and arms and block b has exactly two members, this sufficient audit implies that $O_{b,0}$ and $O_{b,1}$ are global $\mathcal{G}$ -orbits. The audit is not necessary: a coupled-swap subgroup may also generate the same orbit. Absent the audit, the pair-cell notation alone makes no global-orbit claim. Pair-specific propositions and examples are invoked after this sufficient audit; a product specialization such as S_2^B verifies every required pure local swap separately.

Inputs. $\mathcal{G}; \mathcal{V}_b; S_2; v_{b,r}; O_{b,a}; D_i$.

Definition 3 (def:observable-replicated-experiment). Defined object. *observable replicated matched-randomization experiment*

Construction. For each $i \in \{0, \dots, M\}$, draw L_i , draw $Z_i \sim \nu$, set $Y_{iv}^{\text{obs}} = Y_{iv}(Z_{iv})$, and form $O_i = \{(X_{iv}, Z_{iv}, Y_{iv}^{\text{obs}}) : v \in \mathcal{V}\}$. Draw the one common $T \sim q_{\text{tar}}$, observe X_{0T} after selection, and use neither $Y_{0T}(0)$ nor $Y_{0T}(1)$ in fitting or calibration.

Inputs. $L_i; Z_i; \nu; O_i; T; q_{\text{tar}}; X_{0T}$.

Definition 4 (def:invariant-laws). Defined object. $\mathcal{P}_{\mathcal{G},K}$

Construction.

$$\mathcal{P}_{\mathcal{G},K} = \{P : P \circ (gL_i)^{-1} = P \quad \forall g \in \mathcal{G}, \quad \max_{v \in \mathcal{V}, a \in \{0,1\}} |Y_v(a)| \leq K\}$$

Member properties. *ass:finite-group-latent-invariance; ass:bounded-potential-outcomes.*

Definition 5 (def:observable-selector-mixture). Defined object. *observable common selector*

Construction. For every $i \in \{1, \dots, M\}$ and $a \in \{0,1\}$, set $\kappa_{w,a} = \sum_{h \in \mathcal{H}} w_h \kappa_{h,a}$, draw $J_{i,a} \sim \kappa_{w,a}(\cdot \mid Z_i)$, set $U_{i,a} = (J_{i,a}, a)$, and set $q_{w,a}^{\text{obs}} = \sum_{h \in \mathcal{H}} w_h q_{h,a}^{\text{obs}}$.

Inputs. $w; \mathcal{H}; \kappa_{h,a}; J_{i,a}; U_{i,a}; q_{w,a}^{\text{obs}}$.

Definition 6 (def:joint-orbit-balance). Defined object. *joint orbit balance*

Construction.

$$\{w \in \Delta(\mathcal{H}) : q_{w,a}^{\text{obs}}(O) = q_{\text{tar},a}(O) \ \forall a \in \{0,1\}, \ O \in \mathcal{O}\}$$

Inputs. $w; q_{w,a}^{\text{obs}}; q_{\text{tar},a}; \mathcal{O}$.

Definition 7 (def:common-selector-feasibility-lp). Defined object. *stacked common-selector linear program*

Construction.

$$\{w \in \mathbb{R}^{|\mathcal{H}|} : Aw = b_{\text{LP}}, \quad \sum_{h \in \mathcal{H}} w_h = 1, \quad w_h \geq 0 \quad \forall h \in \mathcal{H}\}$$

Inputs. $A; b_{\text{LP}}; w; \mathcal{H}$.

Definition 8 (def:rank-inversion). Defined object. *unweighted observable rank inversion*

Construction. Prespecify $\varepsilon = (\varepsilon_0, \varepsilon_1)$ with $\varepsilon_a > 0$ and $\varepsilon_0 + \varepsilon_1 \leq \alpha$, and set $k_a = \lceil (M+1)(1 - \varepsilon_a) \rceil$. For each $a \in \{0,1\}$, compute $S_{i,a} = |Y_{iJ_{i,a}}^{\text{obs}} - \hat{m}_a(X_{iJ_{i,a}})|$, take $r_{a,(k_a)}$, set

$$I_{a,w}^{\varepsilon} = [\hat{m}_a(X_{0T}) - r_{a,(k_a)}, \hat{m}_a(X_{0T}) + r_{a,(k_a)}] \cap [-K, K]$$

, and return

$$\mathcal{C}_w^{\varepsilon} = I_{1,w}^{\varepsilon} - I_{0,w}^{\varepsilon}$$

. If $k_a = M+1$, set $r_{a,(k_a)} = +\infty$, so $I_{a,w}^{\varepsilon} = [-K, K]$. For proof ranks only, append independent continuous uniforms and use lexicographic ranks; reported closed intervals use weak inclusion. Under $\varepsilon_0 = \varepsilon_1 = \alpha/2$, define $k = k_0 = k_1$, $r_{a,(k)} = r_{a,(k_a)}$, $I_{a,w} = I_{a,w}^{\varepsilon}$, and $\mathcal{C}_w = \mathcal{C}_w^{\varepsilon}$, recovering the symmetric construction.

Inputs. $M; \alpha; k; S_{i,a}; \hat{m}_a; r_{a,(k)}; I_{a,w}; \mathcal{C}_w; T; \rho_{M,\alpha}$.

Definition 9 (def:weighted-rank-inversion). Defined object. *randomization-integrated orbit-weighted rank inversion*

Construction. Prespecify $\varepsilon = (\varepsilon_0, \varepsilon_1)$ with $\varepsilon_a > 0$ and $\varepsilon_0 + \varepsilon_1 \leq \alpha$. For each arm a , place mass $\lambda_{a,O_{i,a}}$ at each $S_{i,a}$, place mass $\lambda_{a,O_{T,a}}$ at $+\infty$, normalize the total mass, and take its left-continuous $(1 - \varepsilon_a)$ -quantile $r_{a,w}^{\text{W}}$. Set

$$I_{a,w}^{\text{W}} = [\hat{m}_a(X_{0T}) - r_{a,w}^{\text{W}}, \hat{m}_a(X_{0T}) + r_{a,w}^{\text{W}}] \cap [-K, K]$$

, and return $\mathcal{C}_w^{\text{W}} = I_{1,w}^{\text{W}} - I_{0,w}^{\text{W}}$. Under $\varepsilon_0 = \varepsilon_1 = \alpha/2$, this is the previous symmetric weighted construction. Thus the declared estimand functional $\mathcal{C}_w^{\text{W}}$ now denotes the arm-specific weighted procedure while retaining its equal-split specialization.

Inputs. $\lambda_{a,O}; O_{i,a}; O_{T,a}; S_{i,a}; r_{a,w}^{\text{W}}; I_{a,w}^{\text{W}}; \mathcal{C}_w^{\text{W}}; \alpha$.

Definition 10 (def:orbit-separating-law). Defined object. *bounded orbit-separating construction*

Construction. For $a \in \{0,1\}$ and $O \in \mathcal{O}$, set $\hat{m}_0 = \hat{m}_1 = 0$, set $Y_v(a) = K \mathbf{1}\{\iota_a(v) \in O\}$, set $Y_v(1-a) = 0$, and randomize the orbit-constant prototype by a uniform $g \in \mathcal{G}$ before drawing $Z_i \sim \nu$.

Inputs. $a; O; K; \hat{m}_a; Y_v(a); \iota_a; \mathcal{G}; Z_i; \nu$.

Definition 11 (def:disjoint-orbit-law). Defined object. *bounded disjoint-orbit construction*

Construction. Fix $a \in \{0,1\}$. Set $\hat{m}_0 = \hat{m}_1 = 0$, set $Y_v(a) = K$ on target-supported arm- a orbits, set $Y_v(a) = 0$ on every state with positive calibration mass and on remaining states, set $Y_v(1-a) = 0$, and randomize the orbit-constant prototype by a uniform $g \in \mathcal{G}$ before drawing $Z_i \sim \nu$.

Inputs. $K; q_{w,a}^{\text{obs}}; q_{\text{tar},a}; \iota_a; \mathcal{G}; Z_i; \nu$.

Definition 12 (def:coverage-frontier). Defined object. *finite common-target orbit-constant joint-inclusion certificate frontier*

Construction. For prespecified $\varepsilon = (\varepsilon_0, \varepsilon_1)$, define

$$\Gamma_{M,\alpha,\varepsilon}(w) = \max_{\substack{s_a: \mathcal{O} \rightarrow [0,K] \\ a \in \{0,1\}}} [(1-\alpha) - \Pr\{s_1(O_{T,1}) \leq Q_{k_1}(s_1(O_{1,1}), \dots, s_1(O_{M,1})) \text{ and } s_0(O_{T,0}) \leq Q_{k_0}(s_0(O_{1,0}), \dots, s_0(O_{M,0}))\}]_+.$$

Here $O_{i,a} \sim q_{w,a}^{\text{obs}}$, the target orbits are the two lifts of the one common $T \sim q_{\text{tar}}$, and the calibration draws use **ass:selector-auxiliary-randomization**, including their within-replicate cross-arm coupling. The functional is restricted to bounded orbit-constant score maps. Its probability object is the joint armwise rank-inclusion event, so $\Gamma_{M,\alpha,\varepsilon}(w)$ is the exact worst joint-armwise-inclusion certificate deficit over that score class, not the actual effect-set coverage deficit. Under $\varepsilon_0 = \varepsilon_1 = \alpha/2$, set $\Gamma_{M,\alpha}(w) = \Gamma_{M,\alpha,\varepsilon}(w)$, recovering the symmetric joint-inclusion certificate frontier.

Inputs. $M; \alpha; \varepsilon; k_a; w; s_a; Q_{k_a}; O_{i,a}; O_{T,a}; q_{w,a}^{\text{obs}}; q_{\text{tar},a}; T$.

Definition 13 (def:finite-rank-error). Defined object. *exact finite-rank error*

Construction. For $N = M + 1$, prespecified $\varepsilon = (\varepsilon_0, \varepsilon_1)$, and $k_a = \lceil N(1 - \varepsilon_a) \rceil$, define

$$\rho_{M,\varepsilon_a} = \frac{N - k_a}{N} = \frac{\lfloor N\varepsilon_a \rfloor}{N}, \quad a \in \{0, 1\}.$$

Under $\varepsilon_0 = \varepsilon_1 = \alpha/2$, write their common value as

$$\rho_{M,\alpha} = \frac{M + 1 - k}{M + 1} = \frac{\lfloor (M + 1)\alpha/2 \rfloor}{M + 1}.$$

Inputs. $M; \alpha; k; \rho_{M,\alpha}$.

Definition 14 (def:minimum-imbalance-common-selector). Defined object. *minimum-imbalance common selector*

Construction. For every finite selector library define $\Delta(\mathcal{H}) = \{w \in \mathbb{R}^{|\mathcal{H}|} : w_h \geq 0 \ \forall h, \sum_{h \in \mathcal{H}} w_h = 1\}$; in particular $\Delta(\emptyset) = \emptyset$. If $\mathcal{H} = \emptyset$, no selector mixture, implemented kernel, selector-indexed ordinary or weighted procedure, D_* , $D_{*,\mathcal{A}}$, $\mathcal{W}_*$, $\mathcal{W}_{*,\mathcal{A}}$, w_* , or $w_{*,\mathcal{A}}$ exists. If $\mathcal{H}$ is finite and nonempty, retain the joint two-arm diagnostics

$$D_* = \min_{w \in \Delta(\mathcal{H})} \frac{1}{2} \sum_{a=0}^1 \sum_{O \in \mathcal{O}} |(Aw - b_{\text{LP}})_{a,O}|, \quad \mathcal{W}_* = \arg \min_{w \in \Delta(\mathcal{H})} \frac{1}{2} \sum_{a=0}^1 \sum_{O \in \mathcal{O}} |(Aw - b_{\text{LP}})_{a,O}|.$$

Their absolute-value linear program minimizes $\frac{1}{2} \sum_{a=0}^1 \sum_{O \in \mathcal{O}} t_{a,O}$ subject to $-t_{a,O} \leq (Aw - b_{\text{LP}})_{a,O} \leq t_{a,O}$, $t_{a,O} \geq 0$, $w \in \Delta(\mathcal{H})$, and their dual is

$$\max_{\|y\|_\infty \leq 1/2} \left\{ \min_{h \in \mathcal{H}} y^\top A_{\cdot h} - y^\top b_{\text{LP}} \right\}.$$

For the prespecified armwise ranks define $\mathcal{A}_\varepsilon = \{a \in \{0, 1\} : k_a \leq M\}$ and

$$D_{*,\mathcal{A}} = \min_{w \in \Delta(\mathcal{H})} \frac{1}{2} \sum_{a \in \mathcal{A}_\varepsilon} \sum_{O \in \mathcal{O}} |(Aw - b_{\text{LP}})_{a,O}|, \quad \mathcal{W}_{*,\mathcal{A}} = \arg \min_{w \in \Delta(\mathcal{H})} \frac{1}{2} \sum_{a \in \mathcal{A}_\varepsilon} \sum_{O \in \mathcal{O}} |(Aw - b_{\text{LP}})_{a,O}|.$$

The active-row linear program minimizes $\frac{1}{2} \sum_{a \in \mathcal{A}_\varepsilon} \sum_{O \in \mathcal{O}} t_{a,O}$ subject to the same two-sided constraints only for $a \in \mathcal{A}_\varepsilon$, together with $t_{a,O} \geq 0$ and $w \in \Delta(\mathcal{H})$. Its dual is

$$\max_{\substack{\|y\|_\infty \leq 1/2 \\ y_{a,O} = 0 \text{ for } a \notin \mathcal{A}_\varepsilon}} \left\{ \min_{h \in \mathcal{H}} y^\top A_{\cdot h} - y^\top b_{\text{LP}} \right\}.$$

If $\mathcal{A}_\epsilon = \emptyset$, define $D_{*,\mathcal{A}} = 0$ and $\mathcal{W}_{*,\mathcal{A}} = \Delta(\mathcal{H})$, and choose $w_{*,\mathcal{A}}$ without solving an LP. The full objects D_* and $\mathcal{W}_*$ remain the joint two-arm balance diagnostics; $D_{*,\mathcal{A}} = 0$ characterizes only active-row balance unless both arms are active. Primal and dual equality, attainment, and the coverage certificates are proved in *thm:active-arm-minimum-imbalance-selector-certificate*.

Inputs. $\mathcal{H}$; A ; b_{LP} ; $\Delta_a(w)$; ϵ ; k_a ; $\mathcal{A}_\epsilon$; D_* ; $D_{*,\mathcal{A}}$; $\mathcal{W}_*$; $\mathcal{W}_{*,\mathcal{A}}$; w_* ; $w_{*,\mathcal{A}}$; w .

Definition 15 (def:simplex-vertex-key-prepass). Defined object. *simplex vertex, edge, face, and rank-one exact-key prepass*

Construction. For a finite set Π_* of probability mass functions on $\{0,1\}^2$, first set

$$\Pi_{\text{vert}} = \{\pi \in \Pi_* : |\text{supp}(\pi)| = 1\}, \quad \Pi_{\text{edge}} = \{\pi \in \Pi_* \setminus \Pi_{\text{vert}} : |\text{supp}(\pi)| = 2\}.$$

For $\pi \in \Pi_{\text{vert}}$, let $u = (x, y)$ be its unique support point and return $\mathbf{1}\{Mx < k_0, My < k_1\}$. For $\pi \in \Pi_{\text{edge}}$, order its two support points $u = (u_0, u_1)$ and $v = (v_0, v_1)$ deterministically; both possible orientations may be compared, with the smaller certified charge chosen and lexicographic tie-breaking. Put $p = \pi_v \in (0, 1)$, let $N \sim \text{Bin}(M, p)$, and define $c_a = Mu_a$ and $d_a = v_a - u_a$ for $a \in \{0, 1\}$. Set

$$L_\pi = \max(\{0\} \cup \{c_a - k_a + 1 : d_a = -1\}), \quad U_\pi = \min(\{M\} \cup \{k_a - c_a - 1 : d_a = 1\}).$$

If $d_a = 0$ and $c_a \geq k_a$ for some a , declare $[L_\pi, U_\pi]$ empty. Otherwise return

$$D_{M,k_0,k_1}(\pi) = \Pr\{L_\pi \leq N \leq U_\pi\}.$$

For a nonempty interval define

$$\eta_\pi = \min\{\min\{L_\pi, M - L_\pi\} + U_\pi - L_\pi + 1, L_\pi + M - U_\pi, \min\{U_\pi + 1, M - U_\pi\} + \min\{L_\pi, M - L_\pi + 1\}\},$$

and set $\eta_\pi = 0$ for an empty interval. Direct interval summation, complementary left-and-right tails, and a difference of two shorter binomial tails are three exact alternatives; choose the least charge deterministically and write $E_M(\pi) = 1 + \lceil \log_2 M \rceil + \eta_\pi$. The edge evaluation uses $O(E_M(\pi))$ exact arithmetic operations and $O(1)$ auxiliary workspace.

For every remaining support-three key and every corner orientation $\sigma \in \{L, U\}^2$, let π^σ be the transformed four-cell law and let $t_a^\sigma = k_a$ under L and $t_a^\sigma = M - k_a + 1$ under U . Define

$$\Sigma_{\text{face}}(\pi) = \{\sigma : \pi_{11}^\sigma = 0, t_0^\sigma + t_1^\sigma > M\}, \quad \Pi_{\text{face}} = \{\pi \in \Pi_* \setminus (\Pi_{\text{vert}} \cup \Pi_{\text{edge}}) : \Sigma_{\text{face}}(\pi) \neq \emptyset\}.$$

For $\sigma \in \Sigma_{\text{face}}(\pi)$, let T_a^σ be the transformed terminal count and set

$$J_\sigma = 1 - \Pr\{T_0^\sigma \geq t_0^\sigma\} - \Pr\{T_1^\sigma \geq t_1^\sigma\}.$$

Reconstruct $D_{M,k_0,k_1}(\pi)$ from J_σ by the LL, LU, UL, or UU corner identity. Each required reconstruction marginal is the complement of one of the two tails already evaluated. With

$$\ell_M(t) = 1 + \lceil \log_2 M \rceil + \min\{t, M - t + 1\}, \quad F_M(\pi) = 1 + \min_{\sigma \in \Sigma_{\text{face}}(\pi)} \{\ell_M(t_0^\sigma) + \ell_M(t_1^\sigma)\},$$

choose a minimizing orientation and break ties lexicographically. The face evaluation uses $O(F_M(\pi))$ exact arithmetic operations and $O(1)$ auxiliary workspace.

For every remaining support-four key, test the exact memoized-cell identity

$$\pi_{00}\pi_{11} = \pi_{01}\pi_{10}.$$

Define

$$\Pi_{\text{rank1}} = \{\pi \in \Pi_* \setminus (\Pi_{\text{vert}} \cup \Pi_{\text{edge}} \cup \Pi_{\text{face}}) : |\text{supp}(\pi)| = 4, \pi_{00}\pi_{11} = \pi_{01}\pi_{10}\}.$$

For $\pi \in \Pi_{\text{rank1}}$, put

$$\theta_0 = \pi_{10} + \pi_{11}, \quad \theta_1 = \pi_{01} + \pi_{11}, \quad \bar{k}_a = \min\{k_a, M - k_a + 1\},$$

and evaluate

$$D_{M,k_0,k_1}(\pi) = H_{M,k_0}(\theta_0)H_{M,k_1}(\theta_1), \quad H_{M,k}(\theta) = \Pr\{\text{Bin}(M, \theta) < k\}.$$

With

$$R_{M,k_0,k_1} = 1 + 2\lceil \log_2 M \rceil + \bar{k}_0 + \bar{k}_1,$$

the rank-one evaluation uses $O(R_{M,k_0,k_1})$ exact arithmetic operations and $O(1)$ auxiliary workspace, with deterministic exact testing and no approximation. Only keys in

$$\Pi_\star \setminus (\Pi_{\text{vert}} \cup \Pi_{\text{edge}} \cup \Pi_{\text{face}} \cup \Pi_{\text{rank1}})$$

enter the remaining rolling, multinomial-sum, or positive-corner dispatcher.

Inputs. $\Pi_\star$; M ; k_0 ; k_1 .

6 Main results

Theorem 1 (thm:orbit-balance-iff). *Conditional on $\mathcal{T}$, joint orbit balance is equivalent to equality in law of $S_{i,a}$ and $S_{0,a}$, for each $a \in \{0, 1\}$, uniformly over $P \in \mathcal{P}_{\mathcal{G},K}$ and arbitrary fixed measurable trained predictor maps $\hat{m}_a : \mathbf{X} \rightarrow \mathbb{R}$. Under balance, conditional on $\mathcal{T}$, $S_{1,a}, \dots, S_{M,a}, S_{0,a}$ are iid for each arm. The assertion is armwise and makes no joint-exchangeability claim across arms. If an orbit mass differs, **def:orbit-separating-law** gives a bounded invariant law and fixed predictor maps with unequal score laws.*

Proof. Fix $a \in \{0, 1\}$ and condition throughout on $\mathcal{T}$. By **lem:orbit-score-factorization**, there are orbit-indexed laws $F_{a,O}$ such that, for every Borel set $C \subseteq [0, \infty)$,

$$\Pr\{S_{i,a} \in C \mid \mathcal{T}\} = \sum_{O \subseteq \mathcal{U}_a} q_{a,O} F_{a,O}(C), \quad \Pr\{S_{0,a} \in C \mid \mathcal{T}\} = \sum_{O \subseteq \mathcal{U}_a} p_{a,O} F_{a,O}(C). \quad (1)$$

Suppose first that joint orbit balance holds. By **def:joint-orbit-balance**, $q_{a,O} = p_{a,O}$ for every arm a and every orbit O . The two mixtures in (1) are therefore equal for every Borel C , every $P \in \mathcal{P}_{\mathcal{G},K}$, and every fixed measurable predictor $\hat{m}_a$. This proves equality in law of $S_{i,a}$ and $S_{0,a}$. The factorization lemma also gives that the M calibration orbit-score pairs are iid and independent of the target orbit-score pair. Since their score marginals now agree, $S_{1,a}, \dots, S_{M,a}, S_{0,a}$ are iid conditional on $\mathcal{T}$.

Conversely, suppose joint orbit balance fails. Then there are an arm a and an orbit $O_\star \subseteq \mathcal{U}_a$ for which

$$q_{a,O_\star} \neq p_{a,O_\star}. \quad (2)$$

Apply **def:orbit-separating-law** to $(a, O_\star)$, with $\hat{m}_0 = \hat{m}_1 = 0$. Its prototype satisfies

$$Y_v(a) = K \mathbf{1}\{(v, a) \in O_\star\}, \quad Y_v(1-a) = 0. \quad (3)$$

All potential outcomes in (3) lie in $[-K, K]$, as required by **ass:bounded-potential-outcomes**. If $L^\star$ denotes the prototype and G_0 is uniform on the finite group $\mathcal{G}$, then, for every $h \in \mathcal{G}$, the arrays $hG_0L^\star$ and $G_0L^\star$ have the same law because left multiplication by h permutes the uniform group law. Because $O_\star$ is a $\mathcal{G}$ -orbit, relabelling preserves the indicator in (3). Thus the symmetrized latent law is $\mathcal{G}$ -invariant and, by **def:invariant-laws**, belongs to $\mathcal{P}_{\mathcal{G},K}$. Taking independent copies and then drawing assignments from the known randomization law gives the experiment in **def:observable-replicated-experiment**.

On every assignment z with $\nu(z) > 0$, **ass:observable-selector-kernel** assigns positive selector mass only to members whose realized arm is a . Hence $Y_{iJ_{i,a}}^{\text{obs}} = Y_{iJ_{i,a}}(a)$, and (3) yields

$$S_{i,a} = K \mathbf{1}\{U_{i,a} \in O_\star\}, \quad S_{0,a} = K \mathbf{1}\{T^a \in O_\star\}. \quad (4)$$

Because $K > 0$, (2)–(4) imply

$$\Pr\{S_{i,a} = K \mid \mathcal{T}\} = q_{a,O_\star} \neq p_{a,O_\star} = \Pr\{S_{0,a} = K \mid \mathcal{T}\}. \quad (5)$$

Thus, whenever an orbit mass differs, the bounded invariant construction supplies fixed predictor maps and unequal score laws. Consequently, equality of the armwise score laws uniformly over the declared invariant-law class and fixed measurable predictors forces joint orbit balance.

Finally, invariance of ν implies that its positive-mass support is invariant under the group action, since $\nu(gz) = \nu(z) > 0$ whenever $\nu(z) > 0$. Assignments outside that support have zero integrated weight in $q_{h,a}^{\text{obs}}$ and hence in every mixture $q_{w,a}^{\text{obs}}$; arbitrary total kernel extensions there therefore do not affect those mixtures. The proof fixes one arm at a time and makes no joint-exchangeability claim across arms. $\square$

Justification. The theorem gives a universal finite-orbit mass characterization for equality of assignment-selected calibration and deployment score laws over iid replicated invariant arrays, together with an orbit-separating necessity witness. Hemerik–Goeman and Ramdas et al. are finite permutation/group-testing comparators [2, 3], Dobriban–Yu is a group-symmetry prediction comparator [7], and Lei–Candès and Wang–Li–Yu are causal/design comparators [1, 5]; no priority or absence claim is made. *Gap.* The comparison concerns different targets and mechanisms: testing under group transformations or predictive validity under symmetry does not itself impose equality of the calibration-selector and deployment-target orbit mixtures. The proof uses normalized realized-arm selector kernels and conditional selector draws, not selector equivariance. *Consumer.* Use orbit-mass equality as the exact armwise audit for universal score-law equality in the replicated-array model; if a mass differs, use the bounded orbit-separating witness.

Theorem 2 (thm:common-selector-lp). *Adopt $\Delta(\emptyset) = \emptyset$ and $\text{conv}(\emptyset) = \emptyset$. There exists $w \in \Delta(\mathcal{H})$ jointly orbit-balancing both arms if and only if $b_{\text{LP}} \in \text{conv}\{A_{\cdot h} : h \in \mathcal{H}\}$. Equivalently, the stacked LP in `def:common-selector-feasibility-lp` is feasible. If $\mathcal{H} = \emptyset$, infeasibility is certified by $\sum_{h \in \emptyset} w_h = 0 \neq 1$; no selector mixture, implemented kernel, selector-indexed ordinary or weighted procedure, D_* , $\mathcal{W}_*$, or w_* exists. If $\mathcal{H} \neq \emptyset$ and the condition fails, a strict Farkas hyperplane separates b_{LP} from the nonempty convex hull. Fix a prespecified $\varepsilon = (\varepsilon_0, \varepsilon_1)$ with $\varepsilon_a > 0$ and $\varepsilon_0 + \varepsilon_1 \leq \alpha$. Conditional on $\mathcal{T}$, a fixed feasible common w , iid arrays, the independent common target T , and selector auxiliary draws imply, for each arm,*

$$\Pr(S_{0,a} > r_{a,(k_a)} \mid \mathcal{T}) \leq \frac{M + 1 - k_a}{M + 1} = \rho_{M,\varepsilon_a} \leq \varepsilon_a.$$

Clipping to $[-K, K]$ and Bonferroni therefore give

$$\Pr(\tau_T \in \mathcal{C}_w^\varepsilon \mid \mathcal{T}) \geq 1 - \rho_{M,\varepsilon_0} - \rho_{M,\varepsilon_1} \geq 1 - \varepsilon_0 - \varepsilon_1 \geq 1 - \alpha,$$

without arm independence. Under $\varepsilon_0 = \varepsilon_1 = \alpha/2$, one has $k_0 = k_1 = k$, $I_{a,w}^\varepsilon = I_{a,w}$, and $\mathcal{C}_w^\varepsilon = \mathcal{C}_w$, so the symmetric rank and coverage formulas follow as a corollary.

Proof. For any $w \in \Delta(\mathcal{H})$, the definitions of A , $q_{w,a}^{\text{obs}}$, and b_{LP} give, row by row,

$$(Aw)_{a,O} = \sum_{h \in \mathcal{H}} w_h q_{h,a}^{\text{obs}}(O) = q_{w,a}^{\text{obs}}(O), \quad (b_{\text{LP}})_{a,O} = q_{\text{tar},a}(O). \quad (1)$$

Hence $Aw = b_{\text{LP}}$ is exactly joint orbit balance. When $\mathcal{H} \neq \emptyset$, the set of all Aw with $w \in \Delta(\mathcal{H})$ is $\text{conv}\{A_{\cdot h} : h \in \mathcal{H}\}$, proving the convex-hull and LP equivalences. If $\mathcal{H} = \emptyset$, the adopted simplex is empty and the normalization equation reads $0 = 1$; all claimed nonexistence conclusions then follow directly from the definitions.

For completeness, suppose $\mathcal{H} \neq \emptyset$ and b_{LP} is outside the convex hull C . The finite hull is compact. Let $c_\star \in C$ minimize Euclidean distance to b_{LP} , and put $y = b_{\text{LP}} - c_\star \neq 0$. For any $c \in C$, minimizing the squared distance along the segment from $c_\star$ to c gives $y^\top(c - c_\star) \leq 0$. Consequently

$$y^\top b_{\text{LP}} - \sup_{c \in C} y^\top c \geq y^\top (b_{\text{LP}} - c_\star) = \|y\|_2^2 > 0,$$

which is the asserted strict Farkas hyperplane.

Now fix a feasible common w and condition on $\mathcal{T}$. By `thm:orbit-balance-iff`, for each arm the $M + 1$ scores $S_{1,a}, \dots, S_{M,a}, S_{0,a}$ are iid. Append independent continuous uniforms and order lexicographically. The target rank is uniform on $\{1, \dots, M + 1\}$. If $k_a \leq M$, the event $S_{0,a} > r_{a,(k_a)}$ implies that the target lexicographic rank exceeds k_a ; if $k_a = M + 1$, the event is empty by definition. Thus in both cases

$$\Pr(S_{0,a} > r_{a,(k_a)} \mid \mathcal{T}) \leq \frac{M + 1 - k_a}{M + 1} = \rho_{M,\varepsilon_a}. \quad (2)$$

Writing $N = M + 1$, the identity $\lceil N - N\varepsilon_a \rceil = N - \lfloor N\varepsilon_a \rfloor$ yields $\rho_{M,\varepsilon_a} = \lfloor N\varepsilon_a \rfloor / N \leq \varepsilon_a$.

Whenever $S_{0,a} \leq r_{a,(k_a)}$, the bounded outcome $Y_{0T}(a) \in [-K, K]$ belongs to the clipped interval $I_{a,w}^\varepsilon$. Simultaneous inclusion of the two potential outcomes implies inclusion of their difference in $\mathcal{C}_w^\varepsilon$. The union bound, without any arm-independence premise, and (2) therefore give

$$\Pr(\tau_T \in \mathcal{C}_w^\varepsilon \mid \mathcal{T}) \geq 1 - \rho_{M,\varepsilon_0} - \rho_{M,\varepsilon_1} \geq 1 - \varepsilon_0 - \varepsilon_1 \geq 1 - \alpha.$$

On setting $\varepsilon_0 = \varepsilon_1 = \alpha/2$, the definitions give $k_0 = k_1 = k$, the symmetric intervals, and the symmetric effect set. $\square$

Justification. The convex-hull and Farkas characterization turns joint two-arm orbit balance with one common mixture into a finite, computable feasibility certificate; its asymmetric coverage conclusion uses the exact sum of the two armwise staircase errors. *Gap.* Separate armwise mixtures would not implement the declared common selector design, and a symmetric-only conclusion hides unequal rank allocations. *Consumer.* The protocol either obtains a feasible common mixture or a separating hyperplane explaining why none exists.

Theorem 3 (thm:weighted-orbit-conformal). *Assume that $\mathcal{H}$ is finite and nonempty, and let $w \in \Delta(\mathcal{H})$ be fixed or prespecified. Let $\epsilon_0, \epsilon_1 > 0$ be prespecified and satisfy $\epsilon_0 + \epsilon_1 \leq \alpha$, and let arm a 's weighted threshold in **def:weighted-rank-inversion** use its left-continuous $(1 - \epsilon_a)$ -quantile. Conditional on $\mathcal{T}$, orbit overlap implies*

$$\Pr_P(\tau_T \in \mathcal{C}_w^W \mid \mathcal{T}) \geq 1 - \epsilon_0 - \epsilon_1 \geq 1 - \alpha$$

for every $P \in \mathcal{P}_{\mathcal{G},K}$ and arbitrary fixed measurable trained predictor maps. Invariance makes score laws coincide within an orbit, so the orbit likelihood ratios suffice even with member-level selector singularity. Under joint balance, every relevant $\lambda_{a,O} = 1$, and $r_{a,w}^W = r_{a,(k_a)}$ for $k_a = \lceil (M+1)(1 - \epsilon_a) \rceil$, including the $+\infty$ clause when $k_a = M+1$; hence weighted inversion recovers ordinary arm-specific rank inversion. Taking $\epsilon_0 = \epsilon_1 = \alpha/2$ gives $k_0 = k_1 = \lceil (M+1)(1 - \alpha/2) \rceil$ and recovers the original symmetric statement and formulas. The one-arm rank inequality is the finite-support likelihood-ratio weighted-exchangeability validity argument; the design-specific deductions are the orbit-quotient likelihood-ratio reduction, the sufficiency of quotient overlap under member-level singularity, and its integration with one common target, one common two-arm selector mixture, clipping, and Bonferroni.

Proof. Condition on $\mathcal{T}$ and fix an arm a . By **lem:orbit-score-factorization**, define laws on orbit-score pairs by

$$Q_a(\{O\} \times C) = q_{a,O} F_{a,O}(C), \quad P_a(\{O\} \times C) = p_{a,O} F_{a,O}(C) \quad (1)$$

for Borel $C \subseteq [0, \infty)$. Orbit overlap gives $P_a \ll Q_a$. On every Q_a -supported orbit, a version of the likelihood ratio is

$$\frac{dP_a}{dQ_a}(O, s) = \lambda_{a,O}. \quad (2)$$

Indeed, for $q_{a,O} > 0$, the common conditional factor in (1) cancels and the ratio is $p_{a,O}/q_{a,O}$; if $q_{a,O} = 0$, overlap gives $p_{a,O} = 0$. Thus quotient weights suffice even if the member-level laws are singular. The factorization lemma also gives iid calibration pairs from Q_a , independently of the target pair from P_a .

Apply **lem:finite-support-weighted-rank-validity** on the orbit-score space with score map $R(O, S) = S$ and the likelihood ratio in (2), namely $\lambda_{a,O}$. Its augmented empirical measure is exactly the measure in **def:weighted-rank-inversion**: calibration observation i has mass $\lambda_{a,O_{i,a}}$ at $S_{i,a}$, and the target has mass $\lambda_{a,O_{T,a}}$ at $+\infty$. Hence

$$\Pr\{S_{0,a} > r_{a,w}^W \mid \mathcal{T}\} \leq \epsilon_a. \quad (3)$$

Because $Y_{0T}(a) \in [-K, K]$, clipping preserves target membership and

$$S_{0,a} \leq r_{a,w}^W \iff Y_{0T}(a) \in I_{a,w}^W. \quad (4)$$

If (4) holds in both arms, then

$$\tau_T = Y_{0T}(1) - Y_{0T}(0) \in I_{1,w}^W - I_{0,w}^W = \mathcal{C}_w^W. \quad (5)$$

The union bound applied to (3)–(5), without cross-arm independence, proves

$$\Pr_P(\tau_T \in \mathcal{C}_w^W \mid \mathcal{T}) \geq 1 - \epsilon_0 - \epsilon_1 \geq 1 - \alpha.$$

Under joint balance, $p_{a,O} = q_{a,O}$, so every orbit that can be sampled has weight one. The normalized augmented measure then has M equal atoms at the calibration scores and one equal atom at $+\infty$. With $N = M + 1$ and $k_a = \lceil N(1 - \epsilon_a) \rceil$, its left-continuous quantile is the k_a -th calibration order statistic when $k_a \leq M$, and is $+\infty$ when $k_a = N$. Therefore $r_{a,w}^W = r_{a,(k_a)}$. The equal split $\epsilon_0 = \epsilon_1 = \alpha/2$ gives the stated symmetric specialization. $\square$

Justification. The general weighted-exchangeability quantile theorem with test mass at $+\infty$ is inherited from Tibshirani, Barber, Candès, and Ramdas [4]. The design-specific step is reduction to finite orbit likelihood ratios, sufficiency of quotient overlap under member-level singularity, and combination of two armwise guarantees for one common causal target by clipping and Bonferroni. Lei-Candès, Wang-Li-Yu, and Barber et al. delimit adjacent target, design, and nonexchangeability scopes [1, 5, 6] without a priority claim. *Gap.* The theorem does not introduce a new generic weighted-conformal principle and does not remove quotient overlap. Its contribution is the matched-randomization orbit reduction and common-target two-arm integration under normalized realized-arm selector kernels. *Consumer.* Use orbit weights when exact selector balance fails but every target-supported orbit has calibration mass; combine the two prespecified arm levels only through clipping and Bonferroni.

Proposition 1 (prop:rank-boundary). *If $k = M + 1$, then $r_{0,(k)} = r_{1,(k)} = +\infty$, $I_{0,w} = I_{1,w} = [-K, K]$, $\mathcal{C}_w = [-2K, 2K]$, and $\Pr_P(\tau_T \in \mathcal{C}_w \mid \mathcal{T}) = 1$ for every law and trained predictor maps.*

Proof. By **def:rank-inversion**, when $k = M + 1$ the reported order statistic is defined to be $+\infty$ in each arm. Hence

$$r_{0,(k)} = r_{1,(k)} = +\infty.$$

For either arm a , the untrimmed interval with this radius is all of $\mathbb{R}$, and clipping therefore gives

$$I_{a,w} = \mathbb{R} \cap [-K, K] = [-K, K].$$

Taking the Minkowski difference yields

$$\mathcal{C}_w = [-K, K] - [-K, K] = [-2K, 2K].$$

By **ass:bounded-potential-outcomes**, both $Y_{0T}(1)$ and $Y_{0T}(0)$ belong to $[-K, K]$. Consequently

$$-2K \leq Y_{0T}(1) - Y_{0T}(0) = \tau_T \leq 2K$$

almost surely, for every latent law and every choice of the trained predictor maps. Thus $\Pr_P(\tau_T \in \mathcal{C}_w \mid \mathcal{T}) = 1$. $\square$

Justification. The symmetric coverage-one boundary is the both-arm branch of a complete asymmetric case split. *Gap.* The equal split cannot display a one-arm boundary. *Consumer.* A report can distinguish one infinite threshold from two.

Proposition 2 (prop:disjoint-orbit-witness). *Suppose $k \leq M$, equivalently $\alpha \geq 2/(M + 1)$, and, for one arm a , every target-supported orbit has zero $q_{w,a}^{\text{obs}}$ mass. Then **def:disjoint-orbit-law** belongs to $\mathcal{P}_{\mathcal{G},K}$, all adverse-arm calibration scores equal 0, and $r_{a,(k)} = 0$. The other arm's scores and threshold equal 0, so $\mathcal{C}_w = \{0\}$. The common target effect equals K for $a = 1$ and $-K$ for $a = 0$, hence $\Pr_P(\tau_T \in \mathcal{C}_w \mid \mathcal{T}) = 0$.*

Proof. Put $N = M + 1$. Since $k = \lceil N(1 - \alpha/2) \rceil$, the elementary equivalence $\lceil x \rceil \leq M$ if and only if $x \leq M$ gives

$$k \leq M \iff N(1 - \alpha/2) \leq M \iff \alpha \geq \frac{2}{M + 1}.$$

Fix the arm a in the premise. Under **def:disjoint-orbit-law**, the arm- a outcome is K on every orbit having positive target mass and is zero on every orbit having positive calibration mass. These prescriptions do not conflict because every target-supported orbit has zero calibration mass. The other-arm outcome and both trained predictors are identically zero. All outcomes lie in $[-K, K]$. Moreover, the prototype is constant on each $\mathcal{G}$ -orbit; hence applying any $g \in \mathcal{G}$ leaves it unchanged. The stated uniform group randomization therefore yields a law satisfying **ass:finite-group-latent-invariance**, and boundedness gives membership in $\mathcal{P}_{\mathcal{G},K}$ by **def:invariant-laws**.

For a calibration replicate, Z_i lies in $\text{supp}(\nu)$ almost surely. On that support, **ass:observable-selector-kernel** selects a member observed in the requested arm. By **ass:selector-auxiliary-randomization**, the selected arm- a state has law $q_{w,a}^{\text{obs}}$. It can therefore lie only in an orbit on which the constructed arm- a outcome is zero, up to a null event. Thus $S_{i,a} = 0$. In the other arm both the outcome and predictor are zero, so $S_{i,1-a} = 0$ as well. Notice that arbitrary total extensions of the kernels off $\text{supp}(\nu)$ cannot change this conclusion, because those assignments have

ν -mass zero. Since $k \leq M$, the k -th order statistic in each arm equals zero. The target-centered predictors also equal zero, and clipping preserves the singleton, whence

$$I_{0,w} = I_{1,w} = \{0\}, \quad \mathcal{C}_w = \{0\}.$$

By **ass:prespecified-target**, the target lies in a target-supported orbit almost surely. Hence $Y_{0T}(a) = K$ and $Y_{0T}(1-a) = 0$. Consequently $\tau_T = K$ when $a = 1$ and $\tau_T = -K$ when $a = 0$. Since $K > 0$, neither value belongs to $\{0\}$, and therefore

$$\Pr_P(\tau_T \in \mathcal{C}_w \mid \mathcal{T}) = 0.$$

□

Justification. The zero-coverage construction is formulated on actual $\mathcal{G}$ -orbits and remains valid without any pair assumption. *Gap.* A pair cell instantiates this witness only after its orbit status is established from the group action; the pure local-swap check is one sufficient route. *Consumer.* Compute actual target and calibration orbit support and report whether $k \leq M$ before invoking zero coverage.

Proposition 3 (prop:s2-reduction). *Fix a pair block b with $\mathcal{V}_b = \{v_{b,0}, v_{b,1}\}$ and suppose the local swap subgroup satisfies $\langle s_b \rangle \leq \mathcal{G}$. Then $s_b(v_{b,r}, a) = (v_{b,1-r}, a)$ for $r, a \in \{0, 1\}$, with s_b fixing all outside members and relabelling complete records as in **def:group-action**, and the global $\mathcal{G}$ -orbits in block b are $O_{b,0} = \{(v_{b,0}, 0), (v_{b,1}, 0)\}$ and $O_{b,1} = \{(v_{b,0}, 1), (v_{b,1}, 1)\}$. If the target arm-1 lifted law is supported on $O_{b,1}$, an exclusive mate-control analogue supported on the realized mate's control state in $O_{b,0}$ has a different orbit-mass vector, cannot satisfy exact joint balance, and is not a legal arm-1 selector. Separately, if a legal implemented arm-1 selector satisfies $q_{w,1}^{\text{obs}}(O_{b,1}) = 0$, then arm 1 satisfies the disjoint-orbit premise. Its zero-coverage conclusion applies when $k \leq M$; for $k = M + 1$, coverage is one by **prop:rank-boundary**.*

Proof. By **def:pair-action**, the nonidentity element s_b interchanges $v_{b,0}$ and $v_{b,1}$, fixes every member outside block b , preserves the arm coordinate, and acts on complete records through **def:group-action**. Thus

$$s_b(v_{b,r}, a) = (v_{b,1-r}, a), \quad r, a \in \{0, 1\}.$$

Every element of $\mathcal{G}$ preserves both blocks and arms. Therefore the orbit of $(v_{b,r}, a)$ is contained in $O_{b,a}$. Since the subgroup $\langle s_b \rangle$ reaches both elements of that cell, the reverse inclusion holds, and the global orbit is exactly

$$O_{b,a} = \{(v_{b,0}, a), (v_{b,1}, a)\}.$$

This proves the two asserted orbit identities.

If the arm-1 target lift is supported on $O_{b,1}$, its row in the target orbit vector assigns mass one to $O_{b,1}$ and zero to $O_{b,0}$. An exclusive mate-control analogue is instead supported on a state in $O_{b,0}$, so its two displayed orbit masses are reversed. Hence its orbit vector cannot equal the target arm-1 vector and cannot satisfy the corresponding equations in **def:joint-orbit-balance**, equivalently the relevant rows of **def:common-selector-feasibility-lp**. It is also not a legal arm-1 selector: on every supported assignment on which the selected mate is a control, that member has $z_v = 0 \neq 1$, and **ass:observable-selector-kernel** forces its arm-1 selection probability to be zero. Off-support kernel values are irrelevant because their ν -weights vanish.

Now suppose a legal implemented arm-1 selector has $q_{w,1}^{\text{obs}}(O_{b,1}) = 0$. The target arm-1 law is supported on that single orbit, so every target-supported orbit has zero calibration mass. This is precisely the arm-1 premise of **prop:disjoint-orbit-witness**. If $k \leq M$, that proposition gives the asserted zero-coverage conclusion. If $k = M + 1$, **prop:rank-boundary** instead gives $\mathcal{C}_w = [-2K, 2K]$ and conditional coverage one. □

Justification. The two-member-block and pure local-swap hypotheses are a conservative sufficient certificate that both arm-specific pair cells are global orbits. *Gap.* The certificate is not necessary, since a coupled-swap subgroup may generate the same orbit; mate-control selection remains illegal for the opposite arm. *Consumer.* Verify the displayed sufficient audit, then check the orbit row, selector legality, and finite-rank boundary.

Proposition 4 (prop:two-pair-sanity-check). *Let $\mathcal{V} = \{v_{1,0}, v_{1,1}, v_{2,0}, v_{2,1}\}$ and $\mathcal{G} = S_2 \times S_2$, with the generators in **def:pair-action**. The four arm-specific orbits are $O_{1,0}, O_{1,1}, O_{2,0}, O_{2,1}$. If q_{tar} concentrates on $v_{1,0}$, its arm 1 orbit vector has mass 1 on $O_{1,1}$. An exclusive mate-control analogue concentrated on $(v_{1,1}, 0)$ has mass 1 on $O_{1,0}$, so the relevant stacked LP equations cannot both hold and the analogue is not a legal arm 1 selector. If a legal implemented selector has $q_{w,1}^{\text{obs}}(O_{1,1}) = 0$, then for $k \leq M$ the bounded witness has $\mathcal{C}_w = \{0\}$ and coverage 0; for $k = M + 1$, it has $I_{0,w} = I_{1,w} = [-K, K]$, $\mathcal{C}_w = [-2K, 2K]$, and coverage 1.*

Proof. The product group $S_2 \times S_2$ contains the pure swap in block 1 and the pure swap in block 2. Applying `def:pair-action` separately to the two blocks shows that every group orbit stays within one block and one arm and contains both members there. Hence the four and only four arm-specific orbits are

$$O_{1,0}, O_{1,1}, O_{2,0}, O_{2,1}.$$

If q_{tar} is the point mass at $v_{1,0}$, its arm-1 lift belongs to $O_{1,1}$, so the target arm-1 orbit vector assigns mass one to that row. The proposed mate-control point $(v_{1,1}, 0)$ belongs to $O_{1,0}$, so the analogue assigns mass zero to the $O_{1,1}$ row and mass one to the $O_{1,0}$ row. The relevant stacked equations in `def:common-selector-feasibility-lp` therefore cannot both equal the target vector. In addition, an arm-1 kernel cannot select a realized control on any assignment of positive ν -mass by `ass:observable-selector-kernel`; arbitrary off-support extensions do not affect the integrated selector law.

If a legal selector instead satisfies $q_{w,1}^{\text{obs}}(O_{1,1}) = 0$, the target-supported arm-1 orbit is disjoint from calibration support. For $k \leq M$, `prop:disjoint-orbit-witness` gives $\mathcal{C}_w = \{0\}$ and coverage zero. For $k = M + 1$, `prop:rank-boundary` gives

$$I_{0,w} = I_{1,w} = [-K, K], \quad \mathcal{C}_w = [-2K, 2K],$$

and conditional coverage one. $\square$

Justification. The declared $S_2 \times S_2$ product contains both required pure local swaps, so the sufficient audit certifies all four cells as global orbits. *Gap.* This finite regression test does not cover an arbitrary four-member group; such a group requires its own orbit computation. *Consumer.* Use the four certified orbit rows to test selector-LP construction and both symmetric rank regimes.

Lemma 1 (`lem:finite-rank-staircase`). *Let $N = M + 1$, $k = \lceil N(1 - \alpha/2) \rceil$, and let $\rho_{M,\alpha}$ be defined by `def:finite-rank-error`. Then*

$$\rho_{M,\alpha} = \frac{N - k}{N} = \frac{\lfloor N\alpha/2 \rfloor}{N} \leq \frac{\alpha}{2}.$$

For every integer $j \geq 0$ whose displayed interval intersects $(0, 1)$, $\rho_{M,\alpha} = j/N$ exactly when $\alpha \in [2j/N, 2(j+1)/N)$. In particular, $k = M + 1$ if and only if $\rho_{M,\alpha} = 0$, equivalently $0 < \alpha < 2/(M + 1)$.

Proof. Specialize `lem:armwise-finite-rank-staircase` to $\varepsilon_0 = \varepsilon_1 = \alpha/2$ and put $N = M + 1$. The diagonal identities in that lemma give $k_0 = k_1 = k$ and $\rho_{M,\varepsilon_0} = \rho_{M,\varepsilon_1} = \rho_{M,\alpha}$. Its armwise formula therefore becomes

$$\rho_{M,\alpha} = \frac{N - k}{N} = \frac{\lfloor N\alpha/2 \rfloor}{N} \leq \frac{\alpha}{2}.$$

Likewise, $\rho_{M,\varepsilon_a} = j/N$ exactly for $\varepsilon_a \in [j/N, (j+1)/N)$; substituting $\varepsilon_a = \alpha/2$ gives

$$\rho_{M,\alpha} = \frac{j}{N} \iff \alpha \in \left[\frac{2j}{N}, \frac{2(j+1)}{N} \right),$$

for every indicated integer j . Finally, the armwise boundary condition $k_a = M + 1$ exactly when $0 < \varepsilon_a < 1/N$ becomes

$$k = M + 1 \iff 0 < \alpha < \frac{2}{N}.$$

On the same interval $\lfloor N\alpha/2 \rfloor = 0$, and outside it that floor is at least one. Thus the boundary condition is also equivalent to $\rho_{M,\alpha} = 0$, proving all claims. $\square$

Lemma 2 (`lem:armwise-finite-rank-staircase`). *Let $\varepsilon = (\varepsilon_0, \varepsilon_1)$ be prespecified with $\varepsilon_a > 0$ and $\varepsilon_0 + \varepsilon_1 \leq \alpha$, put $N = M + 1$, and define $k_a = \lceil N(1 - \varepsilon_a) \rceil$. Then, for each $a \in \{0, 1\}$,*

$$\rho_{M,\varepsilon_a} = \frac{N - k_a}{N} = \frac{\lfloor N\varepsilon_a \rfloor}{N} \leq \varepsilon_a.$$

For every integer $j \geq 0$ whose displayed interval intersects $(0, 1)$, $\rho_{M,\varepsilon_a} = j/N$ exactly when $\varepsilon_a \in [j/N, (j+1)/N)$. Thus the two-dimensional staircase has coordinate elbows at $\varepsilon_a = j/(M + 1)$, and $k_a = M + 1$ exactly when $0 < \varepsilon_a < 1/(M + 1)$. Under $\varepsilon_0 = \varepsilon_1 = \alpha/2$, one has $k_0 = k_1 = k$ and $\rho_{M,\varepsilon_0} = \rho_{M,\varepsilon_1} = \rho_{M,\alpha}$.

Proof. Put $N = M + 1$. Since N is an integer, the elementary identity $\lceil N - x \rceil = N - \lfloor x \rfloor$ gives, for each $a \in \{0, 1\}$,

$$k_a = \lceil N - N\varepsilon_a \rceil = N - \lfloor N\varepsilon_a \rfloor.$$

Consequently, by `def:finite-rank-error`,

$$\rho_{M,\varepsilon_a} = \frac{N - k_a}{N} = \frac{\lfloor N\varepsilon_a \rfloor}{N} \leq \varepsilon_a. \quad (1)$$

For an integer $j \geq 0$, the equality $\lfloor N\varepsilon_a \rfloor = j$ holds if and only if $j \leq N\varepsilon_a < j + 1$, or equivalently

$$\varepsilon_a \in [j/N, (j + 1)/N). \quad (2)$$

This proves the coordinatewise staircase and locates its elbows at $\varepsilon_a = j/(M + 1)$. Moreover, $k_a = N$ if and only if $\lfloor N\varepsilon_a \rfloor = 0$. Because $\varepsilon_a > 0$, this is equivalent to $0 < \varepsilon_a < 1/N$.

Finally, if $\varepsilon_0 = \varepsilon_1 = \alpha/2$, then `def:rank-inversion` gives

$$k_0 = k_1 = \lceil N(1 - \alpha/2) \rceil = k,$$

and (1), together with `def:finite-rank-error`, gives

$$\rho_{M,\varepsilon_0} = \rho_{M,\varepsilon_1} = \frac{\lfloor N\alpha/2 \rfloor}{N} = \rho_{M,\alpha}.$$

Thus the symmetric staircase is exactly the diagonal section of the two-dimensional staircase. $\square$

Justification. This lemma exposes the full two-dimensional finite-rank staircase generated by a prespecified asymmetric Bonferroni budget. *Gap.* The frozen symmetric staircase hides one-arm boundary regimes. *Consumer.* An analyst can allocate unequal armwise error while reading the exact finite-rank elbows.

Lemma 3 (lem:exact-binomial-shorter-tail-evaluator). *Let $M \geq 1$, $1 \leq k \leq M$, and $\vartheta \in [0, 1]$. The probability $H_{M,k}(\vartheta) = \Pr\{\text{Bin}(M, \vartheta) < k\}$ can be evaluated exactly using $O(\log M + \min\{k, M - k + 1\})$ arithmetic operations and $O(1)$ auxiliary memory. At $\vartheta = 0$, one has $H_{M,k}(0) = 1$, and at $\vartheta = 1$, one has $H_{M,k}(1) = 0$; these endpoint evaluations use no division.*

Proof. Assume first that $0 < \vartheta < 1$. The lower-tail endpoint term is

$$t_0 = \Pr\{\text{Bin}(M, \vartheta) = 0\} = (1 - \vartheta)^M.$$

It is computed by exponentiation by squaring in $O(\log M)$ multiplications. Successive lower-tail terms obey

$$t_{r+1} = t_r \frac{M - r}{r + 1} \frac{\vartheta}{1 - \vartheta}, \quad 0 \leq r < k - 1. \quad (1)$$

Thus $H_{M,k}(\vartheta) = \sum_{r=0}^{k-1} t_r$ is computed in $O(\log M + k)$ operations. Only the current term and running sum are stored.

Alternatively, begin at the upper endpoint

$$u_M = \Pr\{\text{Bin}(M, \vartheta) = M\} = \vartheta^M,$$

again computed by exponentiation by squaring, and use the downward recurrence

$$u_{r-1} = u_r \frac{r}{M - r + 1} \frac{1 - \vartheta}{\vartheta}, \quad k < r \leq M. \quad (2)$$

Then

$$H_{M,k}(\vartheta) = 1 - \sum_{r=k}^M u_r,$$

which takes $O(\log M + M - k + 1)$ operations and constant memory. Choosing (1) when $k \leq M - k + 1$ and (2) otherwise gives

$$O(\log M + \min\{k, M - k + 1\})$$

operations and $O(1)$ auxiliary memory. All denominators in (1) and (2) are nonzero under $0 < \vartheta < 1$. If $\vartheta = 0$, the binomial variable equals zero almost surely, and $k \geq 1$ gives $H_{M,k}(0) = 1$. If $\vartheta = 1$, it equals M almost surely, and $k \leq M$ gives $H_{M,k}(1) = 0$. These two cases require no division. $\square$

Lemma 4 (lem:corner-oriented-four-cell-rank-evaluator). *Let $M \geq 1$, let $1 \leq k_0, k_1 \leq M$, and let π be a probability mass function on $\{0, 1\}^2$. For iid $(B_{i0}, B_{i1}) \sim \pi$, put $R = \sum_i B_{i0}$, $S = \sum_i B_{i1}$, $h_a = M - k_a + 1$, and $\bar{k}_a = \min\{k_a, h_a\}$. For $\sigma_a \in \{L, U\}$, replace B_{ia} by itself under L and by $1 - B_{ia}$ under U , set $t_a = k_a$ under L and $t_a = h_a$ under U , and let $J_{\sigma_0 \sigma_1}$ be the probability that both transformed counts are strictly below their thresholds. A rolling lower-left recurrence for the correspondingly permuted four-cell law computes $J_{\sigma_0 \sigma_1}$ in*

$$O\left(\sum_{i=0}^{M-1} \min\{t_0, i+1\} \min\{t_1, i+1\}\right)$$

arithmetic operations and $O(t_0 t_1)$ memory. Moreover, $\Pr\{R < k_0, S < k_1\}$ equals J_{LL} , $\Pr\{R < k_0\} - J_{LU}$, $\Pr\{S < k_1\} - J_{UL}$, or $1 - \Pr\{R \geq k_0\} - \Pr\{S \geq k_1\} + J_{UU}$, respectively. Choosing each orientation to minimize t_a , and evaluating required binomial marginals exactly, yields time

$$O\left(\sum_{i=0}^{M-1} \min\{\bar{k}_0, i+1\} \min\{\bar{k}_1, i+1\}\right)$$

and recurrence workspace $O(\bar{k}_0 \bar{k}_1)$.

For a support-three law, write π^σ for the four-cell law after orientation σ , let T_a^σ be the corresponding transformed terminal counts, and use the thresholds t_a^σ above. Define

$$\Sigma_{\text{face}}(\pi) = \left\{ \sigma \in \{L, U\}^2 : \pi_{11}^\sigma = 0, \quad t_0^\sigma + t_1^\sigma > M \right\}.$$

For every $\sigma \in \Sigma_{\text{face}}(\pi)$, one has $T_0^\sigma + T_1^\sigma \leq M$ almost surely, the two events $\{T_0^\sigma \geq t_0^\sigma\}$ and $\{T_1^\sigma \geq t_1^\sigma\}$ are disjoint, and therefore

$$J_\sigma = 1 - \Pr\{T_0^\sigma \geq t_0^\sigma\} - \Pr\{T_1^\sigma \geq t_1^\sigma\}.$$

Put

$$\ell_M(t) = 1 + \lceil \log_2 M \rceil + \min\{t, M - t + 1\}, \quad F_M(\pi) = 1 + \min_{\sigma \in \Sigma_{\text{face}}(\pi)} \{\ell_M(t_0^\sigma) + \ell_M(t_1^\sigma)\}.$$

Each marginal tail is evaluated by the exact shorter binomial-tail recurrence. The marginal quantities required by the LL, LU, UL, or UU reconstruction identity are complements of tails already evaluated for J_σ , so no duplicated marginal recurrence is charged. Deterministic charge minimization followed by lexicographic tie-breaking therefore evaluates the original lower-lower probability in $O(F_M(\pi))$ exact arithmetic operations and $O(1)$ auxiliary workspace.

For every remaining support-four key satisfying $\pi_{00}\pi_{11} = \pi_{01}\pi_{10}$, define $\theta_0 = \pi_{10} + \pi_{11}$, $\theta_1 = \pi_{01} + \pi_{11}$, $\bar{k}_a = \min\{k_a, M - k_a + 1\}$, and $R_{M,k_0,k_1} = 1 + 2\lceil \log_2 M \rceil + \bar{k}_0 + \bar{k}_1$. Exact determinant testing and the factorization

$$\pi_{00} + \pi_{10}x + \pi_{01}y + \pi_{11}xy = \{(1 - \theta_0) + \theta_0 x\} \{(1 - \theta_1) + \theta_1 y\}$$

give

$$D_{M,k_0,k_1}(\pi) = H_{M,k_0}(\theta_0)H_{M,k_1}(\theta_1)$$

in $O(R_{M,k_0,k_1})$ exact arithmetic operations and $O(1)$ auxiliary workspace. Such a key is evaluated after the vertex, edge, and incompatible-face strata and before any rolling, multinomial-sum, or positive-corner branch.

Proof. Fix an orientation $\sigma = (\sigma_0, \sigma_1)$ and write $B_{ia}^\sigma = B_{ia}$ under L and $B_{ia}^\sigma = 1 - B_{ia}$ under U . This transformation permutes the four atoms of π ; denote the transformed law by π^σ . For partial counts $T_{ia} = \sum_{j=1}^i B_{ja}^\sigma$, initialize $d_0(0, 0) = 1$ and, for $0 \leq r < t_0$ and $0 \leq s < t_1$, update

$$d_{i+1}(r, s) = \sum_{x,y \in \{0,1\}} \pi_{xy}^\sigma d_i(r-x, s-y). \quad (1)$$

Induction on i gives $d_i(r, s) = \Pr\{T_{i0} = r, T_{i1} = s\}$. A state discarded above either threshold cannot return because both increments are nonnegative. At stage $i+1$, at most $\min\{t_0, i+1\} \min\{t_1, i+1\}$ states are retained. Summing this quantity for $i = 0, \dots, M-1$ proves the rolling time bound, and two rolling rectangles give $O(t_0 t_1)$ memory.

Under LL , the retained event is exactly $\{R < k_0, S < k_1\}$. Under LU , the second transformed count is $M - S$, and

$$M - S < M - k_1 + 1 \iff S \geq k_1.$$

Therefore

$$\Pr\{R < k_0, S < k_1\} = \Pr\{R < k_0\} - J_{LU}. \quad (2)$$

The UL identity is symmetric. Under UU , $J_{UU} = \Pr\{R \geq k_0, S \geq k_1\}$, and inclusion–exclusion gives the fourth identity. Choosing L when $k_a \leq h_a$ and U otherwise gives $t_a = \bar{k}_a$. Each required marginal is binomial; `lem:exact-binomial-shorter-tail-evaluator` computes it in

$$O(\log M + \min\{\bar{k}_a, M - \bar{k}_a + 1\}) = O(\log M + \bar{k}_a)$$

operations and constant memory. The rolling sum has M nonzero stage charges and absorbs these marginal charges, proving the oriented rolling certificate.

The positive-base coefficient alternative follows from `lem:positive-corner-coefficient-recurrence`. For every orientation with $\pi_{00}^\sigma > 0$, it evaluates the same J_σ exactly and then uses (2) and its three companion identities. Since the four cells sum to one, at least one atom is positive and can be moved to 00; hence the minimization over positive-base orientations is nonempty. This proves the zero-cell-safe comparison with the coefficient branch.

It remains to prove the zero-corner face reduction. Let π have support three and let $\sigma \in \Sigma_{\text{face}}(\pi)$. The equality $\pi_{11}^\sigma = 0$ implies

$$B_{i0}^\sigma + B_{i1}^\sigma \leq 1 \quad \text{almost surely for every } i.$$

Summing over i gives

$$T_0^\sigma + T_1^\sigma \leq M. \quad (3)$$

If both failure events occurred, then (3) would imply $M \geq T_0^\sigma + T_1^\sigma \geq t_0^\sigma + t_1^\sigma > M$, a contradiction. Thus they are disjoint, and complementation yields

$$J_\sigma = 1 - \Pr\{T_0^\sigma \geq t_0^\sigma\} - \Pr\{T_1^\sigma \geq t_1^\sigma\}. \quad (4)$$

The two transformed marginals are binomial with success probabilities

$$\vartheta_0^\sigma = \pi_{10}^\sigma + \pi_{11}^\sigma, \quad \vartheta_1^\sigma = \pi_{01}^\sigma + \pi_{11}^\sigma.$$

For each a , the tail in (4) is $1 - H_{M, t_a^\sigma}(\vartheta_a^\sigma)$. The exact shorter-tail lemma therefore gives the stated $\ell_M(t_a^\sigma)$ charge and constant workspace, including the cases $\vartheta_a^\sigma \in \{0, 1\}$, where no division is used.

No marginal is charged twice. Under LU , for example, the reconstruction needs $\Pr\{R < k_0\}$, while the first tail already computed in (4) is $\Pr\{R \geq k_0\}$; one subtraction gives the former. Under UL the same argument applies to S . Under UU , the first transformed tail is

$$\Pr\{M - R \geq M - k_0 + 1\} = \Pr\{R < k_0\},$$

and similarly for S , so the upper marginals in inclusion–exclusion are again obtained by complementation. The LL identity needs no marginal. Hence deterministic minimization over the finite set $\Sigma_{\text{face}}(\pi)$, followed by lexicographic tie-breaking, has charge $O(F_M(\pi))$ and $O(1)$ auxiliary workspace, and reconstructs the original lower–lower probability exactly.

It remains to justify the rank-one prepass. For a support-four key put $p_{xy} = \pi_{xy}$, $\theta_0 = p_{10} + p_{11}$, and $\theta_1 = p_{01} + p_{11}$. If $p_{00}p_{11} = p_{01}p_{10}$, then

$$\theta_0\theta_1 = p_{10}p_{01} + p_{11}(p_{10} + p_{01} + p_{11}) = p_{00}p_{11} + p_{11}(1 - p_{00}) = p_{11}. \quad (R1)$$

Thus $p_{10} = \theta_0(1 - \theta_1)$, $p_{01} = (1 - \theta_0)\theta_1$, and $p_{00} = (1 - \theta_0)(1 - \theta_1)$. Therefore

$$p_{00} + p_{10}x + p_{01}y + p_{11}xy = \{(1 - \theta_0) + \theta_0x\}\{(1 - \theta_1) + \theta_1y\}. \quad (R2)$$

Raising (R2) to the M -th power proves that the two terminal counts are independent binomials and hence that

$$D_{M, k_0, k_1}(\pi) = H_{M, k_0}(\theta_0)H_{M, k_1}(\theta_1). \quad (R3)$$

The determinant test consists of two exact multiplications and an exact comparison. Applying `lem:exact-binomial-shorter-tail-evaluator` separately to the two factors in (R3) costs

$$O(1 + 2\lceil \log_2 M \rceil + \bar{k}_0 + \bar{k}_1)$$

exact operations. The factors are processed sequentially, so only the current recurrence term, its partial sum, and the first completed factor are stored. This is $O(1)$ auxiliary workspace. $\square$

Justification. Corner complementation gives the smallest rolling rectangle, while the new positive-base coefficient identity supplies an additional exact branch for every four-cell key. *Gap.* A width-minimizing corner may have zero base mass, so coefficient division cannot be forced on that orientation. *Consumer.* Minimize the coefficient charge only over positive-base orientations, reconstruct the original event with exact corner identities, and compare against the zero-cell-safe rolling and multinomial branches.

Lemma 5 (lem:positive-corner-coefficient-recurrence). *Let $M \geq 1$, let $1 \leq t_0, t_1 \leq M + 1$, and let $\tilde{\pi} = (a, c, b, d)$ be a probability mass function on $\{0, 1\}^2$, ordered as 00, 01, 10, 11, with $a > 0$. Define $p_{rs} = [x^r y^s](a + bx + cy + dxy)^M$, with negative-index coefficients equal to zero. Then $p_{00} = a^M$, $p_{0s} = p_{0,s-1}(M - s + 1)c/(sa)$, and, for $r \geq 1$,*

$$p_{rs} = \frac{(M - r + 1)(bp_{r-1,s} + dp_{r-1,s-1}) - crp_{r,s-1}}{ar}.$$

Consequently $\sum_{r < t_0, s < t_1} p_{rs}$ is the exact lower-left corner probability and is computable in $O(\log M + t_0 t_1)$ exact arithmetic operations and $O(\min\{t_0, t_1\})$ rolling memory after optional transposition. Given ranks $1 \leq k_0, k_1 \leq M$, set $h_a = M - k_a + 1$, $\bar{k}_a = \min\{k_a, h_a\}$, and set $t_a^\sigma = k_a$ under L and $t_a^\sigma = h_a$ under U . For an arbitrary four-cell probability mass function π , let the M pairs (B_{i0}, B_{i1}) be iid from π and put $R = \sum_i B_{i0}$ and $S = \sum_i B_{i1}$. At least one coordinate-complementation orientation σ has $\pi_{00}^\sigma > 0$. For each such orientation, the original lower-lower probability is recovered by the four exact corner identities and any required binomial marginals. Thus, writing $\mathcal{A}_{LL} = \emptyset$, $\mathcal{A}_{LU} = \{0\}$, $\mathcal{A}_{UL} = \{1\}$, $\mathcal{A}_{UU} = \{0, 1\}$, its positive-corner coefficient-branch charge is

$$\mathcal{C}_+(\pi) = \min_{\sigma: \pi_{00}^\sigma > 0} \left\{ \log M + t_0^\sigma t_1^\sigma + \sum_{a \in \mathcal{A}_\sigma} (\log M + \bar{k}_a) \right\},$$

with coefficient workspace $O(\min\{t_0^\sigma, t_1^\sigma\})$ for the selected orientation and constant additional marginal workspace. This branch remains exact when the independently width-minimizing orientation has zero base atom.

Proof. Let

$$P(x, y) = (a + bx + cy + dxy)^M = \sum_{r, s \geq 0} p_{rs} x^r y^s.$$

Because the constant term is a , $p_{00} = a^M$, computable by exponentiation by squaring in $O(\log M)$ operations. At $x = 0$, $P(0, y) = (a + cy)^M$, so comparison of consecutive binomial coefficients gives

$$p_{0s} = p_{0,s-1} \frac{M - s + 1}{s} \frac{c}{a}, \quad 1 \leq s < t_1. \quad (1)$$

Differentiate with respect to x . The polynomial identity

$$(a + bx + cy + dxy)P_x = M(b + dy)P \quad (2)$$

holds identically. Equating coefficients of $x^{r-1}y^s$ in (2) yields

$$arp_{rs} + b(r-1)p_{r-1,s} + crp_{r,s-1} + d(r-1)p_{r-1,s-1} = Mbp_{r-1,s} + Mdp_{r-1,s-1}.$$

Since $a > 0$ and $r \geq 1$, division by ar is valid and rearrangement gives exactly the asserted recurrence. Formula (1) initializes row zero; thereafter row r uses only row $r - 1$ and the immediately preceding entry in row r . Thus one array of length t_1 , updated in place with the saved previous-row diagonal, suffices. If $t_0 < t_1$, transpose x and y , which exchanges b, c and the two thresholds, to obtain $O(\min\{t_0, t_1\})$ memory. There are $t_0 t_1$ coefficients and a running sum adds no asymptotic storage, proving the time and memory bounds. The multinomial expansion of P shows that p_{rs} is exactly the probability of transformed counts (r, s) , so their rectangle sum is the desired corner probability.

Coordinate complementation permutes the four atoms. Because their sum is one, at least one atom is positive and can be moved to the 00 position, proving nonemptiness of the minimization. Under LL , the original event is the computed corner itself. Under LU , the computed corner is $\Pr\{R < k_0, S \geq k_1\}$, so subtracting it from $\Pr\{R < k_0\}$ recovers the lower-lower event. The UL case is symmetric, and the UU case follows by two-event inclusion-exclusion. These are exactly the marginal sets $\mathcal{A}_\sigma$ in the statement. By **lem:exact-binomial-shorter-tail-evaluator**, each required marginal costs $O(\log M + \bar{k}_a)$ operations and constant workspace. Adding those costs to the coefficient recurrence and minimizing over positive-base orientations proves the final charge. No division is attempted at a zero base atom, and the existence argument shows that this restriction never removes every orientation. $\square$

Lemma 6 (lem:weighted-exchangeable-infinity-quantile-comparator). *For weighted-exchangeable observations and an arbitrary conformity score, the weighted empirical score distribution augmented by the test-point weight at $+\infty$ has its stated lower-quantile coverage guarantee. This is the general likelihood-ratio weighted-conformal result of Tibshirani, Barber, Candès, and Ramdas; the finite orbit quotient used in this note is a specialization, not a new general weighted-exchangeability theorem.*

Justification. This cited leaf makes explicit that the general weighted-quantile validity theorem is inherited. *Gap.* The note contributes the orbit-quotient reduction and causal matched-block integration, not the general theorem. *Consumer.* Use it as the comparator when interpreting quotient likelihood-ratio weighting.

Lemma 7 (lem:finite-support-weighted-rank-validity). *Let Q and P be probability laws on a measurable space with $P \ll Q$, and let $\lambda = dP/dQ$. Let $X_1, \dots, X_M$ be iid from Q , independently of $X_0 \sim P$, and let R be a measurable map into $[0, \infty)$. Put $W = \sum_{\ell=0}^M \lambda(X_\ell)$. On $\{W > 0\}$, for every $j \in \{0, \dots, M\}$, define the random probability measure*

$$\mu_j = W^{-1} \left\{ \sum_{\ell \neq j} \lambda(X_\ell) \delta_{R(X_\ell)} + \lambda(X_j) \delta_{+\infty} \right\}$$

and, for fixed $\epsilon \in (0, 1)$, define

$$q_j = \inf \{t \in [0, +\infty] : \mu_j((-\infty, t]) \geq 1 - \epsilon\}.$$

Then $W > 0$ almost surely under the target experiment and

$$\Pr\{R(X_0) > q_0\} \leq \epsilon.$$

Proof. Put $N = M + 1$. First consider a deterministic vector $x = (x_0, \dots, x_M)$, write $\lambda_j = \lambda(x_j)$, $r_j = R(x_j)$, and suppose $W = \sum_j \lambda_j > 0$. Let q_j be the quantile obtained when coordinate j is replaced by the atom at $+\infty$, and define

$$F(x) = \{j : r_j > q_j\}.$$

We claim the deterministic inequality

$$\sum_{j \in F(x)} \lambda_j \leq \epsilon W. \quad (1)$$

If $F(x) = \emptyset$, this is immediate. Otherwise choose $j_* \in F(x)$ with minimal r_{j_*} . Since $q_{j_*} < r_{j_*}$ and the weighted empirical measure has finite support, the definition of its left-continuous quantile implies that at least $(1 - \epsilon)W$ weight lies at finite scores strictly below r_{j_*} . The coordinate j_* itself is at $+\infty$ in that measure, so

$$\sum_{\ell: r_\ell < r_{j_*}} \lambda_\ell \geq (1 - \epsilon)W. \quad (2)$$

Consequently,

$$\sum_{\ell: r_\ell \geq r_{j_*}} \lambda_\ell \leq \epsilon W. \quad (3)$$

Every member of $F(x)$ has score at least r_{j_*} , by the choice of j_* . Thus (3) proves (1), including arbitrary score ties.

It remains to translate (1) into the sampling statement. Introduce an auxiliary random label J , uniform on $\{0, \dots, M\}$. Conditional on $J = j$, draw coordinate j from P and all other coordinates independently from Q . Relative to Q^N and the uniform label law, the joint density at (x, j) is $\lambda(x_j)$. Hence, whenever $W(x) > 0$,

$$\Pr(J = j \mid X = x) = \frac{\lambda_j}{W}. \quad (4)$$

The set $\{W = 0\}$ has zero probability under this mixture, because its mixture density relative to Q^N is W/N . Conditional on $X = x$, equations (1) and (4) give

$$\Pr\{R(X_J) > q_J \mid X = x\} = \frac{1}{W} \sum_{j \in F(x)} \lambda_j \leq \epsilon. \quad (5)$$

By coordinate permutation symmetry, for every j the failure probability conditional on $J = j$ in (5) is the same as the original experiment's probability $\Pr\{R(X_0) > q_0\}$. Averaging these equal probabilities over the uniform label and then using (5) proves the desired inequality. Finally, $P\{\lambda = 0\} = \int \mathbf{1}\{\lambda = 0\} \lambda dQ = 0$, so the actual target satisfies $\lambda(X_0) > 0$ almost surely and therefore $W > 0$ almost surely. $\square$

Proposition 5 (prop:asymmetric-rank-boundary). *For each arm a , if $k_a = M + 1$, then $r_{a,(k_a)} = +\infty$ and $I_{a,w}^\epsilon = [-K, K]$. If $k_0 = M + 1$ and $k_1 \leq M$, then $\mathcal{C}_w^\epsilon = I_{1,w}^\epsilon - [-K, K]$; if $k_1 = M + 1$ and $k_0 \leq M$, then $\mathcal{C}_w^\epsilon = [-K, K] - I_{0,w}^\epsilon$. If both $k_0 = k_1 = M + 1$, then $\mathcal{C}_w^\epsilon = [-2K, 2K]$ and its conditional coverage is one for every law and trained predictor maps. On the symmetric diagonal, either both arms are at the boundary or neither is, and this reduces to **prop:rank-boundary**.*

Proof. Fix $a \in \{0, 1\}$ and suppose $k_a = M + 1$. By **def:rank-inversion**, $r_{a,(k_a)} = +\infty$. The un-clipped interval is therefore $\mathbb{R}$, and clipping gives

$$I_{a,w}^\epsilon = [-K, K]. \quad (1)$$

By **ass:bounded-potential-outcomes**, $Y_{0T}(a) \in [-K, K]$ almost surely, so a boundary arm cannot cause effect-set miscoverage. If $k_0 = M + 1$ and $k_1 \leq M$, then $\mathcal{C}_w^\epsilon = I_{1,w}^\epsilon - [-K, K]$; if $k_1 = M + 1$ and $k_0 \leq M$, then $\mathcal{C}_w^\epsilon = [-K, K] - I_{0,w}^\epsilon$. If both ranks equal $M + 1$, applying (1) twice yields

$$\mathcal{C}_w^\epsilon = [-K, K] - [-K, K] = [-2K, 2K].$$

Boundedness implies $-2K \leq \tau_T \leq 2K$ almost surely, proving conditional coverage one. On the symmetric diagonal, **lem:armwise-finite-rank-staircase** gives $k_0 = k_1 = k$. Thus either both arms are boundary arms or neither is, and the conclusion reduces to **prop:rank-boundary**. $\square$

Justification. It separates exactly-one-arm and both-arm infinite-rank regimes after support clipping. *Gap.* The symmetric boundary statement cannot exhibit a one-arm boundary. *Consumer.* An analysis plan can report the exact support-clipped set for each boundary pattern.

Lemma 8 (lem:four-cell-multinomial-sum-alternative). *Let $M \geq 1$, $1 \leq k_0, k_1 \leq M$, and let π be a probability mass function on $\{0, 1\}^2$. For M iid draws, the retained-rectangle probability is*

$$\begin{aligned} Q_{M,k_0,k_1}(\pi) &= \sum_{r=0}^{k_0-1} \sum_{s=0}^{k_1-1} \sum_{t=\max\{0, r+s-M\}}^{\min\{r,s\}} \frac{M!}{(M-r-s+t)!(s-t)!(r-t)!t!} \\ &\quad \times \pi_{00}^{M-r-s+t} \pi_{01}^{s-t} \pi_{10}^{r-t} \pi_{11}^t. \end{aligned}$$

Empty ranges contribute zero, $0^0 = 1$, and $0^j = 0$ for $j > 0$. With

$$\mathsf{T}_{M,k_0,k_1} = \sum_{r < k_0} \sum_{s < k_1} [\min\{r, s\} - \max\{0, r + s - M\} + 1]_+,$$

one has $\mathsf{T}_{M,k_0,k_1} \leq k_0 k_1 \min\{k_0, k_1\}$. Shared factorial preprocessing costs $O(M)$; a multinomial key costs $O(M + \mathsf{T}_{M,k_0,k_1})$ time and $O(M)$ memory, while the rolling alternative costs $O(M k_0 k_1)$ time and $O(k_0 k_1)$ memory.

For a finite key set Π , first partition

$$\Pi = \Pi_{\text{vert}} \dot{\cup} \Pi_{\text{edge}} \dot{\cup} \Pi_{\text{face}} \dot{\cup} \Pi_{\text{rank1}} \dot{\cup} \Pi_{\text{MS}} \dot{\cup} \Pi_{\text{DP}},$$

where the first two strata have support sizes one and two, respectively, Π_{face} contains exactly the remaining support-three keys admitted by the oriented zero-corner incompatibility prepass, Π_{rank1} contains exactly the remaining support-four keys satisfying $\pi_{00}\pi_{11} = \pi_{01}\pi_{10}$, and every key in the last two strata lies outside all four prepasses. Vertices use the deterministic count formula. Edges use the exact binomial interval formula and charge $\mathsf{E}_M(\pi)$ from **def:simplex-vertex-key-prepass**. Face keys use the exact charge $\mathsf{F}_M(\pi)$ from **lem:corner-oriented-four-cell-rank-evaluator**. For rank-one keys put $\bar{k}_a = \min\{k_a, M - k_a + 1\}$ and use $\mathsf{R}_{M,k_0,k_1} = 1 + 2\lceil \log_2 M \rceil + \bar{k}_0 + \bar{k}_1$ with constant workspace. Only keys outside all four prepasses are assigned between the multinomial and rolling branches. Sequential processing has aggregate operation charge

$$O\left(|\Pi_{\text{vert}}| + \sum_{\pi \in \Pi_{\text{edge}}} \mathsf{E}_M(\pi) + \sum_{\pi \in \Pi_{\text{face}}} \mathsf{F}_M(\pi) + \sum_{\pi \in \Pi_{\text{rank1}}} \mathsf{R}_{M,k_0,k_1} + M \mathbf{1}\{\Pi_{\text{MS}} \neq \emptyset\} + \sum_{\pi \in \Pi_{\text{MS}}} (M + \mathsf{T}_{M,k_0,k_1}) + |\Pi_{\text{DP}}| M k_0 k_1\right).$$

The factorial table is built if and only if $\Pi_{\text{MS}} \neq \emptyset$, probability-specific power tables are rebuilt for each multinomial key, and retention of one scalar per key uses

$$O(|\Pi| + \max\{1, M \mathbf{1}\{\Pi_{\text{MS}} \neq \emptyset\}, k_0 k_1 \mathbf{1}\{\Pi_{\text{DP}} \neq \emptyset\}\})$$

memory. The vertex, edge, oriented zero-corner face, and support-four rank-one prepasses precede the downstream three-way dispatcher, so none of those strata enters a bivariate rolling, multinomial-sum, or positive-corner coefficient branch. Face and rank-one keys use only $O(1)$ auxiliary workspace and do not trigger factorial preprocessing.

Proof. Let N_{xy} count draws equal to (x, y) , and put

$$R = N_{10} + N_{11}, \quad S = N_{01} + N_{11}.$$

If $R = r$, $S = s$, and $N_{11} = t$, then necessarily

$$(N_{00}, N_{01}, N_{10}, N_{11}) = (M - r - s + t, s - t, r - t, t). \quad (1)$$

Conversely, (1) is a nonnegative integer vector of total M exactly when

$$\max\{0, r + s - M\} \leq t \leq \min\{r, s\}. \quad (2)$$

The multinomial probability of (1) is the displayed summand. Summing over (2), then over $r < k_0$ and $s < k_1$, proves the formula. The conventions $0^0 = 1$ and $0^j = 0$ for $j > 0$ make it valid on every simplex face.

For each (r, s) , the number of feasible t 's is at most $\min\{r, s\} + 1 \leq \min\{k_0, k_1\}$. Hence

$$\mathsf{T}_{M, k_0, k_1} \leq k_0 k_1 \min\{k_0, k_1\}. \quad (3)$$

Factorials through $M!$ require $O(M)$ shared arithmetic and memory. For each key, its four probability-specific power tables require $O(M)$ time and memory, and each feasible summand thereafter has constant cost. This proves the multinomial charge. The retained convolution in `lem:corner-oriented-four-cell-rank-evaluator` proves the rolling alternative and its stated memory.

We now justify the refined partition. A support-one key has deterministic counts by `def:simplex-vertex-key-prepass`. A support-two key has the exact affine-binomial interval formula there, with charge $\mathsf{E}_M(\pi)$ and constant workspace. For a remaining support-three key, the face set is selected precisely when some orientation satisfies $\pi_{11}^o = 0$ and $t_0^o + t_1^o > M$. The last part of `lem:corner-oriented-four-cell-rank-evaluator` then gives its exact value with charge $\mathsf{F}_M(\pi)$ and constant workspace. After those three strata, the determinant calculation in `lem:corner-oriented-four-cell-rank-evaluator` selects exactly the remaining support-four keys with $\pi_{00}\pi_{11} = \pi_{01}\pi_{10}$, evaluates them with charge R_{M, k_0, k_1} , and uses constant workspace. Therefore no vertex, edge, face, or rank-one key is assigned to either generic branch. Every key assigned to Π_{MS} or Π_{DP} lies outside all four prepasses.

Sequentially adding the exact per-key charges gives the displayed aggregate operation bound. The shared factorial table is built exactly when $\Pi_{\text{MS}} \neq \emptyset$; neither an edge, a face, nor a rank-one computation uses it. Probability-specific power tables are rebuilt for each multinomial key, as claimed. Retaining one scalar answer per key and taking the maximum live workspace gives the displayed memory bound because vertices, edges, face keys, and rank-one keys each use $O(1)$ auxiliary storage. The same argument permits the remaining nonface keys to be compared with the positive-corner coefficient evaluator without changing any probability.

Adding $\sum_{\pi \in \Pi_{\text{rank}1}} \mathsf{R}_{M, k_0, k_1}$ to the sequential branch charges proves the proposed aggregate time formula. Because the rank-one branch is constant-space and has no factorial table, the displayed memory maximum and the factorial-trigger condition are unchanged. $\square$

Justification. The feasible count-vector bijection gives a zero-cell-safe exact evaluator and a reference branch against which the coefficient and rolling recurrences are compared. *Gap.* Probability-specific power tables must be rebuilt per multinomial key even though factorial preprocessing is shared. *Consumer.* Process exact keys sequentially, share factorials, and use $M + \mathsf{T}_{M, \bar{k}_0, \bar{k}_1}$ only as one branch of the three-way time comparison.

Lemma 9 (`lem:four-cell-rank-event-dynamic-program`). *Let $\mathcal{E}_0$ and $\mathcal{E}_1$ be finite sets, let R be a probability mass function on $\mathcal{E}_0 \times \mathcal{E}_1$, and let $b_a : \mathcal{E}_a \rightarrow \{0, 1\}$. Define $\pi_{xy} = \sum_{e_0, e_1} R(e_0, e_1) \mathbf{1}\{b_0(e_0) = x, b_1(e_1) = y\}$. For $1 \leq k_0, k_1 \leq M + 1$, if both ranks are at most M , the retained recurrence*

$$d_0(0, 0) = 1, \quad d_{i+1}(r, s) = \sum_{x, y \in \{0, 1\}} \pi_{xy} d_i(r - x, s - y)$$

satisfies

$$\sum_{r < k_0, s < k_1} d_M(r, s) = \Pr \left\{ \sum_i b_0(E_{i,0}) < k_0, \sum_i b_1(E_{i,1}) < k_1 \right\}.$$

After R is tabulated, this generic recurrence costs $O(|\mathcal{E}_0||\mathcal{E}_1| + Mk_0k_1)$ operations and $O(k_0k_1)$ memory. With exactly one boundary rank, the other one-dimensional recurrence costs $O(|\mathcal{E}_0||\mathcal{E}_1| + Mk_c)$ operations and $O(k_c)$ memory; with two boundary ranks the event is certain.

For a finite collection Π of four-cell keys, first evaluate $\Pi_{\text{vert}} = \{\pi : |\text{supp}(\pi)| = 1\}$ by

$$D_{M,k_0,k_1}(\pi) = \mathbf{1}\{Mx < k_0, My < k_1\}$$

at its unique support point (x, y) . Next evaluate $\Pi_{\text{edge}} = \{\pi \notin \Pi_{\text{vert}} : |\text{supp}(\pi)| = 2\}$ as follows. For a deterministically oriented support $\{u, v\}$, put $p = \pi_v$, $N \sim \text{Bin}(M, p)$, $c_a = Mu_a$, and $d_a = v_a - u_a$. Then the two terminal counts are

$$R_M = c_0 + d_0N, \quad S_M = c_1 + d_1N,$$

and their strict inequalities are equivalent to the explicit interval $[L_\pi, U_\pi]$ in **def:simplex-vertex-key-prepass**, with the stated empty-interval convention. Hence

$$D_{M,k_0,k_1}(\pi) = \Pr\{L_\pi \leq N \leq U_\pi\}.$$

The vertex branch costs $O(1)$ exact time and workspace; the edge branch costs $O(E_M(\pi))$ exact arithmetic operations and $O(1)$ workspace. Deterministic orientation, evaluator selection, and tie-breaking preserve exactness. For every remaining support-three key, inspect the four corner orientations. If $\pi_{11}^\sigma = 0$ and $t_0^\sigma + t_1^\sigma > M$, then $T_0^\sigma + T_1^\sigma \leq M$, so the two transformed failure events are disjoint and

$$J_\sigma = 1 - \Pr\{T_0^\sigma \geq t_0^\sigma\} - \Pr\{T_1^\sigma \geq t_1^\sigma\}.$$

The exact shorter binomial-tail recurrences and the existing corner identities evaluate the original lower-lower probability in $O(F_M(\pi))$ exact arithmetic operations and $O(1)$ workspace; reconstruction reuses complements of the two evaluated tails. Deterministic charge minimization and lexicographic tie-breaking preserve exactness. Only two-finite-rank support-at-least-three keys outside this face prepass can require a generic bivariate evaluator.

For every remaining support-four key satisfying $\pi_{00}\pi_{11} = \pi_{01}\pi_{10}$, define $\theta_0 = \pi_{10} + \pi_{11}$, $\theta_1 = \pi_{01} + \pi_{11}$, $\bar{k}_a = \min\{k_a, M - k_a + 1\}$, and $R_{M,k_0,k_1} = 1 + 2\lceil \log_2 M \rceil + \bar{k}_0 + \bar{k}_1$. Exact determinant testing and the factorization

$$\pi_{00} + \pi_{10}x + \pi_{01}y + \pi_{11}xy = \{(1 - \theta_0) + \theta_0x\}\{(1 - \theta_1) + \theta_1y\}$$

give

$$D_{M,k_0,k_1}(\pi) = H_{M,k_0}(\theta_0)H_{M,k_1}(\theta_1)$$

in $O(R_{M,k_0,k_1})$ exact arithmetic operations and $O(1)$ auxiliary workspace. Such a key is evaluated after the vertex, edge, and incompatible-face strata and before any rolling, multinomial-sum, or positive-corner branch.

Thus only two-finite-rank keys outside the vertex, edge, incompatible-face, and rank-one strata can require a generic bivariate evaluator.

Proof. Put $R_i = \sum_{j=1}^i b_0(E_{j,0})$ and $S_i = \sum_{j=1}^i b_1(E_{j,1})$. At $i = 0$, $d_0(0, 0) = 1$. If the retained entries at stage i equal the corresponding probabilities, independence of draw $i + 1$ and the definition of π give

$$\Pr\{R_{i+1} = r, S_{i+1} = s\} = \sum_{x,y \in \{0,1\}} \pi_{xy} \Pr\{R_i = r - x, S_i = s - y\}. \quad (1)$$

Both increments are nonnegative, so a state discarded after reaching either strict threshold cannot return. Induction proves the recurrence and the terminal rectangle sum. Forming π scans R once; each of M stages has at most k_0k_1 retained states and two rolling arrays suffice. This proves the generic time and memory bounds. A rank $M + 1$ is never reached by a count of M Bernoulli increments. Deleting one such coordinate gives the one-dimensional recurrence, and deleting both makes the event certain.

If π has support one at $u = (x, y)$, every increment equals u , so

$$(R_M, S_M) = (Mx, My)$$

almost surely, proving the vertex formula. If its support is $\{u, v\}$, orient it deterministically and let I_i indicate the atom v . Then $N = \sum_i I_i \sim \text{Bin}(M, \pi_v)$, and

$$R_M = Mu_0 + (v_0 - u_0)N, \quad S_M = Mu_1 + (v_1 - u_1)N. \quad (2)$$

For coordinate a , the inequality is an upper bound on N when $d_a = 1$, a lower bound when $d_a = -1$, automatic or impossible when $d_a = 0$. Intersecting those conditions with $0 \leq N \leq M$ gives exactly $[L_\pi, U_\pi]$, including its empty convention. The three interval evaluators in `def:simplex-vertex-key-prepass`, together with `lem:exact-binomial-shorter-tail-evaluator`, prove the exact charge $E_M(\pi)$ and constant workspace. Reversing the support orientation replaces N by $M - N$, so deterministic least-charge choice and tie-breaking preserve the answer.

Finally let a remaining support-three key have an admissible face orientation. Since $\pi_{11}^\sigma = 0$, every transformed increment satisfies $B_{i0}^\sigma + B_{i1}^\sigma \leq 1$, and hence $T_0^\sigma + T_1^\sigma \leq M$. The inequality $t_0^\sigma + t_1^\sigma > M$ makes the two transformed upper-tail events disjoint. Thus

$$J_\sigma = 1 - \Pr\{T_0^\sigma \geq t_0^\sigma\} - \Pr\{T_1^\sigma \geq t_1^\sigma\}. \quad (3)$$

The two marginals are binomial. Exact shorter-tail evaluation, complementation, and the four corner identities give the original lower-lower probability with the charge $F_M(\pi)$ and constant workspace proved in `lem:corner-oriented-four-cell-rank-evaluator`. Reconstruction introduces no duplicate marginal evaluation. Therefore only support-at-least-three keys not selected by the face prepass can reach a generic bivariate evaluator.

It remains to justify the rank-one prepass. For a support-four key put $p_{xy} = \pi_{xy}$, $\theta_0 = p_{10} + p_{11}$, and $\theta_1 = p_{01} + p_{11}$. If $p_{00}p_{11} = p_{01}p_{10}$, then

$$\theta_0\theta_1 = p_{10}p_{01} + p_{11}(p_{10} + p_{01} + p_{11}) = p_{00}p_{11} + p_{11}(1 - p_{00}) = p_{11}. \quad (R1)$$

Thus $p_{10} = \theta_0(1 - \theta_1)$, $p_{01} = (1 - \theta_0)\theta_1$, and $p_{00} = (1 - \theta_0)(1 - \theta_1)$. Therefore

$$p_{00} + p_{10}x + p_{01}y + p_{11}xy = \{(1 - \theta_0) + \theta_0x\}\{(1 - \theta_1) + \theta_1y\}. \quad (R2)$$

Raising (R2) to the M -th power proves that the two terminal counts are independent binomials and hence that

$$D_{M,k_0,k_1}(\pi) = H_{M,k_0}(\theta_0)H_{M,k_1}(\theta_1). \quad (R3)$$

The determinant test consists of two exact multiplications and an exact comparison. Applying `lem:exact-binomial-shorter-tail-evaluator` separately to the two factors in (R3) costs

$$O(1 + 2\lceil \log_2 M \rceil + \bar{k}_0 + \bar{k}_1)$$

exact operations. The factors are processed sequentially, so only the current recurrence term, its partial sum, and the first completed factor are stored. This is $O(1)$ auxiliary workspace.

Applying this exact test only after the support-one, support-two, and eligible support-three strata makes the four prepasses disjoint. Thus a key reaching a generic bivariate routine lies outside the rank-one stratum as well as the earlier three strata, proving the final assertion. $\square$

Justification. The retained-state induction proves exactness on every simplex face and supplies the rolling branch of the three-way evaluator. *Gap.* The coarse Mk_0k_1 charge hides both corner complementation and positive-corner coefficient savings. *Consumer.* Use the least exact operation count among the rolling corner, multinomial sum, and positive-corner coefficient branches, but charge the workspace of that same selected branch.

Lemma 10 (`lem:orbit-score-factorization`). *Conditional on $\mathcal{T}$, for every arm a and every fixed measurable predictor $\hat{m}_a$, there are probability laws $F_{a,O}$ on $[0, \infty)$, indexed by $O \in \mathcal{O}$ with $O \subseteq \mathcal{U}_a$, such that for every Borel set C ,*

$$\Pr(O_{i,a} = O, S_{i,a} \in C \mid \mathcal{T}) = q_{a,O}F_{a,O}(C), \quad \Pr(O_{T,a} = O, S_{0,a} \in C \mid \mathcal{T}) = p_{a,O}F_{a,O}(C).$$

The M calibration orbit-score pairs are iid and independent of the target pair, armwise.

Proof. Fix an arm $a \in \{0, 1\}$ and work conditionally on $\mathcal{T}$. By `ass:external-observed-training`, conditioning fixes $\hat{m}_a$ without changing the law of the arrays, assignments, target, or auxiliary draws. For $v \in \mathcal{V}$, define

$$R_{i,v,a} = |Y_{iv}(a) - \hat{m}_a(X_{iv})|.$$

Let $F_{a,v}$ be its conditional law. This law is independent of i by **ass:independent-replicated-arrays**. If (v, a) and (v', a) lie in the same orbit, then $v' = \pi_g(v)$ for some $g \in \mathcal{G}$. By **def:group-action**, the v' -coordinate of gL_i is the v -coordinate of L_i ; hence **ass:finite-group-latent-invariance** gives

$$(X_{iv'}, Y_{iv'}(a)) \stackrel{d}{=} (X_{iv}, Y_{iv}(a)).$$

Applying the same fixed measurable map $\hat{m}_a$ gives $F_{a,v'} = F_{a,v}$. Thus $F_{a,O} := F_{a,v}$, for any $(v, a) \in O$, is a well-defined probability law on $[0, \infty)$. The mixture kernel inherits the kernel and realized-arm support conditions on every z with $\nu(z) > 0$. Thus $J_{i,a} = v$ implies $Z_{iv} = a$ almost surely, and then $S_{i,a} = R_{i,v,a}$. The assignment is independent of L_i by **ass:known-matched-randomization**, and the auxiliary selector draw depends on the record only through Z_i by **ass:selector-auxiliary-randomization**. Therefore, for every Borel $C \subseteq [0, \infty)$,

$$\Pr\{U_{i,a} = (v, a), S_{i,a} \in C \mid \mathcal{T}\} = \sum_{z: \nu(z) > 0} \nu(z) \kappa_{w,a}(v \mid z) F_{a,v}(C) = q_{w,a}^{\text{obs}}(v, a) F_{a,v}(C). \quad (1)$$

The last equality uses **lem:null-support-kernel-extensions**: the unrestricted defining sum has zero contribution from every off-support assignment. The same lemma shows that $q_{w,a}^{\text{obs}}$ is a probability law supported on $\mathcal{U}_a$. Summing (1) over the members whose arm lift belongs to O gives

$$\Pr\{O_{i,a} = O, S_{i,a} \in C \mid \mathcal{T}\} = q_{a,O} F_{a,O}(C). \quad (2)$$

For the target, **ass:prespecified-target** makes T independent of L_0 and $\mathcal{T}$. Consequently

$$\Pr\{T^a = (v, a), S_{0,a} \in C \mid \mathcal{T}\} = q_{\text{tar}}(v) F_{a,v}(C),$$

and summing over an orbit yields

$$\Pr\{O_{T,a} = O, S_{0,a} \in C \mid \mathcal{T}\} = p_{a,O} F_{a,O}(C). \quad (3)$$

Finally, the replicated arrays and assignments are iid, and the auxiliary-selector law is a product over replicates. Hence the M calibration orbit–score pairs are iid. The target pair is a function of (L_0, T) , which is independent of all calibration records and their auxiliary draws. This proves the asserted armwise independence and factorization. $\square$

Justification. This lemma isolates the common within-orbit score kernel and proves that supported observable selection changes only orbit mixing weights. *Gap.* Without the factorization, orbit balance would be a combinatorial condition with no demonstrated connection to score laws. *Consumer.* Use the factorization to compare calibration and target experiments through their finite orbit-mass vectors; off-support kernel values are irrelevant.

Proposition 6 (prop:finite-rank-orbit-gap-bound). *For every fixed or prespecified w , conditional on $\mathcal{T}$,*

$$\Pr_P(\tau_T \in \mathcal{C}_w \mid \mathcal{T}) \geq \max\{0, 1 - 2\rho_{M,\alpha} - \Delta_0(w) - \Delta_1(w)\}$$

for every $P \in \mathcal{P}_{\mathcal{G},K}$ and arbitrary fixed measurable trained predictor maps. Here $\rho_{M,\alpha} = j/(M+1)$ on each staircase interval $\alpha \in [2j/(M+1), 2(j+1)/(M+1))$, so the elbows occur at $\alpha = 2j/(M+1)$. Consequently $|\delta_{a,O}(w)| \leq \Delta_a(w)$ for every $a \in \{0, 1\}$ and $O \in \mathcal{O}$.

Proof. Set $\varepsilon_0 = \varepsilon_1 = \alpha/2$. By **lem:finite-rank-staircase**, $k_0 = k_1 = k$ and $\rho_{M,\varepsilon_0} = \rho_{M,\varepsilon_1} = \rho_{M,\alpha}$. By **def:rank-inversion**, $\mathcal{C}_w^\varepsilon = \mathcal{C}_w$. Substitution into **prop:asymmetric-finite-rank-orbit-gap-bound** proves the coverage inequality. The same staircase lemma gives $\rho_{M,\alpha} = j/(M+1)$ precisely for $\alpha \in [2j/(M+1), 2(j+1)/(M+1))$. Finally, both orbit-mass vectors are probability vectors, so $\sum_O \delta_{a,O}(w) = 0$. Their total positive and negative gap masses each equal $\frac{1}{2} \sum_O |\delta_{a,O}(w)| = \Delta_a(w)$, proving $|\delta_{a,O}(w)| \leq \Delta_a(w)$. $\square$

Proposition 7 (prop:asymmetric-finite-rank-orbit-gap-bound). *For every fixed or prespecified w , conditional on $\mathcal{T}$, the asymmetric ordinary rank construction satisfies*

$$\Pr_P\{\tau_T \in \mathcal{C}_w^\varepsilon \mid \mathcal{T}\} \geq \max\{0, 1 - \rho_{M,\varepsilon_0} - \rho_{M,\varepsilon_1} - \Delta_0(w) - \Delta_1(w)\}$$

for every $P \in \mathcal{P}_{\mathcal{G},K}$ and arbitrary fixed measurable trained predictor maps. The coordinate elbows are $\varepsilon_a = j/(M+1)$, and $|\delta_{a,O}(w)| \leq \Delta_a(w)$ for every arm and orbit. On the diagonal $\varepsilon_0 = \varepsilon_1 = \alpha/2$, this becomes the symmetric bound $\max\{0, 1 - 2\rho_{M,\alpha} - \Delta_0(w) - \Delta_1(w)\}$.

Proof. Condition on $\mathcal{T}$. The factorization argument in `lem:orbit-score-factorization` gives $Q_a^S = \sum_O q_{a,O} F_{a,O}$ and $P_a^S = \sum_O p_{a,O} F_{a,O}$. Exactly as in (1) of the active-arm proposition,

$$\sup_C |P_a^S(C) - Q_a^S(C)| \leq \Delta_a(w). \quad (1)$$

Replace only the independent target score by a draw from Q_a^S . Integrating the rank-event section, a $[0, 1]$ -valued function, against the signed measure in (1) changes its probability by at most $\Delta_a(w)$. Under the replacement law the $M + 1$ scores are iid, and lexicographic randomized ranks give

$$\Pr\{S_{0,a} > r_{a,(k_a)} \mid \mathcal{T}\} \leq \rho_{M,\varepsilon_a} + \Delta_a(w). \quad (2)$$

When $k_a = M + 1$, the left side is zero, so (2) remains true. Boundedness shows that clipping preserves membership of $Y_{0T}(a)$, and simultaneous armwise membership implies membership of the difference. Bonferroni applied to (2) therefore gives

$$\Pr(\tau_T \in \mathcal{C}_w^\varepsilon \mid \mathcal{T}) \geq \max\{0, 1 - \rho_{M,\varepsilon_0} - \rho_{M,\varepsilon_1} - \Delta_0(w) - \Delta_1(w)\}.$$

By `lem:armwise-finite-rank-staircase`, $\rho_{M,\varepsilon_a} = j/(M+1)$ exactly on $\varepsilon_a \in [j/(M+1), (j+1)/(M+1))$, giving the elbows. Since the signed orbit gaps sum to zero, each absolute coordinate is at most the common positive or negative total $\Delta_a(w)$. Finally, setting $\varepsilon_0 = \varepsilon_1 = \alpha/2$ gives $k_0 = k_1 = k$, the symmetric construction, and the displayed diagonal bound. $\square$

Justification. The proposition is the sharp boundary-regime version of the finite-rank total-variation certificate: only finite-rank arms can fail after clipping. *Gap.* Charging $\Delta_a(w)$ to a boundary arm understates coverage even though that arm's interval is already $[-K, K]$. *Consumer.* Compute $\mathcal{A}_\varepsilon$ from the chosen arm budgets and sum $\rho_{M,\varepsilon_a} + \Delta_a(w)$ only over that set.

Proposition 8 (`prop:asymmetric-disjoint-orbit-witness`). *Fix an arm a and suppose every target-supported arm- a orbit has zero $q_{w,a}^{\text{obs}}$ mass. Under `def:disjoint-orbit-law`, the latent law belongs to $\mathcal{P}_{\mathcal{G},K}$, every selected calibration score in both arms is zero, and the common target effect is K if $a = 1$ and $-K$ if $a = 0$. If $k_0 \leq M$ and $k_1 \leq M$, then $I_{0,w}^\varepsilon = I_{1,w}^\varepsilon = \{0\}$, $\mathcal{C}_w^\varepsilon = \{0\}$, and coverage is zero. If exactly one $k_c = M + 1$, then the boundary arm interval is $[-K, K]$, the other is $\{0\}$, $\mathcal{C}_w^\varepsilon = [-K, K]$, and this witness has coverage one. If both $k_0 = k_1 = M + 1$, then $\mathcal{C}_w^\varepsilon = [-2K, 2K]$ and this witness again has coverage one. Moreover $k_c \leq M$ is equivalent to $\varepsilon_c \geq 1/(M+1)$. The symmetric choice recovers `prop:disjoint-orbit-witness`.*

Proof. The membership and score calculation in `prop:disjoint-orbit-witness` does not use equality of the two rank indices. Under `def:disjoint-orbit-law`, the orbit-constant bounded prototype is $\mathcal{G}$ -invariant, all calibration-selected outcomes in the adverse arm are zero, all other-arm outcomes are zero, and both predictors are zero. Selection occurs on $\text{supp}(\nu)$ almost surely, where the corrected kernel condition guarantees realized-arm support; total off-support extensions carry zero integrated mass. Thus, in both arms,

$$S_{1,c} = \dots = S_{M,c} = 0, \quad c \in \{0, 1\}.$$

The common target effect is K for $a = 1$ and $-K$ for $a = 0$, exactly as in the symmetric witness.

If $k_0, k_1 \leq M$, each finite order statistic is zero, so

$$I_{0,w}^\varepsilon = I_{1,w}^\varepsilon = \{0\}, \quad \mathcal{C}_w^\varepsilon = \{0\}.$$

Since $K > 0$, coverage is zero. Suppose exactly one rank, say k_c , equals $M+1$. By `prop:asymmetric-rank-boundary`, the corresponding interval is $[-K, K]$; the other rank is finite and its interval is $\{0\}$. Either possible Minkowski difference is therefore $[-K, K]$. The witness effect is one of the endpoints K and $-K$, so its coverage is one. If both ranks equal $M + 1$, the same boundary proposition gives $\mathcal{C}_w^\varepsilon = [-2K, 2K]$, which again contains the witness effect surely.

Finally, with $N = M + 1$,

$$k_c = \lceil N(1 - \varepsilon_c) \rceil \leq M \iff N(1 - \varepsilon_c) \leq M \iff \varepsilon_c \geq \frac{1}{M+1}.$$

On setting $\varepsilon_0 = \varepsilon_1 = \alpha/2$, the two ranks coincide with k , and the finite-rank part reduces to **prop:disjoint-orbit-witness**, while the common boundary part reduces to **prop:rank-boundary**. $\square$

Justification. It separates catastrophic unsupported-orbit failure from the support-trivializing protection of an infinite conformal radius. *Gap.* A zero-coverage warning stated without the rank regime is false when a boundary interval already covers $[-K, K]$. *Consumer.* Flag unsupported target orbits, then report zero coverage only when both arm ranks are finite.

Proposition 9 (prop:asymmetric-s2-reduction). *The pair-swap and mate-control legality conclusions of **prop:s2-reduction** remain unchanged under the asymmetric budget. If the target arm-1 lift is supported on $O_{b,1}$ and a legal implemented arm-1 selector has $q_{w,1}^{\text{obs}}(O_{b,1}) = 0$, the asymmetric disjoint-orbit witness has zero coverage when both $k_0, k_1 \leq M$, has effect set $[-K, K]$ and coverage one when exactly one rank equals $M + 1$, and has effect set $[-2K, 2K]$ and coverage one when both ranks equal $M + 1$. The symmetric choice recovers **prop:s2-reduction**.*

Proof. The orbit calculation, stacked orbit-mass mismatch, and arm-1 selector illegality in **prop:s2-reduction** depend only on the group action and on **ass:observable-selector-kernel**, not on the conformal error allocation. They therefore remain unchanged. In particular, target support on $O_{b,1}$ together with $q_{w,1}^{\text{obs}}(O_{b,1}) = 0$ is the arm-1 disjoint-orbit premise. Applying **prop:asymmetric-disjoint-orbit-witness** gives, according to the three exhaustive rank patterns,

(k_0, k_1)	$\mathcal{C}_w^\varepsilon$	coverage
$k_0, k_1 \leq M$	$\{0\}$	0
exactly one $k_a = M + 1$	$[-K, K]$	1
$k_0 = k_1 = M + 1$	$[-2K, 2K]$	1.

The boundary rows are also consistent with **prop:asymmetric-rank-boundary**. On the equal-split diagonal, **lem:armwise-finite-rank-staircase** gives $k_0 = k_1 = k$; hence the middle row cannot occur and the remaining two rows are exactly **prop:s2-reduction**. $\square$

Justification. The pair reduction carries the asymmetric rank regimes without changing the selector-legality certificate. *Gap.* Mate-control leakage and one-arm conformal boundaries have not been combined in this finite orbit calculation. *Consumer.* A matched-pair protocol can audit selector legality and boundary behavior together.

Proposition 10 (prop:asymmetric-two-pair-sanity-check). *In the two-pair instance of **prop:two-pair-sanity-check**, the four arm-specific orbits and the mate-control LP and legality failures are unchanged under the asymmetric budget. If a legal selector has $q_{w,1}^{\text{obs}}(O_{1,1}) = 0$, the bounded witness has $\mathcal{C}_w^\varepsilon = \{0\}$ and coverage zero when both ranks are finite, $\mathcal{C}_w^\varepsilon = [-K, K]$ and coverage one when exactly one rank is at the boundary, and $\mathcal{C}_w^\varepsilon = [-2K, 2K]$ and coverage one when both ranks are at the boundary. The symmetric choice recovers the frozen two-pair calculation.*

Proof. Changing the armwise error budget changes neither $\mathcal{G} = S_2 \times S_2$ nor its action, the target orbit vector, the stacked selector equations, or the realized-arm support restriction. Thus the four-orbit computation and mate-control LP and legality failures proved in **prop:two-pair-sanity-check** remain valid. If $q_{w,1}^{\text{obs}}(O_{1,1}) = 0$, the target-supported arm-1 orbit has zero calibration mass. The three cases in **prop:asymmetric-disjoint-orbit-witness** therefore give respectively

$$(\mathcal{C}_w^\varepsilon, \text{coverage}) = (\{0\}, 0), \quad ([-K, K], 1), \quad ([-2K, 2K], 1),$$

when both ranks are finite, exactly one is a boundary rank, or both are boundary ranks. On the symmetric diagonal, **lem:armwise-finite-rank-staircase** gives $k_0 = k_1 = k$, so only the first and third cases remain, exactly reproducing **prop:two-pair-sanity-check**. $\square$

Justification. This is a finite implementation check for all three asymmetric rank patterns. *Gap.* The symmetric four-member test omits the exactly-one-boundary set. *Consumer.* It is a unit test for selector LP and armwise rank code.

Proposition 11 (prop:active-arm-finite-rank-orbit-gap-bound). *For a prespecified $\varepsilon = (\varepsilon_0, \varepsilon_1)$, define the active-arm set $\mathcal{A}_\varepsilon = \{a \in \{0, 1\} : k_a \leq M\}$. For every fixed or prespecified w , conditional on $\mathcal{T}$,*

$$\Pr_P\{\tau_T \in \mathcal{C}_w^\varepsilon \mid \mathcal{T}\} \geq \max\left\{0, 1 - \sum_{a \in \mathcal{A}_\varepsilon} (\rho_{M, \varepsilon_a} + \Delta_a(w))\right\}$$

for every $P \in \mathcal{P}_{G, K}$ and arbitrary fixed measurable trained predictor maps. If $a \notin \mathcal{A}_\varepsilon$, then $I_{a, w}^\varepsilon = [-K, K]$ contains $Y_{0T}(a)$ surely, so arm a contributes neither a rank penalty nor an orbit-gap penalty. The coordinate elbows are $\varepsilon_a = j/(M+1)$, and $|\delta_{a, O}(w)| \leq \Delta_a(w)$ for every arm and orbit. If exactly one arm is at the boundary, only the other arm appears in the sum. If both arms are at the boundary, the effect set is $[-2K, 2K]$ and coverage is one. On the equal-split diagonal with $k \leq M$, the bound is $\max\{0, 1 - 2\rho_{M, \alpha} - \Delta_0(w) - \Delta_1(w)\}$; with $k = M+1$, coverage is one.

Proof. Work conditionally on $\mathcal{T}$. Fix an arm a . By `lem:orbit-score-factorization`, its calibration- and target-score laws are

$$Q_a^S = \sum_O q_{a, O} F_{a, O}, \quad P_a^S = \sum_O p_{a, O} F_{a, O}.$$

Put $d_O = p_{a, O} - q_{a, O}$. Since $\sum_O d_O = 0$ and $0 \leq F_{a, O}(C) \leq 1$, every Borel C satisfies

$$|P_a^S(C) - Q_a^S(C)| = \left| \sum_O d_O F_{a, O}(C) \right| \leq \sum_{O: d_O > 0} d_O = \frac{1}{2} \sum_O |d_O| = \Delta_a(w). \quad (1)$$

Suppose $a \in \mathcal{A}_\varepsilon$, so $k_a \leq M$. Let $\tilde{S}_{0, a} \sim Q_a^S$ be independent of the calibration scores. For the rank-failure event, integrate its calibration-section indicator as a function $f(s) \in [0, 1]$. Because $P_a^S - Q_a^S$ has total mass zero, (1) implies

$$\Pr(S_{0, a} > r_{a, (k_a)} \mid \mathcal{T}) \leq \Pr(\tilde{S}_{0, a} > r_{a, (k_a)} \mid \mathcal{T}) + \Delta_a(w). \quad (2)$$

Under the comparison law all $M+1$ scores are iid. Continuous lexicographic tie-breakers make the target rank uniform, and strict failure implies rank greater than k_a . Thus

$$\Pr(S_{0, a} > r_{a, (k_a)} \mid \mathcal{T}) \leq \rho_{M, \varepsilon_a} + \Delta_a(w). \quad (3)$$

If $a \notin \mathcal{A}_\varepsilon$, then $k_a = M+1$; by `prop:asymmetric-rank-boundary`, $I_{a, w}^\varepsilon = [-K, K]$, which contains $Y_{0T}(a)$ surely. For an active arm, boundedness gives $Y_{0T}(a) \notin I_{a, w}^\varepsilon$ if and only if $S_{0, a} > r_{a, (k_a)}$. If both outcomes lie in their intervals, their difference lies in $\mathcal{C}_w^\varepsilon$. A union bound over active arms in (3) therefore proves

$$\Pr_P(\tau_T \in \mathcal{C}_w^\varepsilon \mid \mathcal{T}) \geq \max\left\{0, 1 - \sum_{a \in \mathcal{A}_\varepsilon} (\rho_{M, \varepsilon_a} + \Delta_a(w))\right\}.$$

The staircase and its elbows follow from `lem:armwise-finite-rank-staircase`. Also $\sum_O \delta_{a, O}(w) = 0$, so the total positive and negative gap masses both equal $\Delta_a(w)$; hence $|\delta_{a, O}(w)| \leq \Delta_a(w)$. The one- and two-boundary conclusions follow from the definition of the active set and `prop:asymmetric-rank-boundary`. On the equal-split diagonal, both arms are active exactly when $k \leq M$, and the asserted symmetric formula follows; if $k = M+1$, coverage is one. $\square$

Justification. It is the coverage-facing sharpening: orbit-kernel total-variation contraction applies to every finite-rank arm, while bounded support makes every boundary arm surely covered. *Gap.* Charging $\Delta_a(w)$ to an arm with $k_a = M+1$ understates the available certificate because that arm cannot fail. *Consumer.* Form $\mathcal{A}_\varepsilon$ from the prespecified ranks and sum rank and imbalance penalties only over that set.

Lemma 11 (lem:null-support-kernel-extensions). *Let κ_a and $\tilde{\kappa}_a$ be total Markov kernels on the full assignment space such that $\kappa_a(v \mid z) = \tilde{\kappa}_a(v \mid z)$ for every $v \in \mathcal{V}$ and every z with $\nu(z) > 0$. Then $\sum_z \nu(z) \kappa_a(v \mid z) = \sum_z \nu(z) \tilde{\kappa}_a(v \mid z)$ for every v . Under `ass:observable-selector-kernel`, each unrestricted integrated law $q_{h, a}^{\text{obs}}$, and every mixture $q_{w, a}^{\text{obs}}$, is a probability law supported on $\mathcal{U}_a$. Thus arbitrary total extensions of the selector kernels off $\text{supp}(\nu)$ cannot affect any integrated selector law or orbit mass.*

Proof. For every v , subtract the two unrestricted finite sums. Agreement on the support gives

$$\sum_z \nu(z) \{ \kappa_a(v \mid z) - \tilde{\kappa}_a(v \mid z) \} = \sum_{z: \nu(z)=0} \nu(z) \{ \kappa_a(v \mid z) - \tilde{\kappa}_a(v \mid z) \} = 0.$$

For a library kernel, nonnegativity on supported assignments implies $q_{h,a}^{\text{obs}}(v, a) \geq 0$, and normalization gives

$$\sum_{v \in \mathcal{V}} q_{h,a}^{\text{obs}}(v, a) = \sum_{z: \nu(z) > 0} \nu(z) \sum_v \kappa_{h,a}(v \mid z) = \sum_{z: \nu(z) > 0} \nu(z) = 1.$$

If $z_v \neq a$, realized-arm support makes its supported summand zero, so the integrated law is supported on $\mathcal{U}_a$. Convex mixtures preserve these properties. Every orbit mass is a finite sum of integrated member masses, so it is likewise unchanged by an off-support extension. $\square$

Lemma 12 (lem:one-boundary-orbit-frontier-evaluator). *Suppose exactly one arm $b \in \{0, 1\}$ has $k_b = M + 1$, and put $c = 1 - b$, where $k_c \leq M$. In `def:coverage-frontier`, the boundary-arm rank event is identically true, so its score map, total preorder, target lift, and calibration-orbit coordinate are irrelevant. For a fixed nonboundary score map s_c , define $\theta(O_t) = \sum_{O \in \mathcal{O}_c: s_c(O) < s_c(O_t)} q_{c,O}$. Then the joint rank-event probability equals*

$$\sum_{O_t \in \mathcal{O}_c} p_{c,O_t} \Pr\{\text{Bin}(M, \theta(O_t)) < k_c\}.$$

For ordered nonempty tie classes $C_1 \prec \dots \prec C_d$, define $q_j = \sum_{O \in C_j} q_{c,O}$, $p_j = \sum_{O \in C_j} p_{c,O}$, and $\theta_j = \sum_{\ell < j} q_\ell$. The probability is exactly

$$\sum_{j=1}^d p_j \Pr\{\text{Bin}(M, \theta_j) < k_c\},$$

and q_j , p_j , and θ_j are obtained by one cumulative ordered-class scan. For $0 < \theta < 1$, the lower-tail terms start at $t_0 = (1 - \theta)^M$ and satisfy

$$t_{r+1} = t_r \frac{M - r}{r + 1} \frac{\theta}{1 - \theta}, \quad 0 \leq r < k_c - 1,$$

whereas the upper-tail terms start at $u_M = \theta^M$ and satisfy

$$u_{r-1} = u_r \frac{r}{M - r + 1} \frac{1 - \theta}{\theta}, \quad k_c < r \leq M.$$

The initial power is computed by exponentiation by squaring in $O(\log M)$ arithmetic operations, and summing the shorter of the lower tail and the complementary upper tail uses $O(\min\{k_c, M - k_c + 1\})$ further operations. At $\theta = 0$ the probability is 1, and at $\theta = 1$ it is 0, without division. Consequently, after both $q_c = (q_{c,O} : O \in \mathcal{O}_c)$ and $p_c = (p_{c,O} : O \in \mathcal{O}_c)$ are tabulated, exact maximization in the unit-cost arithmetic model enumerates only the F_{m_c} total preorders of $\mathcal{O}_c$ and uses

$$O(F_{m_c} m_c [1 + \log M + \min\{k_c, M - k_c + 1\}])$$

arithmetic operations and $O(1)$ auxiliary CDF memory. The bound contains no factor F_{m_1-c} , $|\mathcal{V}|$, or per-case $m_0 m_1$ scan.

Proof. Because $k_b = M + 1$, the convention in `def:rank-inversion` gives $Q_{k_b} = +\infty$. Hence, for every realization and every boundary-arm score map,

$$s_b(O_{T,b}) \leq Q_{k_b}(s_b(O_{1,b}), \dots, s_b(O_{M,b})) = +\infty. \quad (1)$$

Thus the boundary-arm event is the entire sample space. Removing it from the conjunction in `def:coverage-frontier` proves that the boundary-arm score map, preorder, target lift, and calibration-orbit coordinate are irrelevant.

Fix s_c and condition on $O_{T,c} = O_t$. The elementary order-statistic identity gives

$$s_c(O_t) \leq Q_{k_c}(s_c(O_{1,c}), \dots, s_c(O_{M,c})) \iff N_t := \sum_{i=1}^M \mathbf{1}\{s_c(O_{i,c}) < s_c(O_t)\} < k_c. \quad (2)$$

Indeed, the k_c -th order statistic is strictly below the target score exactly when at least k_c calibration scores are strictly below it. By `ass:known-matched-randomization`, `ass:selector-auxiliary-randomization`, and `def:observable-selector-mixture`, the nonboundary calibration orbits are iid with marginal mass function q_c . They are independent of the target by `ass:prespecified-target`. Therefore, conditional on $O_{T,c} = O_t$, the summands in N_t are iid Bernoulli variables with success probability

$$\theta(O_t) = \sum_{O \in \mathcal{O}_c: s_c(O) < s_c(O_t)} q_{c,O}. \quad (3)$$

Consequently $N_t \sim \text{Bin}(M, \theta(O_t))$. Averaging (2) over the target-orbit law proves

$$\sum_{O_t \in \mathcal{O}_c} p_{c,O_t} \Pr\{\text{Bin}(M, \theta(O_t)) < k_c\}. \quad (4)$$

The strict inequality in (3) shows explicitly that (4) handles ties.

Fix one total preorder and write its ordered nonempty tie classes as $C_1 \prec C_2 \prec \dots \prec C_d$, where $d \leq m_c$. Set

$$q_j = \sum_{O \in C_j} q_{c,O}, \quad p_j = \sum_{O \in C_j} p_{c,O}, \quad \theta_j = \sum_{\ell < j} q_\ell. \quad (5)$$

Every target orbit in C_j has the same strict-prefix mass θ_j , so (4) becomes

$$\sum_{j=1}^d p_j \Pr\{\text{Bin}(M, \theta_j) < k_c\}. \quad (6)$$

Initialize $\theta_1 = 0$. In class C_j , scan its members once to form q_j and p_j , evaluate the current binomial probability, and update $\theta_{j+1} = \theta_j + q_j$. Each orbit is visited exactly once, so the aggregation cost is $O(m_c)$.

For $0 < \theta_j < 1$, the lower-tail endpoint term is $t_0 = (1 - \theta_j)^M$, and consecutive terms satisfy

$$t_{r+1} = t_r \frac{M-r}{r+1} \frac{\theta_j}{1-\theta_j}, \quad 0 \leq r < k_c - 1. \quad (7)$$

Thus the desired probability is $\sum_{r=0}^{k_c-1} t_r$. Alternatively, the upper-tail endpoint term is $u_M = \theta_j^M$, with downward recurrence

$$u_{r-1} = u_r \frac{r}{M-r+1} \frac{1-\theta_j}{\theta_j}, \quad k_c < r \leq M, \quad (8)$$

and the desired probability is $1 - \sum_{r=k_c}^M u_r$. Exponentiation by squaring initializes either endpoint in $O(\log M)$ operations. Choosing the shorter recurrence, as formalized in `lem:exact-binomial-shorter-tail-evaluator`, costs

$$O(\log M + \min\{k_c, M - k_c + 1\}) \quad (9)$$

operations and constant auxiliary memory. All denominators in (7) and (8) are nonzero. At $\theta_j = 0$, the probability in (6) is 1; at $\theta_j = 1$, it is 0, because $1 \leq k_c \leq M$. Neither endpoint case uses division.

Combining the one-pass aggregation with (9), one preorder costs

$$O(m_c [1 + \log M + \min\{k_c, M - k_c + 1\}]). \quad (10)$$

The event depends on s_c only through its total preorder. Conversely, any total preorder with d ordered classes has a representative in $[0, K]$: assign score 0 when $d = 1$, and otherwise assign $(j-1)K/(d-1)$ to C_j . Hence enumeration of all F_{m_c} preorders gives the stated total time and $O(1)$ auxiliary distribution-function memory after p_c and q_c are tabulated. Formula (6) uses only the nonboundary marginals, so no factor F_{m_1-c} , member-level sum $|\mathcal{V}|$, or per-preorder $m_0 m_1$ scan occurs. $\square$

Justification. A boundary rank makes one arm's event certain, reducing the exact joint-inclusion certificate to a one-arm binomial mixture over ordered tie classes. *Gap.* A two-arm evaluator would retain an irrelevant preorder factor and obscure the exact one-dimensional reduction. *Consumer.* When exactly one rank equals $M + 1$, enumerate only the nonboundary arm's total preorders and evaluate its strict binomial tails.

Lemma 13 (lem:lebrun-rectangular-probability-comparator). *Lebrun computes arbitrary rectangular probabilities for multinomial, multivariate hypergeometric, and multivariate Polya distributions with a time--space algorithm and compares accuracy, cost, and storage against prior exact methods.*

Lemma 14 (lem:timans-max-rank-comparator). *Max-rank exploits dependence among simultaneous conformal tests to control family-wise error and improve on Bonferroni.*

Lemma 15 (lem:schlembach-multi-target-allocation-comparator). *Schlembach, Smirnov, and Winands optimize per-target error allocation for axis-aligned hyperrectangle volume while retaining joint coverage under exchangeability.*

Lemma 16 (lem:structured-volume-dynamic-program-comparator). *Gao, Shan, Srinivas, and Vijayaraghavan give a dynamic-programming conformal algorithm for near-optimal volume within structured families such as unions of k intervals. Its structured-set volume objective differs from the coherent common-target joint-armwise-inclusion certificate frontier evaluated in this note. The latter is a simultaneous armwise-inclusion certificate, not the actual causal effect-set coverage frontier.*

Justification. This cited leaf records Gao et al.'s structured-set volume algorithm as the closest verified computational comparator while making explicit that this note evaluates a coherent common-target joint-armwise-inclusion certificate frontier, which is sufficient for but not identical to causal effect-set coverage. *Gap.* Gao et al. optimize structured-set volume. This note instead evaluates a coherent common-target joint-armwise-inclusion certificate deficit, a simultaneous sufficient-event certificate rather than the actual causal effect-set coverage frontier. *Consumer.* Use this comparator to delimit the computational contribution: Gao et al. optimize structured-set volume, whereas this note exactly evaluates a coherent common-target joint-armwise-inclusion certificate frontier that lower-certifies, but is not identical to, causal effect-set coverage.

Lemma 17 (lem:coherent-two-finite-rank-frontier-evaluator). *Suppose $k_0, k_1 \leq M$. Define the arm-orbit sets $\mathcal{O}_a$, their sizes m_a , the joint calibration-orbit law*

$$R_w(O_0, O_1) = \sum_z \nu(z) \kappa_{w,0}(O_0 \mid z) \kappa_{w,1}(O_1 \mid z),$$

and the coherent common-target law

$$P_T(O_0, O_1) = \sum_{v \in \mathcal{V}} q_{\text{tar}}(v) \mathbf{1}\{O_0(v) = O_0, O_1(v) = O_1\}.$$

Let $L = |\text{supp}(P_T)| \leq \min\{|\mathcal{V}|, m_0 m_1\}$. For every preorder pair $C = (c_0, c_1)$, let P_T^C , L_C , the strict-prefix keys Π_C , U_C , and the local weights ω_C be the corresponding pushforwards. Then $U_C \leq L_C \leq \min\{L, d_0 d_1\}$, and

$$\Pr\{\text{both rank events under } C\} = \sum_{\pi \in \Pi_C} \omega_C(\pi) D_{M, k_0, k_1}(\pi).$$

A global memo is keyed exactly by $(M, k_0, k_1, \pi_{00}, \pi_{01}, \pi_{10})$; the fourth cell is determined by normalization, and local weights never enter the key. Thus the unchanged R_w , unchanged coherent P_T^C , common selector, common target, and within-replicate cross-arm dependence are preserved.

Let $\Pi_\star = \bigcup_C \Pi_C$, $G = |\Pi_\star|$, and partition it first into the support-one stratum Π_{vert} , then the support-two stratum Π_{edge} , then the eligible support-three face stratum Π_{face} , then the remaining support-four determinant-zero stratum Π_{rank1} , and finally the residual strata $\Pi_{\text{roll}}, \Pi_{\text{MS}}, \Pi_+$ chosen by deterministic least-charge comparison. Put

$$h_a = M - k_a + 1, \quad \bar{k}_a = \min\{k_a, h_a\}, \quad C_{M, k_0, k_1} = \sum_{i=0}^{M-1} \min\{\bar{k}_0, i+1\} \min\{\bar{k}_1, i+1\}.$$

Vertices use the deterministic formula; edges use the exact binomial interval charge $E_M(\pi)$; eligible support-three face keys use $F_M(\pi)$; rank-one support-four keys satisfying $\pi_{00}\pi_{11} = \pi_{01}\pi_{10}$ use $R_{M,k_0,k_1} = 1 + 2\lceil \log_2 M \rceil + \bar{k}_0 + \bar{k}_1$; and only residual keys enter the rolling, zero-cell-safe multinomial, or positive-corner branches. With prespecified tie-breaking,

$$\Pi_\star = \Pi_{\text{vert}} \dot{\cup} \Pi_{\text{edge}} \dot{\cup} \Pi_{\text{face}} \dot{\cup} \Pi_{\text{rank1}} \dot{\cup} \Pi_{\text{roll}} \dot{\cup} \Pi_{\text{MS}} \dot{\cup} \Pi_+,$$

and

$$\begin{aligned} B = & |\Pi_{\text{vert}}| + \sum_{\pi \in \Pi_{\text{edge}}} E_M(\pi) + \sum_{\pi \in \Pi_{\text{face}}} F_M(\pi) + \sum_{\pi \in \Pi_{\text{rank1}}} R_{M,k_0,k_1} + M \mathbf{1}\{\Pi_{\text{MS}} \neq \emptyset\} + \sum_{\pi \in \Pi_{\text{roll}}} C_{M,k_0,k_1} \\ & + \sum_{\pi \in \Pi_{\text{MS}}} (M + T_{M,\bar{k}_0,\bar{k}_1}) + \sum_{\pi \in \Pi_+} C_+(\pi). \end{aligned}$$

If $\sigma(\pi)$ minimizes the positive-corner charge, define

$$W = \max\left(\{0\} \cup \{\bar{k}_0\bar{k}_1 : \Pi_{\text{roll}} \neq \emptyset\} \cup \{M : \Pi_{\text{MS}} \neq \emptyset\} \cup \{\min(t_0^{\sigma(\pi)}, t_1^{\sigma(\pi)}) : \pi \in \Pi_+\}\right).$$

Vertices, edges, face keys, and rank-one keys contribute only $O(1)$ workspace, and the factorial preprocessing occurs if and only if $\Pi_{\text{MS}} \neq \emptyset$. The face and rank-one strata contribute no nonconstant term to W . With $\mathfrak{C}$ the $F_{m_0}F_{m_1}$ preorder pairs, exact enumeration after R_w is tabulated costs

$$O\left(|\mathcal{V}| + F_{m_0}F_{m_1}(m_0m_1 + L) + \sum_{C \in \mathfrak{C}} [L_C \log(1 + L_C) + L_C + U_C] + \left(\sum_C U_C\right) \log(1 + G) + B\right),$$

and its uniform version replaces the middle preorder-specific terms by

$$F_{m_0}F_{m_1}[m_0m_1 + L + L_{\max} \log(1 + L_{\max}) + L_{\max} + U_{\max}\{1 + \log(1 + G)\}].$$

Working memory is $O(m_0m_1 + L + U_{\max} + G + W)$.

There is a strict admissible edge improvement. Let $M = 2n \geq 4$, $k_0 = n$, $k_1 = M$, and $\pi = (1/2, 1/2, 0, 0)$ in 00, 01, 10, 11 order. Then

$$D_{M,n,M}(\pi) = 1 - 2^{-M}.$$

The edge evaluator uses $\Theta(\log M)$ exact arithmetic and $O(1)$ workspace, whereas the least-charge former nonvertex dispatcher uses $\Theta(M)$ arithmetic. This key is realized by two audited pair blocks under S_2^2 , independent fair one-treated-one-control assignment in each block, a singleton selector library whose legal equivariant arm-1 rule selects either block equiprobably and then its unique observed arm-1 member, a target supported in block 2, an arm-0 preorder with block 2 lowest, and an arm-1 preorder with block 1 below block 2. Taking

$$\varepsilon_0 = \frac{n+1}{2n+1}, \quad \varepsilon_1 = \frac{1}{2n+1}, \quad \alpha = \frac{n+2}{2n+1}$$

gives the stated ranks and satisfies the error-budget assumptions.

There is also a strict admissible zero-corner face improvement for every $M \geq 2$. Let $k_0 = k_1 = M$ and $\pi = (1/3, 1/3, 1/3, 0)$ in 00, 01, 10, 11 order. Then

$$D_{M,M,M}(\pi) = 1 - 2 \cdot 3^{-M}.$$

The LL face orientation has $\pi_{11} = 0$ and $t_0 + t_1 = 2M > M$, so the face evaluator costs $\Theta(\log M)$ exact arithmetic and $O(1)$ workspace. The rolling, multinomial, and every allowed positive-base coefficient orientation each cost $\Theta(M)$. The key is realized by two pair blocks with the trivial finite group, a known law uniform on three assignment vectors having one treated member per block, deterministic legal observable selector lookups whose strict-prefix pairs are 00, 01, and 10, a singleton common selector library, and a fixed common target in block 2. In each arm, place both block-1 singleton orbits strictly below the target orbit and tie the block-2 singleton orbits with the target. Taking

$$\varepsilon_0 = \varepsilon_1 = \frac{1}{M+1}, \quad \alpha = \frac{2}{M+1}$$

gives $k_0 = k_1 = M$ and satisfies the error-budget assumptions.

There is a strict admissible rank-one support-four improvement. Let $M = 2n$ with $n \geq 3$, $k_0 = n$, $k_1 = n + 2$, and $\pi = (1/4, 1/4, 1/4, 1/4)$. Then

$$D_{2n,n,n+2}(\pi) = H_{2n,n}(1/2)H_{2n,n+2}(1/2).$$

The rank-one prepass uses $\Theta(M)$ exact arithmetic and $O(1)$ workspace, while the least-charge former branch uses $\Theta(M^2)$ arithmetic and $\Theta(M)$ workspace. The key is realized by two pair blocks with the trivial group, independent fair one-treated-one-control assignments, a singleton selector choosing block 1's unique control in arm 0 and block 2's unique treated member in arm 1, the common target $v_{2,0}$, and preorders making one block-1 control singleton orbit and the block-2 mate-treatment singleton orbit strictly below their target orbits. The admissible levels are

$$\varepsilon_0 = \frac{n+1}{2n+1}, \quad \varepsilon_1 = \frac{n-1}{2n+1}, \quad \alpha = \frac{2n}{2n+1}.$$

Proof. Write $S_\nu = \{z : \nu(z) > 0\}$. By **ass:known-matched-randomization**, $\nu(gz) = \nu(z)$, so S_ν is $\mathcal{G}$ -invariant. For $z \in S_\nu$, define the orbit-aggregated kernel

$$\kappa_{w,a}(O \mid z) = \sum_{v:(v,a) \in O} \kappa_{w,a}(v \mid z). \quad (1)$$

By **ass:observable-selector-kernel**, (1) is a probability mass function on the arm-orbit set $\mathcal{O}_a$. Conditional on $Z_i = z$, **ass:selector-auxiliary-randomization** makes the two selector draws independent with these two kernels. Hence their joint orbit law is

$$R_w(O_0, O_1) = \sum_{z \in S_\nu} \nu(z) \kappa_{w,0}(O_0 \mid z) \kappa_{w,1}(O_1 \mid z). \quad (2)$$

This equals the unrestricted sum in the statement because every omitted assignment has zero ν -mass. Thus arbitrary total kernel extensions outside S_ν have zero integrated weight. Independence across replicates makes the calibration orbit pairs iid with law (2). By **ass:prespecified-target**, the independent common target has coherent joint law

$$P_T(O_0, O_1) = \sum_{v \in \mathcal{V}} q_{\text{tar}}(v) \mathbf{1}\{O_0(v) = O_0, O_1(v) = O_1\}. \quad (3)$$

Its support is an image of $\mathcal{V}$, and therefore $L \leq \min\{|\mathcal{V}|, m_0 m_1\}$.

Fix a preorder pair $C = (c_0, c_1)$, having d_a nonempty ordered classes in arm a , and push (3) through the two class maps. Pushforward cannot enlarge support and its range has size $d_0 d_1$, whence

$$L_C \leq \min\{L, d_0 d_1\}. \quad (4)$$

Condition on a target class pair (j, ℓ) in that support and define

$$B_{i0} = \mathbf{1}\{c_0(O_{i,0}) < j\}, \quad B_{i1} = \mathbf{1}\{c_1(O_{i,1}) < \ell\}. \quad (5)$$

For either arm, the elementary order-statistic identity

$$s_a(O_{T,a}) \leq Q_{k_a}(s_a(O_{1,a}), \dots, s_a(O_{M,a})) \iff \sum_{i=1}^M \mathbf{1}\{s_a(O_{i,a}) < s_a(O_{T,a})\} < k_a \quad (6)$$

uses a strict prefix and remains valid under ties. A two-dimensional prefix table for (2) supplies the four-cell law $\pi^C(j, \ell)$ of (5) in constant time. Across replicates these pairs are iid, although their two coordinates need not be independent. Grouping target class pairs that yield the same four-cell vector gives

$$\Pr\{\text{both rank events under } C\} = \sum_{\pi \in \Pi_C} \omega_C(\pi) D_{M,k_0,k_1}(\pi), \quad U_C \leq L_C. \quad (7)$$

The value $D_{M,k_0,k_1}(\pi)$ depends only on the ranks and three independent cells because $\pi_{11} = 1 - \pi_{00} - \pi_{01} - \pi_{10}$. Thus the exact global key is $(M, k_0, k_1, \pi_{00}, \pi_{01}, \pi_{10})$, while the preorder-local weights remain outside that key.

Equations (2), (3), and (7) preserve the common selector, the coherent common target, and within-replicate cross-arm dependence.

Apply the declared exact evaluators key by key. By `lem:four-cell-rank-event-dynamic-program`, support-one and support-two keys have the stated deterministic and affine-binomial formulas. By `lem:corner-oriented-four-cell-rank-evaluator`, every eligible support-three face key has charge $F_M(\pi)$ and constant workspace, and each remaining support-four determinant-zero key factors into two strict binomial distribution functions with charge R_{M,k_0,k_1} and constant workspace. By `lem:four-cell-multinomial-sum-alternative` and `lem:positive-corner-coefficient-recurrence`, every residual key is evaluated exactly by the rolling, zero-cell-safe multinomial, or positive-corner branch. Prespecified least-charge selection and tie-breaking therefore give the displayed seven disjoint strata and exactly the sum B . Only Π_{MS} builds the factorial table. Sequential processing uses the largest live branch workspace, namely W ; the first four strata use only constant auxiliary space.

Constructing (3) costs $O(|V|)$. For each preorder pair, class aggregation, the prefix table for (2), and target pushforward cost $O(m_0 m_1 + L)$. A balanced tree groups its L_C local keys in $O(L_C \log(1 + L_C))$ comparisons and $O(L_C)$ arithmetic operations, and (7) costs $O(U_C)$. The global memo tree adds

$$O\left(\left(\sum_C U_C\right) \log(1 + G)\right). \quad (8)$$

Adding B proves the adaptive time bound. Replacing L_C, U_C by $L_{\max}, U_{\max}$ proves the uniform bound. The joint table, target table, largest local grouping table, G memoized answers, and selected live branch give memory $O(m_0 m_1 + L + U_{\max} + G + W)$.

For the edge audit, use two pair blocks, the group S_2^2 , and independent fair one-treated-one-control assignments. On every positive-mass assignment and in each arm, choose either block with probability $1/2$, then select its unique member observed in the requested arm. This lookup is normalized and has requested realized-arm support; extend it to a total probability kernel outside S_ν , where the extension has zero weight in (2). Put the target in block 2, place block 2 lowest in arm 0, and place block 1 below block 2 in arm 1. Then $B_{i0} = 0$ and B_{i1} is fair, so $\pi = (1/2, 1/2, 0, 0)$ and

$$D_{M,n,M}(\pi) = \Pr\{\text{Bin}(M, 1/2) < M\} = 1 - 2^{-M}. \quad (9)$$

The edge charge is $\Theta(\log M)$ with constant workspace, whereas each former nonvertex charge is $\Theta(M)$. Direct substitution of the displayed positive levels gives ranks n and M , and their sum is α .

For the face audit, take the trivial group on two pair blocks and let ν be uniform on

$$z^A = (1, 0, 1, 0), \quad z^B = (1, 0, 0, 1), \quad z^C = (0, 1, 1, 0), \quad (10)$$

in member order $(v_{1,0}, v_{1,1}, v_{2,0}, v_{2,1})$. Define both selector kernels on every positive-mass assignment as follows: on z^A , select the requested arm in block 2; on z^B , select block 2 in arm 0 and block 1 in arm 1; on z^C , select block 1 in arm 0 and block 2 in arm 1. These are normalized point-mass lookups with requested realized-arm support, and arbitrary total off-support extensions do not affect (2). With target $v_{2,0}$, put both block-one singleton orbits below the target in each arm and tie the block-two singleton orbits with it. The three strict-prefix pairs are 00, 01, 10, so $\pi = (1/3, 1/3, 1/3, 0)$. At $k_0 = k_1 = M$, failure means that all draws are 10 or all are 01; these events are disjoint. Hence

$$D_{M,M,M}(\pi) = 1 - 2 \cdot 3^{-M}. \quad (11)$$

The face charge is $\Theta(\log M)$ with constant workspace. The rolling and multinomial charges are $\Theta(M)$; direct inspection of positive-base orientations gives rectangles of dimensions M by M , M by 1, or 1 by M , so their least charge is also $\Theta(M)$. The displayed levels give $k_0 = k_1 = M$ and sum to α .

For the rank-one audit, retain the trivial group and use all four independent fair one-treated-one-control assignments. On each positive-mass assignment, arm 0 selects block 1's unique control and arm 1 selects block 2's unique treated member. These are normalized requested-arm-supported lookups; off-support extensions again have zero integrated weight. With target $v_{2,0}$, put one block-one control singleton orbit below the arm-zero target orbit and the block-two mate-treatment singleton orbit below the arm-one target orbit. The prefix indicators are independent fair Bernoulli variables, so $\pi = (1/4, 1/4, 1/4, 1/4)$. Its determinant vanishes, and therefore

$$D_{2n,n,n+2}(\pi) = H_{2n,n}(1/2)H_{2n,n+2}(1/2). \quad (12)$$

The displayed levels give ranks $n, n+2$, are positive for $n \geq 3$, and sum to $2n/(2n+1)$. Here $\bar{k}_0 = n$ and $\bar{k}_1 = n-1$, so the rank-one charge is $\Theta(M)$ with constant workspace. Every former branch has two widths of order M , hence least time $\Theta(M^2)$ and workspace $\Theta(M)$. This proves all claims. $\square$

Justification. Exact strict-prefix vectors permit global reuse of four-cell count probabilities while retaining preorder-local coherent target weights and the full within-replicate joint calibration law. *Gap.* Memoizing target weights, factorizing the two-arm calibration law, or minimizing time and memory independently would change the probability object or invalidate the resource certificate. *Consumer.* Memoize only $(M, k_0, k_1, \pi_{00}, \pi_{01}, \pi_{10})$, retain each preorder's ω_C , and assign each exact key to its certified least-charge evaluator.

Theorem 4 (thm:minimum-imbalance-selector-certificate). *Assume that $\mathcal{H}$ is finite and nonempty. The minimum D_* in `def:minimum-imbalance-common-selector` is attained, so $\mathcal{W}_*$ is nonempty, and the displayed absolute-value linear program has value D_* . Its quantitative dual is attained and satisfies the strong-duality identity*

$$D_* = \max_{\|y\|_\infty \leq 1/2} \left\{ \min_{h \in \mathcal{H}} y^\top A_{\cdot h} - y^\top b_{\text{LP}} \right\}.$$

If $D_* > 0$, the negative of any maximizing dual vector is a Farkas separating direction with margin D_* . Every $w_* \in \mathcal{W}_*$ is computable solely from the known randomization law, target rule, and selector library before scores are observed, and hence is prespecified. Moreover, $D_* = 0$ if and only if the exact joint-balance linear program in `def:common-selector-feasibility-lp` is feasible. For every prespecified $\varepsilon = (\varepsilon_0, \varepsilon_1)$ with $\varepsilon_a > 0$ and $\varepsilon_0 + \varepsilon_1 \leq \alpha$, every $P \in \mathcal{P}_{\mathcal{G}, K}$, and arbitrary fixed measurable trained predictor maps, conditional on $\mathcal{T}$,

$$\Pr_P(\tau_T \in \mathcal{C}_{w_*}^\varepsilon \mid \mathcal{T}) \geq \max\{0, 1 - \rho_{M, \varepsilon_0} - \rho_{M, \varepsilon_1} - D_*\}.$$

This numerical certificate weakly dominates the corresponding certificate for every fixed common $w \in \Delta(\mathcal{H})$, and the dominance is strict whenever $D_* < \Delta_0(w) + \Delta_1(w)$ and $1 - \rho_{M, \varepsilon_0} - \rho_{M, \varepsilon_1} - D_* > 0$. On the equal-split diagonal $\varepsilon_0 = \varepsilon_1 = \alpha/2$,

$$\Pr_P(\tau_T \in \mathcal{C}_{w_*} \mid \mathcal{T}) \geq \max\{0, 1 - 2\rho_{M, \alpha} - D_*\}.$$

Proof. The full-row part of `thm:active-arm-minimum-imbalance-selector-certificate` proves attainment of D_* and $\mathcal{W}_*$, equality with the absolute-value LP, the displayed attained dual, the separating-margin conclusion, and the equivalence $D_* = 0$ with exact joint balance. It also shows that a deterministically chosen w_* is computable from A and b_{LP} before scores are observed.

For every w , equation (1) in the proof of `thm:common-selector-lp` gives

$$\frac{1}{2} \sum_{a=0}^1 \sum_{O \in \mathcal{O}} |(Aw - b_{\text{LP}})_{a,O}| = \Delta_0(w) + \Delta_1(w). \quad (1)$$

Thus every $w_* \in \mathcal{W}_*$ satisfies $D_* = \Delta_0(w_*) + \Delta_1(w_*)$. Substitute this equality into `prop:asymmetric-finite-rank-orbit-gap-bound` to obtain

$$\Pr_P(\tau_T \in \mathcal{C}_{w_*}^\varepsilon \mid \mathcal{T}) \geq \max\{0, 1 - \rho_{M, \varepsilon_0} - \rho_{M, \varepsilon_1} - D_*\}. \quad (2)$$

Optimality and (1) imply $D_* \leq \Delta_0(w) + \Delta_1(w)$ for every fixed common w . Monotonicity of truncation at zero proves weak dominance. If the inequality is strict and the untruncated right side of (2) is positive, the optimized truncated bound is strictly larger, proving the stated strictness condition. Finally, on the equal-split diagonal, $\rho_{M, \varepsilon_0} = \rho_{M, \varepsilon_1} = \rho_{M, \alpha}$ and $\mathcal{C}_{w_*}^\varepsilon = \mathcal{C}_{w_*}$, which gives the last display. $\square$

Justification. The theorem converts the active-arm coverage formula into an attained prespecified design optimization with an exact dual audit while retaining the full D_* diagnostic. *Gap.* A yes-or-no feasibility LP and a full two-arm minimum do not identify the best coverage certificate when one or both arm thresholds are infinite. *Consumer.* Solve the active-row absolute-value LP for $w_{*, \mathcal{A}}$; use its primal value in the coverage lower bound and its dual maximizer as the quantitative certificate.

Theorem 5 (thm:active-arm-minimum-imbalance-selector-certificate). Assume that $\mathcal{H}$ is finite and nonempty. The unchanged joint two-arm objects D_* and $\mathcal{W}_*$ are attained, the full absolute-value linear program has value D_* , and

$$D_* = \max_{\|y\|_\infty \leq 1/2} \left\{ \min_{h \in \mathcal{H}} y^\top A_{\cdot h} - y^\top b_{\text{LP}} \right\}.$$

If $D_* > 0$, the negative of every maximizing vector is a Farkas separator with margin D_* , and $D_* = 0$ if and only if the exact joint-balance program is feasible. For a prespecified ϵ , let $\mathcal{A}_\epsilon = \{a : k_a \leq M\}$. The active-row minimum $D_{*,\mathcal{A}}$ and optimizer set $\mathcal{W}_{*,\mathcal{A}}$ in `def:minimum-imbalance-common-selector` are attained, and the active-row absolute-value program has value $D_{*,\mathcal{A}}$. Its attained dual is

$$D_{*,\mathcal{A}} = \max_{\substack{\|y\|_\infty \leq 1/2 \\ y_{a,O} = 0 \text{ for } a \notin \mathcal{A}_\epsilon}} \left\{ \min_{h \in \mathcal{H}} y^\top A_{\cdot h} - y^\top b_{\text{LP}} \right\}.$$

Moreover, $D_{*,\mathcal{A}} = 0$ if and only if one selector balances every active-arm row; this characterizes joint two-arm balance only when $\mathcal{A}_\epsilon = \{0, 1\}$. Every prespecified $w_{*,\mathcal{A}} \in \mathcal{W}_{*,\mathcal{A}}$ is computable before scores are observed and satisfies

$$\Pr_P \left\{ \tau_T \in \mathcal{C}_{w_{*,\mathcal{A}}}^\epsilon \mid \mathcal{T} \right\} \geq \max \left\{ 0, 1 - \sum_{a \in \mathcal{A}_\epsilon} \rho_{M,\epsilon_a} - D_{*,\mathcal{A}} \right\}.$$

This active-row certificate weakly dominates the corresponding certificate for every fixed common w , with strict dominance whenever its active imbalance is strictly smaller and the optimized untruncated lower bound is positive. If exactly one arm is at the boundary, the optimization contains only the other arm's rows. If $\mathcal{A}_\epsilon = \emptyset$, then $D_{*,\mathcal{A}} = 0$, $\mathcal{W}_{*,\mathcal{A}} = \Delta(\mathcal{H})$, no optimization is needed, and coverage is one. On the equal-split finite-rank diagonal, $D_{*,\mathcal{A}} = D_*$ and $\mathcal{W}_{*,\mathcal{A}} = \mathcal{W}_*$; at the two-boundary elbow, coverage is one. The original w_* -certificate and its strict-dominance claim remain valid as consequences of the full two-arm optimum. No selector-indexed object exists when $\mathcal{H} = \emptyset$.

Proof. Assume $\mathcal{H}$ is finite and nonempty. For any row set R , write A_R and b_R for the corresponding restrictions and define

$$d_R = \min_{w \in \Delta(\mathcal{H})} \frac{1}{2} \|A_R w - b_R\|_1. \quad (1)$$

The simplex is compact and nonempty and the objective is continuous, so the minimum and its optimizer set are attained. For fixed w , minimizing the auxiliary variables in the displayed absolute-value LP sets $t_r = |(A_R w - b_R)_r|$. Thus that LP has value d_R .

We next prove the dual identity without importing an optimization hypothesis. Put $S_R = \{A_R w - b_R : w \in \Delta(\mathcal{H})\}$. For every y with $\|y\|_\infty \leq 1/2$, Hölder's inequality gives

$$\min_{s \in S_R} y^\top s \leq \min_{s \in S_R} \frac{1}{2} \|s\|_1 = d_R. \quad (2)$$

If $d_R = 0$, equality is achieved by $y = 0$. Suppose $d_R > 0$. For $0 < r < d_R$, let $B_r = \{u : \|u\|_1/2 \leq r\}$. The compact convex sets S_R and B_r are disjoint. Choose a Euclidean closest pair $s_r \in S_R$, $u_r \in B_r$, put $z_r = s_r - u_r$, and minimize squared distance along line segments. This gives

$$z_r^\top (s - s_r) \geq 0 \quad (s \in S_R), \quad z_r^\top (u - u_r) \leq 0 \quad (u \in B_r). \quad (3)$$

Normalize $y_r = z_r / (2\|z_r\|_\infty)$. The support function of B_r at y_r is r , so (3) yields $\min_{s \in S_R} y_r^\top s \geq r$. The box $\{y : \|y\|_\infty \leq 1/2\}$ is compact. Letting $r \uparrow d_R$ along a convergent subsequence and using continuity gives a y_* with

$\min_{s \in S_R} y_*^\top s \geq d_R$. Together with (2),

$$d_R = \max_{\|y\|_\infty \leq 1/2} \min_{w \in \Delta(\mathcal{H})} y^\top (A_R w - b_R) = \max_{\|y\|_\infty \leq 1/2} \left\{ \min_{h \in \mathcal{H}} y^\top A_{R,h} - y^\top b_R \right\}. \quad (4)$$

Compactness also proves dual attainment.

Take first all rows. Then (1) is D_* , and (4) is the displayed full dual. If $D_* > 0$ and y_* maximizes it, then for every column h ,

$$(-y_*)^\top b_{LP} - (-y_*)^\top A_{\cdot,h} \geq D_*,$$

so $-y_*$ is a separating direction with margin D_* . Moreover, $D_* = 0$ exactly when some w satisfies $Aw = b_{LP}$, which is the exact joint-balance program by `thm:common-selector-lp`.

Now let R contain precisely the rows (a, O) with $a \in \mathcal{A}_\epsilon$. Then (1) and (4) are exactly $D_{*,\mathcal{A}}$ and its displayed dual after padding y by zeros on inactive rows. The equality $D_{*,\mathcal{A}} = 0$ is equivalent to the existence of one w satisfying every active row. It is equivalent to full joint balance only when both arms are active.

By (1), every optimizer satisfies

$$D_{*,\mathcal{A}} = \frac{1}{2} \sum_{a \in \mathcal{A}_\epsilon} \sum_{O \in \mathcal{O}} |(Aw_{*,\mathcal{A}} - b_{LP})_{a,O}| = \sum_{a \in \mathcal{A}_\epsilon} \Delta_a(w_{*,\mathcal{A}}). \quad (5)$$

The matrices and target vector depend only on the known design, target rule, and selector library, so deterministic tie-breaking makes $w_{*,\mathcal{A}}$ prespecified. Substituting (5) in `prop:active-arm-finite-rank-orbit-gap-bound` proves

$$\Pr_P\{\tau_T \in \mathcal{C}_{w_{*,\mathcal{A}}}^\epsilon \mid \mathcal{T}\} \geq \max \left\{ 0, 1 - \sum_{a \in \mathcal{A}_\epsilon} \rho_{M,\epsilon_a} - D_{*,\mathcal{A}} \right\}. \quad (6)$$

For any fixed common w , optimality gives $D_{*,\mathcal{A}} \leq \sum_{a \in \mathcal{A}_\epsilon} \Delta_a(w)$. Since truncation at zero is monotone, (6) weakly dominates its certificate; if the inequality is strict and the optimized untruncated bound is positive, the dominance is strict.

If one arm is at the boundary, only the other arm's rows occur in (1). If both are at the boundary, the active set is empty, the objective is zero for every w , and the definitions give $D_{*,\mathcal{A}} = 0$ and $\mathcal{W}_{*,\mathcal{A}} = \Delta(\mathcal{H})$; coverage is one by `prop:asymmetric-rank-boundary`. On the equal-split finite-rank diagonal both arms are active, so the active and full objectives and optimizer sets coincide. A boundary rank has $\rho_{M,\epsilon_a} = 0$; hence applying the active-arm bound to any full optimizer also yields the original certificate with D_* , and its dominance follows from full-objective optimality. Finally, `def:minimum-imbalance-common-selector` declares every selector-indexed object undefined when $\mathcal{H} = \emptyset$. $\square$

Justification. The active-row LP optimizes the exact coverage-facing orbit-gap certificate and provides an attained dual audit. *Gap.* A full two-arm imbalance objective can waste design effort on a boundary arm that cannot cause miscoverage. *Consumer.* Optimize only rows with $k_a \leq M$, then report the truncated lower bound and its dual certificate.

Theorem 6 (`thm:asymmetric-sharp-orbit-coverage-frontier`). *Within the deterministic bounded orbit-constant-score submodel optimized by `def:coverage-frontier`, the functional $\Gamma_{M,\alpha,\epsilon}(w)$ is the exact finite worst deficit of the joint rank-inclusion event for ranks k_0, k_1 . This exactness is solely over deterministic maps $s_a : \mathcal{O} \rightarrow [0, K]$. For an arbitrary $P \in \mathcal{P}_{G,K}$, a residual score can remain stochastic conditional on its orbit, so $\Gamma_{M,\alpha,\epsilon}(w)$ is not a uniform joint-inclusion certificate and is not a causal effect-set coverage certificate for the full model; no equality with a supremum over $P \in \mathcal{P}_{G,K}$ is asserted. For the ordinary asymmetric rank construction, simultaneous armwise inclusion implies effect-set inclusion pointwise:*

$$\{Y_{0T}(1) \in I_{1,w}^\epsilon, Y_{0T}(0) \in I_{0,w}^\epsilon\} \subseteq \{\tau_T \in \mathcal{C}_w^\epsilon\}, \quad \Pr_P\{\tau_T \in \mathcal{C}_w^\epsilon \mid \mathcal{T}\} \geq \Pr_P\{Y_{0T}(1) \in I_{1,w}^\epsilon, Y_{0T}(0) \in I_{0,w}^\epsilon \mid \mathcal{T}\}.$$

Equality need not hold: with $I_{0,w}^\epsilon = I_{1,w}^\epsilon = \{0\}$ and $Y_{0T}(0) = Y_{0T}(1) = K$, one has $\tau_T = 0 \in \mathcal{C}_w^\epsilon = \{0\}$ although both armwise inclusions fail. This pointwise containment is a separate algebraic fact and does not make $\Gamma_{M,\alpha,\epsilon}(w)$ certify its premise for arbitrary invariant laws. Within the deterministic submodel, if both ranks are finite, the maximum is attained by enumerating the $F_{m_0}F_{m_1}$ total-preorder pairs, every preorder has a representative in $[0, K]$ with at most m_a levels, and the coherent evaluator in `lem:coherent-two-finite-rank-frontier-evaluator` applies.

In that evaluator, retain the unchanged R_w , coherent P_T^C , local ω_C , exact-key sets Π_C , union $\Pi_\star$, and memo key $(M, k_0, k_1, \pi_{00}, \pi_{01}, \pi_{10})$. Partition

$$\Pi_\star = \Pi_{\text{vert}} \dot{\cup} \Pi_{\text{edge}} \dot{\cup} \Pi_{\text{face}} \dot{\cup} \Pi_{\text{rank1}} \dot{\cup} \Pi_{\text{roll}} \dot{\cup} \Pi_{\text{MS}} \dot{\cup} \Pi_+,$$

where the first two strata have support sizes one and two, Π_{face} contains eligible support-three zero-corner keys, Π_{rank1} contains the remaining support-four keys satisfying $\pi_{00}\pi_{11} = \pi_{01}\pi_{10}$, and the last three contain only residual keys. Vertices use their deterministic counts; edges use the exact affine-binomial interval evaluator; face keys use the incompatibility evaluator in `def:simplex-vertex-key-prepass`; rank-one keys use the product of two exact strict binomial CDFs with charge R_{M,k_0,k_1} ; the remaining keys use the prespecified least-charge rolling, multinomial-sum, or positive-corner branch. Put

$$\begin{aligned} B = & |\Pi_{\text{vert}}| + \sum_{\pi \in \Pi_{\text{edge}}} E_M(\pi) + \sum_{\pi \in \Pi_{\text{face}}} F_M(\pi) + \sum_{\pi \in \Pi_{\text{rank1}}} R_{M,k_0,k_1} + M \mathbf{1}\{\Pi_{\text{MS}} \neq \emptyset\} + \sum_{\pi \in \Pi_{\text{roll}}} C_{M,k_0,k_1} \\ & + \sum_{\pi \in \Pi_{\text{MS}}} (M + T_{M,\bar{k}_0,\bar{k}_1}) + \sum_{\pi \in \Pi_+} C_+(\pi), \end{aligned}$$

and let W be the maximum of 0, $\bar{k}_0\bar{k}_1$ when the rolling branch is nonempty, M when the multinomial branch is nonempty, and the selected positive-corner rolling widths. There is no additional edge, face, or rank-one term in W . The exact enumeration and uniform time bounds are

$$O\left(|\mathcal{V}| + F_{m_0}F_{m_1}(m_0m_1 + L) + \sum_C [L_C \log(1 + L_C) + L_C + U_C] + \left(\sum_C U_C\right) \log(1 + G) + B\right)$$

and

$$O(|\mathcal{V}| + F_{m_0}F_{m_1}[m_0m_1 + L + L_{\max} \log(1 + L_{\max}) + L_{\max} + U_{\max}\{1 + \log(1 + G)\}] + B),$$

with memory $O(m_0m_1 + L + U_{\max} + G + W)$. The factorial table is built only when a remaining nonface support-at-least-three key is assigned to Π_{MS} ; the face and rank-one strata neither trigger that table nor add nonconstant workspace.

For $M = 2n \geq 4$, $k_0 = n$, $k_1 = M$, and $\pi = (1/2, 1/2, 0, 0)$,

$$D_{M,n,M}(\pi) = 1 - 2^{-M};$$

the edge prepass costs $\Theta(\log M)$ exact arithmetic and $O(1)$ workspace, while the former least-charge nonvertex dispatcher costs $\Theta(M)$. The audited two-pair realization, selector, target, preorders, and levels

$$\varepsilon_0 = \frac{n+1}{2n+1}, \quad \varepsilon_1 = \frac{1}{2n+1}, \quad \alpha = \frac{n+2}{2n+1}$$

are those stated in `lem:coherent-two-finite-rank-frontier-evaluator`.

Within the deterministic submodel, if exactly one rank $k_b = M + 1$, the boundary-arm event is certain and, for $c = 1 - b$, the exact rank-event probability is

$$\sum_{O_t \in \mathcal{O}_c} p_{c,O_t} \Pr \left\{ \text{Bin} \left(M, \sum_{O: s_c(O) < s_c(O_t)} q_{c,O} \right) < k_c \right\}.$$

Enumeration uses only the F_{m_c} nonboundary preorders and costs

$$O(F_{m_c} m_c [1 + \log M + \min\{k_c, M - k_c + 1\}])$$

arithmetic operations and $O(1)$ auxiliary CDF memory after the two marginal vectors are tabulated. If both ranks equal $M + 1$, the deterministic-submodel deficit is zero. Within this submodel, binary attainment remains

false: at $M = 3$, $\alpha = 1/2$, and $\varepsilon_0 = \varepsilon_1 = 1/4$, a four-level objective is 24957/160000, every binary pair is at most 759109/5000000, and the gap is 83189/20000000. If target and calibration orbit supports are disjoint in a finite-rank arm, the deterministic-submodel deficit is $1 - \alpha$. For the full model, the valid guarantees are instead the exact-balance Bonferroni result in **thm:common-selector-lp** and the active-arm total-variation bound in **prop:active-arm-finite-rank-orbit-gap-bound**; neither guarantee is derived from $\Gamma_{M,\alpha,\varepsilon}(w)$. On the equal-split diagonal this theorem yields **thm:sharp-orbit-coverage-frontier**. The computational prepasses alter no active-arm, pair-audit, empty-library, ordinary-rank, or weighted-orbit semantics.

For every $M \geq 2$, the admissible face key $k_0 = k_1 = M$ and $\pi = (1/3, 1/3, 1/3, 0)$ satisfies

$$D_{M,M,M}(\pi) = 1 - 2 \cdot 3^{-M}.$$

The zero-corner face prepass costs $\Theta(\log M)$ exact arithmetic and $O(1)$ workspace, whereas rolling, multinomial summation, and every allowed positive-base coefficient orientation cost $\Theta(M)$. The trivial-group two-pair assignment support, deterministic legal selector lookup, fixed block-2 target, singleton-orbit preorders, and levels

$$\varepsilon_0 = \varepsilon_1 = \frac{1}{M+1}, \quad \alpha = \frac{2}{M+1}$$

are audited in **lem:coherent-two-finite-rank-frontier-evaluator**.

For $M = 2n$ with $n \geq 3$, $k_0 = n$, $k_1 = n + 2$, and $\pi = (1/4, 1/4, 1/4, 1/4)$, the exact determinant test selects the rank-one prepass and

$$D_{2n,n,n+2}(\pi) = H_{2n,n}(1/2)H_{2n,n+2}(1/2).$$

The new branch costs $\Theta(M)$ exact arithmetic and $O(1)$ workspace, while the least-charge former branch costs $\Theta(M^2)$ arithmetic and $\Theta(M)$ workspace. The legal two-block construction and levels

$$\varepsilon_0 = \frac{n+1}{2n+1}, \quad \varepsilon_1 = \frac{n-1}{2n+1}, \quad \alpha = \frac{2n}{2n+1}$$

are audited in the coherent evaluator.

Proof. Put

$$J(s_0, s_1) = \Pr\{s_1(O_{T,1}) \leq Q_{k_1}(s_1(O_{1,1}), \dots, s_1(O_{M,1})) \text{ and } s_0(O_{T,0}) \leq Q_{k_0}(s_0(O_{1,0}), \dots, s_0(O_{M,0}))\}. \quad (1)$$

By **def:coverage-frontier**,

$$\Gamma_{M,\alpha,\varepsilon}(w) = \max_{s_0, s_1} [(1 - \alpha) - J(s_0, s_1)]_+. \quad (2)$$

Thus (2) is exactly the finite worst deficit over the declared deterministic maps $s_a : \mathcal{O} \rightarrow [0, K]$. This is not a full-model assertion. For example, take fixed covariates, zero predictors, iid nondegenerate Bernoulli variables ξ_i , and set $Y_{iv}(a) = K\xi_i$ for every member and arm. The resulting latent law is bounded and invariant, but its residual score is nondegenerate conditional on every orbit. Hence the deterministic domain in (2) is genuinely narrower than $\mathcal{P}_{G,K}$, as asserted.

Let E_{joint} be simultaneous armwise inclusion and E_{eff} effect-set inclusion. On E_{joint} ,

$$\tau_T = Y_{0T}(1) - Y_{0T}(0) \in I_{1,w}^\varepsilon - I_{0,w}^\varepsilon = \mathcal{C}_w^\varepsilon. \quad (3)$$

Therefore $E_{\text{joint}} \subseteq E_{\text{eff}}$, and conditional probabilities give the displayed inequality. If both intervals are $\{0\}$ and both target outcomes equal K , the simultaneous armwise event fails while $\tau_T = 0 \in \mathcal{C}_w^\varepsilon$, proving that equality need not hold.

Assume first that $k_0, k_1 \leq M$. For either arm,

$$s_a(O_{T,a}) \leq Q_{k_a}(s_a(O_{1,a}), \dots, s_a(O_{M,a})) \iff \sum_{i=1}^M \mathbf{1}\{s_a(O_{i,a}) < s_a(O_{T,a})\} < k_a. \quad (4)$$

Thus only the total preorder induced by each score map matters. Conversely, every total preorder with ordered nonempty classes $C_{a,1} \prec \dots \prec C_{a,d_a}$ has a representative in $[0, K]$: use zero if $d_a = 1$, and otherwise assign $(j-1)K/(d_a-1)$ to class j . Hence the maximum is attained among the $F_{m_0}F_{m_1}$ preorder pairs.

For each pair C , **lem:coherent-two-finite-rank-frontier-evaluator** gives

$$J(C) = \sum_{\pi \in \Pi_C} \omega_C(\pi) D_{M,k_0,k_1}(\pi), \quad (5)$$

Finally, `thm:common-selector-lp` supplies the full-model guarantee under exact balance, and `prop:active-arm-finite-rank-orbit-gap-bound` supplies the full-model total-variation guarantee under imbalance; neither is derived from (2). Setting $\varepsilon_0 = \varepsilon_1 = \alpha/2$ gives the equal-split theorem. Because every computational branch evaluates the same probability in (5), none changes the active-arm, pair-audit, empty-library, ordinary-rank, or weighted-orbit semantics. Equations (9), (11), and (12) in the coherent evaluator prove the three displayed admissible strict improvements and their exact levels. $\square$

Justification. The theorem exactly evaluates the worst joint rank-inclusion deficit only within the deterministic bounded orbit-constant-score submodel, for arbitrary prespecified arm budgets. *Gap.* The functional $\Gamma_{M,\alpha,\varepsilon}(w)$ does not cover stochastic conditional residual scores and is neither a uniform full-model joint-inclusion certificate nor a causal effect-set coverage certificate; marginal or arm-independent evaluation would also lose common-target and within-replicate cross-arm coupling. *Consumer.* Use the asymmetric preorder and memoized four-cell evaluator only for the deterministic orbit-score frontier. For full-model effect coverage, use `thm:common-selector-lp` under exact balance or `prop:active-arm-finite-rank-orbit-gap-bound` under imbalance; the event containment remains a separate pointwise implication.

Theorem 7 (`thm:sharp-orbit-coverage-frontier`). *Within the equal-split deterministic bounded orbit-constant-score submodel optimized by `def:coverage-frontier`, the functional $\Gamma_{M,\alpha}(w)$ is the exact finite worst deficit of the joint rank-inclusion event. This exactness is solely over deterministic maps. For an arbitrary $P \in \mathcal{P}_{\mathcal{G},K}$, residual scores can remain stochastic conditional on their orbits, so $\Gamma_{M,\alpha}(w)$ is not a uniform joint-inclusion certificate and is not a causal effect-set coverage certificate for the full model; no equality with actual effect-set coverage or with a supremum over invariant latent laws is asserted. For the symmetric ordinary rank construction,*

$$\{Y_{0T}(1) \in I_{1,w}, Y_{0T}(0) \in I_{0,w}\} \subseteq \{\tau_T \in \mathcal{C}_w\}, \quad \Pr_P\{\tau_T \in \mathcal{C}_w \mid \mathcal{T}\} \geq \Pr_P\{Y_{0T}(1) \in I_{1,w}, Y_{0T}(0) \in I_{0,w} \mid \mathcal{T}\}.$$

Equality need not hold, as shown by $I_{0,w} = I_{1,w} = \{0\}$ and $Y_{0T}(0) = Y_{0T}(1) = K$. This containment is a separate algebraic fact and does not make $\Gamma_{M,\alpha}(w)$ certify its premise for arbitrary invariant laws. The functional is the equal-split diagonal of $\Gamma_{M,\alpha,\varepsilon}(w)$. If $k \leq M$, its maximum is attained by the $F_{m_0}F_{m_1}$ total-preorder pairs, with at most m_a score levels in arm a , while retaining the unchanged R_w , coherent P_T^C , local weights, and exact memo keys.

Partition

$$\Pi_\star = \Pi_{\text{vert}} \dot{\cup} \Pi_{\text{edge}} \dot{\cup} \Pi_{\text{face}} \dot{\cup} \Pi_{\text{rank1}} \dot{\cup} \Pi_{\text{roll}} \dot{\cup} \Pi_{\text{MS}} \dot{\cup} \Pi_+,$$

where Π_{face} contains eligible support-three zero-corner keys, Π_{rank1} contains remaining support-four determinant-zero keys, and the last three strata contain only residual keys. Put $\bar{k} = \min\{k, M - k + 1\}$ and

$$C_{M,k,k} = \sum_{i=0}^{M-1} \min\{\bar{k}, i+1\}^2.$$

Vertices use deterministic counts, edges use the exact affine-binomial interval evaluator, face keys use the incompatibility evaluator, rank-one keys use the product-binomial evaluator, and only the remaining keys enter the three-way dispatcher. Define

$$\begin{aligned} B_{\text{sym}} = & |\Pi_{\text{vert}}| + \sum_{\pi \in \Pi_{\text{edge}}} E_M(\pi) + \sum_{\pi \in \Pi_{\text{face}}} F_M(\pi) + \sum_{\pi \in \Pi_{\text{rank1}}} R_{M,k,k} + M\mathbf{1}\{\Pi_{\text{MS}} \neq \emptyset\} + \sum_{\pi \in \Pi_{\text{roll}}} C_{M,k,k} \\ & + \sum_{\pi \in \Pi_{\text{MS}}} (M + T_{M,\bar{k},\bar{k}}) + \sum_{\pi \in \Pi_+} C_+(\pi). \end{aligned}$$

Let W_{sym} be the maximum of 0, $\bar{k}^2$ when rolling is used, M when the multinomial branch is used, and the selected positive-corner rolling widths. Vertices, edges, face keys, and rank-one keys add no nonconstant workspace. Exact enumeration costs

$$O\left(|\mathcal{V}| + F_{m_0}F_{m_1}(m_0m_1 + L) + \sum_C [L_C \log(1 + L_C) + L_C + U_C] + \left(\sum_C U_C\right) \log(1 + G) + B_{\text{sym}}\right),$$

its uniform version replaces the preorder-specific terms by

$$F_{m_0}F_{m_1}[m_0m_1 + L + L_{\text{max}} \log(1 + L_{\text{max}}) + L_{\text{max}} + U_{\text{max}}\{1 + \log(1 + G)\}],$$

and memory is $O(m_0 m_1 + L + U_{\max} + G + W_{\text{sym}})$. Shared factorial preprocessing occurs if and only if a residual key enters Π_{MS} ; rank-one keys do not trigger it.

For every $M \geq 2$, a diagonal strict edge witness is obtained from the audited two-pair realization by taking $\alpha = 2/(M+1)$, hence $k = M$, and the same key $\pi = (1/2, 1/2, 0, 0)$: then $D_{M,M,M}(\pi) = 1 - 2^{-M}$, the edge branch costs $\Theta(\log M)$, and the former dispatcher costs $\Theta(M)$. The asymmetric audit witness $M = 2n$, $k_0 = n$, $k_1 = M$, together with its exact levels and legality checks, remains the strict off-diagonal certificate for the common evaluator. If $k = M+1$, evaluation is constant and $\Gamma_{M,\alpha}(w) = 0$. Within the deterministic submodel, binary attainment is false with the exact four-level and binary values 24957/160000 and 759109/5000000 at $M = 3$, $\alpha = 1/2$. Within the deterministic submodel, under disjoint target and calibration orbit support in one arm and $k \leq M$, $\Gamma_{M,\alpha}(w) = 1 - \alpha$. For the full model, the valid guarantees are instead the exact-balance Bonferroni result in **thm:common-selector-lp** and the active-arm total-variation bound in **prop:active-arm-finite-rank-orbit-gap-bound**; neither guarantee is derived from $\Gamma_{M,\alpha}(w)$. The prepasses change no common-target, active-arm, pair-audit, empty-library, ordinary-rank, or weighted-orbit semantics.

For every $M \geq 2$, the diagonal face key $\pi = (1/3, 1/3, 1/3, 0)$ at $\alpha = 2/(M+1)$, hence $k = M$, satisfies

$$D_{M,M,M}(\pi) = 1 - 2 \cdot 3^{-M}.$$

It has the trivial-group two-pair realization audited in **lem:coherent-two-finite-rank-frontier-evaluator**. The face prepass costs $\Theta(\log M)$ exact arithmetic and $O(1)$ workspace, whereas each former branch costs $\Theta(M)$.

The admissible off-diagonal audit $M = 2n$, $n \geq 3$, $k_0 = n$, $k_1 = n+2$, and $\pi = (1/4, 1/4, 1/4, 1/4)$ remains a strict certificate for the common evaluator: the rank-one branch costs $\Theta(M)$ time and $O(1)$ workspace, whereas the former least-charge branch costs $\Theta(M^2)$ time and $\Theta(M)$ workspace. The diagonal partition uses the same exact determinant test and the charge $R_{M,k,k} = 1 + 2\lceil \log_2 M \rceil + 2\bar{k}$.

Proof. Set $\varepsilon_0 = \varepsilon_1 = \alpha/2$. By **lem:armwise-finite-rank-staircase**,

$$k_0 = k_1 = k, \quad \Gamma_{M,\alpha,\varepsilon}(w) = \Gamma_{M,\alpha}(w). \quad (1)$$

The deterministic-preorder interpretation, coherent common-target calculation, full-model disclaimer, binary counterexample, disjoint-support value, and comparator scope therefore specialize from **thm:asymmetric-sharp-orbit-coverage-frontier**. In particular, its bounded common-Bernoulli construction proves that the diagonal functional is not a uniform full-model joint-inclusion or causal effect-set coverage certificate.

The pointwise implication specializes to

$$\{Y_{0T}(1) \in I_{1,w}, Y_{0T}(0) \in I_{0,w}\} \subseteq \{\tau_T \in I_{1,w} - I_{0,w}\} = \{\tau_T \in \mathcal{C}_w\}. \quad (2)$$

If both intervals are $\{0\}$ and both outcomes equal K , the right event holds while the left event fails.

When $k \leq M$, substitute $\bar{k}_0 = \bar{k}_1 = \bar{k} = \min\{k, M - k + 1\}$ into the asymmetric seven-way partition. The rolling charge becomes

$$C_{M,k,k} = \sum_{i=0}^{M-1} \min\{\bar{k}, i+1\}^2. \quad (3)$$

Substitution into the asymmetric branch sum gives exactly B_{sym} , including the face and rank-one terms, and substitution into the live-workspace maximum gives W_{sym} . The adaptive, uniform, and memory bounds follow from **lem:coherent-two-finite-rank-frontier-evaluator**. Vertex, edge, face, and rank-one keys use constant auxiliary workspace, and rank-one keys do not trigger factorial preprocessing.

For every $M \geq 2$, $\alpha = 2/(M+1)$ gives $k = M$. The audited edge key $\pi = (1/2, 1/2, 0, 0)$ then satisfies

$$D_{M,M,M}(\pi) = 1 - 2^{-M}, \quad (4)$$

with $\Theta(\log M)$ exact time rather than $\Theta(M)$. The audited face key $\pi = (1/3, 1/3, 1/3, 0)$ satisfies

$$D_{M,M,M}(\pi) = 1 - 2 \cdot 3^{-M}, \quad (5)$$

with the same strict resource separation. The off-diagonal rank-one audit remains a strict certificate for the common evaluator: at $M = 2n$, $k_0 = n$, $k_1 = n+2$, its determinant-zero branch costs $\Theta(M)$ time and constant workspace rather than $\Theta(M^2)$ time and $\Theta(M)$ workspace. On the diagonal, the same determinant test has charge

$$R_{M,k,k} = 1 + 2\lceil \log_2 M \rceil + 2\bar{k}. \quad (6)$$

If $k = M + 1$, both rank events are certain by `prop:rank-boundary`, and therefore $\Gamma_{M,\alpha}(w) = 0$.

Equation (1) also transfers the exact four-level and binary values 24957/160000 and 759109/5000000, their positive gap 83189/20000000, and the value $1 - \alpha$ under disjoint support. The full-model guarantees remain those of `thm:common-selector-lp` and `prop:active-arm-finite-rank-orbit-gap-bound`. The literature comparison likewise specializes only as a comparison of different optimized objects. Consequently, the computational prepasses change no common-target, active-arm, pair-audit, empty-library, ordinary-rank, or weighted-orbit semantics. $\square$

Justification. This is the equal-split specialization of the deterministic asymmetric joint rank-inclusion frontier, with the same coherent common-target semantics and no full-model certificate interpretation. *Gap.* The functional $\Gamma_{M,\alpha}(w)$ does not cover stochastic conditional residual scores and is not a uniform full-model joint-inclusion or causal effect-set coverage certificate; within its deterministic submodel, binary scores need not attain the frontier and boundary ranks require separate treatment. *Consumer.* Use the symmetric evaluator only on the equal-split deterministic orbit-score submodel. For full-model effect coverage, use `thm:common-selector-lp` under exact balance or `prop:active-arm-finite-rank-orbit-gap-bound` under imbalance.

7 Interpretation

For the asymmetric ordinary procedure, simultaneous inclusion of $Y_{0T}(1)$ and $Y_{0T}(0)$ in their arm intervals implies inclusion of τ_T in their Minkowski difference, but the converse can fail. Exact orbit balance makes each arm’s calibration and target scores iid because their orbit mixtures coincide. The observable selector conditions used for this result are normalization and requested realized-arm support on $\text{supp}(\nu)$, together with conditional auxiliary draws; assignments outside the support have zero integrated weight. Quotient weighting corrects unequal orbit masses under quotient overlap. The deterministic frontier answers a different audit question: it computes the worst joint rank-inclusion deficit over deterministic orbit-score maps, preserving the common target and cross-arm calibration coupling. A full-model residual may remain random inside an orbit, so the audit cannot be read as causal effect-set coverage for every invariant law.

Technical internal limitation (diagnostic only)

All probability guarantees use iid replicated latent arrays, independent external training, a prespecified common target, a known invariant assignment law, normalized observable selector kernels supported on the requested realized arm on the assignment support, and independent selector auxiliary draws conditional on assignment. Off-support kernel extensions are immaterial because they receive zero design weight. The weighted theorem additionally requires orbit overlap. The exact frontier is restricted to deterministic bounded orbit-constant score maps and can be combinatorial in the arm-orbit counts; it is neither a supremum over all invariant latent laws nor a uniform full-model joint-inclusion or effect-set coverage certificate.

8 Honest scope

The manuscript proves finite-sample results for the stated replicated matched-block model, not design-based coverage for one fixed Shamba Maisha finite population. Exact balance gives ordinary split-conformal validity; imbalance gives active-arm total-variation bounds and explicit failures; quotient overlap gives weighted validity. The S_2 results require audited pair orbits. Selector kernels need only be normalized and supported on the requested realized arm for positive-mass assignments, together with the declared conditional selector draws; off-support extensions are immaterial. The quantities $\Gamma_{M,\alpha,\varepsilon}(w)$ and $\Gamma_{M,\alpha}(w)$ are exact only for deterministic bounded orbit-constant scores and are presented as audit evaluators, not as new general coverage theorems.

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